Abstract

Institutional response delay and periodic review are distinct operators, not interchangeable clocks, in renewable-resource models. For a declared continuous-feedback baseline, the linearized characteristic equation identifies two mobilising-channel stability crossings, while a small-gain condition yields delay-independent local stability for the protective rule. Interval enclosures certify local characteristic roots at the stated coefficients; the computed periodic-orbit topology remains numerical and is not a proved fold theorem. For periodic review, exact sample-and-hold monodromy and one-step command approximations produce different stability boundaries at the illustrative baseline: the exact mobilising map crosses near a 6.5-year review interval, while Euler steps add artefact crossings; the protective map remains spectrally stable over the tested grid. The 6.5-year value is parameter-sensitive, not a policy threshold: it ranges from 0.87 to 10.67 years across the $\pm 0.5\%$ joint-perturbation corners and is absent in 20 of 64 $\pm 1\%$ corners. A multiplicity-controlled screen of 42 fish stocks finds no robust target-band institutional cycle; a 32-system case search finds no unconfounded cross-sector oscillator under the declared criteria. The supported result is narrow: operator form and rule sign can change computed local stability at specified parameterizations. No coefficients or model topology are empirically identified for a named fishery or institution, and no threshold transfers between continuous delay τ and review interval T_r.

Governance latency: the delay, the clock, and the stability of periodically reviewed renewable resources

Amin Abaee[1]

[1]Independent Researcher, Tehran, Iran

ORCID: 0000-0002-0019-1842

amin_abaee@ut.ac.ir

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Part I — The continuous channel: governance delay as a lag in the control loop

Study scope and principal findings

Delayed feedback can alter stability in renewable-resource systems; ecological and informational delays are well studied. The analysis treats a delay that sits elsewhere — **governance delay**, the lag from observed decline to institutional response. A three-state model — harvested stock, filtered deficit memory, bounded effort — compares two rules: the **mobilising rule**, effort grows as more effort is deployed; and the **protective rule**, a quota-tracking law restoring harvest toward a cap.

For the mobilising rule, intermediate delay stabilises the equilibrium: two characteristic-root crossings, enclosed near 3.7 and 150 yr, bound the local stability window; numerical Lyapunov-coefficient calculations support a subcritical classification, but no validated nonlinear Hopf theorem is claimed. For the protective rule the loop gain stays below one at the stated baseline parameter vector: the equilibrium is exponentially stable at every delay — a no-Hopf theorem. Under periodic review the rules respond oppositely: annual protective review remains stable — its apparent 2.3-yr threshold is a discretisation artefact — while annual mobilising review is unstable, restabilising only above about 6.5 yr at the declared parameter vector through a Neimark–Sacker-type crossing — a value stable across discretisation schemes (6.50–6.73 yr) but not identifiable in the parameters, sweeping 0.87–10.67 yr under a half-percent joint perturbation and vanishing entirely in a third of cases under one percent (Part II, Section 3.4.1). The attractor topology is five-regime: two Krawczyk-enclosed candidates in the finite collocation systems, with continuous-delay fold validation open; the basin-transition and large-amplitude attractor interpretations remain numerical.

The equations form a stock-agnostic model class, but that does not establish transportability to other resource systems. The design coordinates — rule sign, deployment delay, and review cadence — are objects a new application would need to declare and test; the northern-cod timeline and the 32-system search in Part II provide context, not coefficients or validation. At the stated baseline parameter vector, feedback form and timing change local stability. The review interval is a local spectral design variable, not a universal prescription; more frequent review is not always stabilising in the sampled operator.

1. Introduction

1.1 Delays in the governance loop

Delayed feedback has a long history in resource dynamics. The classic results all place the delay inside the ecology itself:

- Hutchinson (1948) showed that a delayed self-limitation term destabilises logistic growth.
- Ezekiel (1938) documented the two-year lag between price signals and supply response in the pork cycle.
- Ludwig, Jones, and Holling (1978) placed delays and hysteresis at the centre of outbreak dynamics.
- Gurney, Blythe, and Nisbet (1980) gave the canonical stability analysis of the single-species delayed logistic equation.
- Costantino et al. (1995) demonstrated delayed-feedback cycles in controlled laboratory populations.
- A large contemporary literature continues this line, with delays placed in the ecological dynamics themselves: maturation and stage structure (Aiello and Freedman, 1990;

Kuang, 1993), and harvested predator–prey systems whose paired ecological delays are Hopf objects in their own right (Zhang, Shen, and Chen, 2013).

The delay studied in this paper sits elsewhere: in the *institutional* loop — **governance delay**, the lag between the institution's observation of a decline and the response that finally acts on it. Stock assessment produces a signal of decline. The signal is filtered into a deficit memory. The memory drives a change in deployed extraction effort, with a deployment delay between decision and action. The behavioural basis for such lags is documented. Decision makers in renewable-resource experiments systematically misperceive the dynamics of the systems they govern, responding too slowly and in the wrong functional form (Moxnes, 1998). The modelling counterpart treats the delay as one of knowledge: delayed knowledge of the harvested population state destabilises bioeconomic equilibria and induces resource-harvester cycles, which harvester forecasting can dampen (Adamson and Hilker, 2020). Institutional design determines which signals reach which actors at which time (Ostrom, 1990). The management record supplies the field-scale counterpart: the northern cod collapse proceeded through years of assessment advice and incremental institutional response before the 1992 moratorium (Hutchings and Myers, 1994; Walters and Maguire, 1996), and the assessment-frequency literature now studies the review cadence itself as a managed design parameter with measured consequences for yield and depletion risk (Li, Bence, and Brenden, 2016; Peterson et al., 2022).

What the dynamical-systems literature has not supplied is an analysis of what this delay *does* to the stability of a harvested-stock system when the delay sits in the governance controller rather than in the ecology. Two questions follow, and the paper takes up both on a declared class of stock–memory–effort models. First, does the sign structure of the feedback law — mobilising versus protective — select between qualitatively different bifurcation behaviour? Second, what happens when the delay is replaced by periodic review (sample-and-hold)?

The institutional loop is not fisheries-specific, and neither is the analysed class. Assessment on a fixed cadence, a rule that converts the assessed signal into a quota or an effort decision, and a deployment lag between decision and action — this structure recurs across renewable-resource institutions, from quota-tracked harvests to capped withdrawals. It is the formal signature of a coupled human–natural system whose human component is a controller with memory. The model class of Section 2 is deliberately stock-agnostic: the state N is a renewable stock in material or biomass units, not a named fishery. Section 9 registers the same governance structure acting on a stock whose own ecology carries a maturation delay, so institutional and ecological delay are analysed within one frame. Fisheries supply the motivating instance — the setting in which periodic review is most literally institutionalised — and the management language of the two rules (effort escalation versus quota tracking) is the language of harvest control.

The overshoot mechanism analysed throughout has a one-sentence single-asset form. A hen produces a sustainable flow of eggs $S(N)$ each year; consumption D above $S(N)$ cannot be conjured, so the shortfall is covered by slaughtering the hen — the asset is destroyed to meet present demand. The loop is self-reinforcing. A deficit raises liquidation pressure, liquidation lowers the stock, the lower stock lowers sustainable yield, and the deficit grows. With the institutional deployment delay of this paper inserted into the loop, the deficit is observed, filtered, and answered with a lag. The hen's loop is then the delayed mobilising channel, whose local bifurcation analysis occupies Sections 2–5; the protective channel is Section 6; sample-and-hold review, Section 7; the global numerics, Section 8. The delay does not create the shortfall; it decides when the response to it arrives. The phase-stabilised window of Section 5.1 is the interval of delays inside which the lag, not the feedback, holds the loop stable.

Not all liquidation channels are the same. Killing the hen removes a reproducing adult at once (culling the standing stock). Eating fertilised eggs suppresses next year's recruits without touching a living adult (recruitment suppression). Eating an apple often leaves the seed unharmed, so consumption and reproduction can be substantially decoupled. The first two

channels are the two-channel mixture registered in Section 8.3 (M3-LC). That is why the channel assignment there requires age-, stage-, or replenishment-specific evidence rather than local spectra alone.

1.2 Contributions

1. **A gated three-state core.** A stock–memory–effort system (Section 2) in which a multiplicative saturation gate enforces the effort boundary exactly, making the admissible box forward invariant (Theorem 2.1). The model is a mathematical parameterisation, not a calibration to a named fishery; two reference parameter sets (Candidates A and B) serve as sensitivity anchors.

2. **The complete Hopf cubic and the even-pairs algebra.** The characteristic quasi-polynomial of the linearisation reduces to a cubic modulus condition with a phase relation (Theorem 4.1). At the interior equilibrium, a filter identity makes the cross term of the cubic vanish identically, so the total algebraic multiplicity of positive roots is even — generically zero or two distinct simple roots (Corollary 4.1). The two-frequency structure of the architecture is thereby an algebraic fact, not a numerical accident.

3. **Interval-certified crossings.** The two imaginary-root crossing delays of the mobilising channel are enclosed by interval Newton iteration on the cubic with outward rounding — near 3.7 and 150 yr at Candidate A, with the 13-digit enclosures displayed in the table of Section 5.1. The certification tier is stated precisely: the enclosures certify the *local spectrum* of the cubic. The method of Church and Lessard (2022), which certifies full Hopf bifurcation for retarded functional differential equations, is the stronger standard and is not claimed here.

4. **The sign separation.** The mobilising law (gain growing with deployed effort, $C_Z > 0$) and the protective quota-tracking law (restoring gain, $C_Z < 0$) yield opposite local mathematics on the same stock–memory block. The mobilising channel carries the Hopf pair, both crossings subcritical (Section 5.2). The protective channel carries a **no-Hopf theorem** — all cubic coefficients positive, Descartes' rule and Routh–Hurwitz exclude every imaginary root, and the loop gain peaks at $0.08011 < 1$ — so exponential stability holds for every delay (Theorem 6.1). The two channels are interpolated, and a weighted small-gain theorem together with a conditional weight threshold show that delay-induced instability requires a sufficiently large mobilising weight (Section 5.4).

5. **The review interval as control.** Under sample-and-hold review with one explicit Euler step per period — the update timing fixed by the flow-then-update (pre-review/post-review) convention of Section 7 — the monodromy (the one-period return map of the reviewed loop) is computed in closed form (Proposition 6.2, Theorem 7.1). The protective channel is stable under annual review (spectral radius below one), and its apparent instability threshold near 2.3 yr is a discretisation artefact of the Euler update — provably not a Hopf of the continuous system (Sections 6.2 and 6.4). The mobilising channel is unstable under annual review (spectral radius just above one under both the Euler and the exact update) and has its restabilising complex unit-circle crossing — the spectral signature of a Neimark–Sacker bifurcation, with nonlinear conditions not verified — at about 6.5 yr under the exact held-measurement update (Proposition 7.1). The Euler-reported half-century crossing and its companion -1 crossing are command-step artefacts. For a slow stock under this review operator, the review interval is a local spectral design parameter, and the continuous-delay recommendation does not transfer to periodic review (Section 7).

6. **Global numerics at declared status.** Continuation, Floquet-multiplier tracking, and basin tests give a five-regime attractor topology at declared certification tiers — and the tiers are stated precisely rather than as a single "certified" claim. Under the pre-registered five-regime campaign of Section 8.2, the lower boundary resolves into a single fold near 5.6 yr of one S-shaped branch — the subcritical-Hopf small arm and the stable large arm meet near a Krawczyk-enclosed Moore–Spence zero of the finite collocation system — and an analogous enclosure occurs near 64.4 yr for the second branch. The latter has recorded finite-dimensional nondegeneracy checks; neither enclosure by itself establishes a fold of the continuous delay equation. For both, the continuum off-grid residual stage and the continuous-delay lift remain **open**; a rigorous saddle-node result for the delay equation itself is not claimed (Beretka and Vas, 2020 covers specific classes; none is claimed for this system). The saddle-node-of-periodic-orbits classification is stated at its certification tier in Section 8.2. The stable returning arm of the upper branch is not generically reachable near the fold, so the capture onset near 149 yr is a basin boundary rather than a fold; the object captured there is a large-amplitude face cycle that no campaign could Fourier-collocate at $m = 64$ or 128, and the link from the stable returning arm's last record to that cycle over an $\approx$ 84-yr gap is an unverified conjecture. The five-regime reading of attractors (iv)–(v) is therefore a numerical reading that combines finite-dimensional fold candidates with an uncollocatable, unverified-capture region, and is reported as such.

1.3 Organization

Section 2 defines the model class and its admissible domain. Section 3 derives equilibria and the characteristic equation. Section 4 develops the Hopf cubic and the even-pairs algebra. Section 5 analyses the mobilising channel, and Section 6 the protective channel; together they carry the sign separation — opposite local mathematics on the same stock–memory block. Section 7 treats sample-and-hold review: the review interval is a spectral design parameter in its own right, and the reviewed loop is a hybrid system, not the delay equation sampled at the review interval. Section 8 reports the global numerics and their certification levels. Section 9 registers the maturation-delayed recruitment system — the applied analogue in which the same governance structure acts on a stock whose own delay is the maturation lag from spawning to recruitment — and records the companion stage-analysis fine-map bands and their nonlinear ground truth. Section 10 unifies the two channels in the loop-gain family. Section 11 discusses design consequences, the generality of the institutional-loop coordinates, the documented institutional timelines that ground the two timing coordinates, the ecological reading of the registered records, the relation to the early-warning literature, and open problems. Section 12 concludes.

1.4 Related work

Delays in population dynamics. That a lag inside a feedback loop can destabilise an otherwise stable equilibrium is classical. Hutchinson (1948) framed it as circular causality; the delay-differential machinery that turns the remark into a theorem — characteristic quasi-polynomials, stability switching, Hopf bifurcation — is textbook (Hale and Verduyn Lunel, 1993; Diekmann et al., 1995; Kuang, 1993; Hassard, Kazarinoff and Niculescu, S.-I., 2007. *Stability and Stabilization of Time-Delay Systems: An Eigenvalue-Based Approach.* SIAM. doi:10.1137/1.9780898718645.

Pyragas, K., 1992. Continuous control of chaos by self-controlling feedback. *Physics Letters A*, 170, 421–428. doi:10.1016/0375-9601(92)90745-8.

Wan, 1981; Hayes, 1950). What that literature delays is, almost without exception, a *biological* process: maturation, gestation, digestion, recruitment. Aiello and Freedman (1990)

and Gurney, Blythe and Nisbet (1980) are representative. The delay analysed here is none of these, and it is not a delay in the stock's own dynamics at all.

Informational and knowledge delay. A second strand delays the *observation* rather than the process. Adamson and Hilker (2020) show that delayed knowledge of the harvested population state can induce resource–harvester cycles, and that forecasting by harvesters can damp them; Li, Bence and Brenden (2016) and Peterson et al. (2022) study the assessment-frequency analogue. This is closer in spirit to what follows, but the mechanism differs: in those models the lag enters the *argument* of a feedback law evaluated on the current state, whereas here it enters the *state* the institution is tracking, and the loop is closed by a rule that carries dynamics of its own.

Management and governance delay as a cost. The empirical and simulation literature on delay in management response treats it as unambiguously harmful. Shertzer and Prager (2007) simulate a spectrum of fisheries and find that increased delay requires larger catch reductions sustained over more years to recover benchmark status, and causes collapse most often under depensatory recruitment or density-dependent catchability. Brown et al. (2012) ask how long management can wait, and find that delays beyond five years raise collapse probability under environmental decline, with the long-run cost of delay exceeding the short-term benefit of overfishing. Karlsson and Gilek (2020) treat delay as a governance pathology to be coped with. A recent review of 101 studies, organised by a temporal Driver–Pressure–State–Impact–Response scheme, reaches the same verdict at cross-domain scale, and draws the methodological moral that response lags must be *decoupled* from ecosystem lags if either is to be understood (Hocherman, Trop and Ghermandi, 2025).

Delay as a stabilising mechanism. The framing of delay as a cost is not the only one the systems literature offers, and the alternative should be named before this paper's finding is compared to either. Delayed feedback is used *deliberately* as a stabilising mechanism: Pyragas's method feeds back the difference between the current and a delayed state to stabilise an unstable orbit without prior knowledge of its location (Pyragas, 1992), and delayed positive feedback is known to stabilise oscillatory systems that undelayed feedback does not (Abdallah, Dorato, Benites-Read and Byrne, 1993); the general question of when a delay stabilises rather than destabilises is the subject of the time-delay stability and stabilisation literature (Niculescu, 2007).

The resemblance to what follows is real and is not minimised here; the difference is in who chooses. In that literature the delay is a *control design parameter* — an engineer introduces it, tunes it, and proves the closed loop stable at the chosen value. Here the delay belongs to the institution, nobody introduces it as a stabiliser, and the finding is that a bounded intermediate range of it happens to stabilise while both shorter and longer latencies do not. That is a statement about the window a management body may find itself inside, not a design prescription, and it is why the result is reported as a bracketed window carrying its perturbation band rather than as a recommendation.

Where this paper departs. Every source above treats delay as a cost: more delay is worse, and the operative question is how much can be tolerated. The central finding here is different in kind, not in degree. For the mobilising rule, *intermediate* governance delay stabilises the equilibrium, and the stabilising set is a bounded window — two interval-enclosed local spectral crossings, with numerically evaluated subcritical Hopf signatures, bracket it (Section 5). Delay is not merely tolerable inside that window; it is the stabilising mechanism. Because the window is bounded above as well as below, the result does not contradict Shertzer and Prager (2007) or Brown et al. (2012): it localises the range over which their monotonicity should be expected to hold, and identifies the range where it should not. As to the review's methodological call, this paper meets it by construction — the models below carry governance delay and no ecological delay, so the two lag families cannot be confounded, and any delay-induced stability is attributable to the institutional loop alone.

On certification. The crossing values reported here are interval-certified rather than floating-point, using the computer-assisted apparatus for Hopf bifurcation in functional differential equations (Church and Lessard, 2022; Church and Queirolo, 2024) over interval arithmetic (Cloud, Moore and Kearfott, 2009). That apparatus is now standard in rigorous dynamics but remains rare in the resource-governance literature, where crossing values are typically reported as point estimates from numerical continuation. The difference matters here because the claimed object is an interval, not a point: a stabilising window is a statement that every delay in a range has a property, and a floating-point sample of that range does not establish it.

2. The Model Class

Notation. Throughout, N is a renewable stock (material or biomass units); Z is a filtered deficit-memory state (stock per time); and E is an extraction effort — a dimensionless institutional deployment intensity, not a conserved material or energy stock. The catchability q has units $(\text{effort}\cdot\text{yr})^{-1}$. The signal references Δ_{ref}, Z_{ref} and the offset δ have stock-per-time units; η and δ_0 carry yr^{-1} and effort-per-time units. These assignments make each equation dimensionally homogeneous without treating effort as physical mass. The delay τ is a discrete action (deployment) delay. The filter timescale τ_m is an ordinary state relaxation time, not a second delay. The two sign channels are called the *mobilising channel* — the feedback law whose gain grows with deployment — and the *protective channel* — the quota-tracking law whose gain restores toward a cap. Periodic-review results use *sample-and-hold*: an Euler-reviewed zero-order-hold sampling of the delayed signal at intervals of length T_r. In plain words, the delayed signal is measured once at each review instant and held fixed between reviews, so the review interval T_r is the cadence at which the institution re-decides.

2.1 The gated three-state core

Let

$$S(N) = rN\left(1 - \frac{N}{K}\right), \qquad \mathrm{sp}_k(s) = \frac{1}{k}\log(1 + e^{ks}), \qquad \Phi_k(s) = \max\left\{0,\ \mathrm{sp}_k(s) - \frac{\log 2}{k} + \delta\right\},$$

where Φ_k is a shifted, non-negative signal map. The boundary-exact (gated) three-state core is

$$\begin{aligned} \dot{N} &= S(N) - qEN, \\ \dot{Z} &= \frac{1}{\tau_m}\big[\Phi_k(qEN - S(N)) - Z\big], \\ \dot{E} &= \left(1 - \frac{E}{E_{\max}}\right)\left[\eta E\left(\frac{Z(t-\tau)}{\Delta_{\text{ref}}} - \frac{E}{E_{\max}}\right) + \delta_0 \frac{Z(t-\tau)}{Z_{\text{ref}} + Z(t-\tau)}\right]. \end{aligned} \tag{1}$$

Three structural features of (1) carry everything that follows:

- **Depletion filtering.** The memory input is a smoothed version of the stock-decline rate: on this core the identity $qEN - S(N) = -\dot{N}$ holds exactly, so Φ_k filters $-\dot{N}$.
- **The multiplicative gate.** The gate $(1 - E/E_{\max})$ is essential to what follows. It is a hard saturation architecture, not a generic effort law, and it enforces $E \in [0, E_{\max}]$ by construction.
- **The institutional delay.** The effort law reads the filtered memory at one fixed lag — the deployment delay τ between decision and action.

The registered parameterisations are:

Parameter	Candidate A	Candidate B	Role
r	0.02	0.02	yr^{-1}; stock renewal
K	100	100	stock units; normalisation
q	0.001	0.001	$(\text{effort}\cdot\text{yr})^{-1}$
η	0.914	2.756	yr^{-1}; effort response
$E_{\max}$	30	26	effort units; saturation boundary
Δ_{ref}	1.0	1.0	stock yr^{-1}; signal scale
δ_0	0.01	0.01	effort yr^{-1}; baseline source
τ_m	5	5	yr; filter relaxation
Z_{ref}	1.0	1.0	stock yr^{-1}; signal scale
δ	$\log 2/10$	$\log 2/10$	stock yr^{-1}; signal offset
k	10	10	$(\text{stock } \mathrm{yr}^{-1})^{-1}$; regularisation
τ	varied	varied	yr; action/deployment delay

These are mathematical parameterisations and sensitivity anchors, not a joint calibration to a named resource or institution. The institutional coefficients have not been independently identified from a field system. Candidates A and B are two points in the effort-response chart, not rescalings of one class. For the reported pair $\delta = \log(2)/k$ the outer floor cancels algebraically ($\Phi_k(s) = \mathrm{sp}_k(s) > 0$ for every finite s), so the floor is inactive on every reported periodic orbit. This identity is parameter-specific; for $\delta \neq \log(2)/k$ floor contact must be checked orbit by orbit.

2.2 Admissibility

Theorem 2.1 (Forward invariance of the admissible region)**.** *For the history class $\varphi \in C([-\tau, 0], \mathbb{R}^3)$ define*

$$\mathcal{D} = \{0 \le N \le K,\ Z \ge 0,\ 0 \le E \le E_{\max}\}.$$

Assume
(H1) the initial history lies in $\mathcal{D}$ — every history value, not only the endpoint.
Then every classical solution of (1) remains in $\mathcal{D}$ for as long as it exists.

Proof. Check the five boundary faces. (i) $N = 0$: $S(0) = 0$ and the harvest qEN vanishes, so $\dot{N} = 0$; the face is invariant. (ii) $N = K$: $S(K) = 0$, so $\dot{N} = -qEK \le 0$, inward. (iii) $Z = 0$: the source $\Phi_k(qEN - S(N)) \ge 0$ by construction of the non-negative signal map, so $\dot{Z} = \Phi_k(\cdot)/\tau_m \ge 0$, inward. (iv) $E = 0$: the gate factor is positive (away from $E = E_{\max}$) and the bracket is $\delta_0 Z_\tau/(Z_{\mathrm{ref}} + Z_\tau) \ge 0$, so $\dot{E} \ge 0$, inward. (v) $E = E_{\max}$: the gate vanishes, $\dot{E} = 0$; the face is invariant. The delay enters only through $Z(t - \tau)$, and the initial history lies in $\mathcal{D}$, so the delayed argument is non-negative throughout the first interval $[0, \tau]$. The vector field is therefore subtangential on the first interval by (i)–(v); the argument repeats by induction over $[n\tau, (n+1)\tau]$ by the method of steps (Hale and Verduyn Lunel, 1993). At $\tau = 0$ the same faces give subtangentiality of the ordinary differential equation directly. □

Corollary 2.1 (Boundedness and global continuation)**.** *Let*

$$\bar{Z} = \max\{\sup_{[-\tau,0]} Z,\ \Phi_k(qE_{\max}K)\}.$$

Then every solution satisfies $0 \le Z(t) \le \bar{Z}$; all three states remain in a bounded set on which the vector field is locally Lipschitz; and the solution continues for all $t \ge 0$.

Proof. The Z-equation reads $\dot{Z} = d(\nu(t) - Z)$ with $d = 1/\tau_m$ and the monotone bounded input $\nu(t) = \Phi_k(qE(t)N(t) - S(N(t))) \le \Phi_k(qE_{\max}K)$. The letter ν is used only in this section, for the filter input; it denotes the Halanay decay rate in Section 10.2. Variation of

constants gives $Z(t) = e^{-dt}Z(0) + d\int_0^t e^{-d(t-s)}\nu(s)\,ds \le e^{-dt}Z(0) + \bar{Z}(1-e^{-dt})$, hence $Z(t) \le \max\{Z(0), \bar{Z}\} \le \bar{Z}$ by the definition of $\bar{Z}$. Nonnegativity of Z is part of Theorem 2.1. With $N \in [0, K]$, $E \in [0, E_{\max}]$, and $Z \in [0, \bar{Z}]$, the state lies in a compact set on which the right-hand side of (1) is locally Lipschitz: the logistic and the polynomial effort law are smooth; the signal map Φ_k is locally Lipschitz in general and smooth where its floor is inactive — in particular, at the registered relation $\delta = \log 2/k$ the floor cancels and $\Phi_k = \mathrm{sp}_k$ is C^1 everywhere. The solution therefore extends for all time. Local Lipschitz regularity suffices for the existence, uniqueness, and continuation arguments; the C^1 statements are needed only where bifurcation coefficients are computed, and there the floor-inactive parameter sets of Section 5.2 are the declared scope. One further observation is recorded here because Section 3 uses it. On the invariant extinction face $N = 0$ the memory input is $\Phi_k(0) = \delta$, so Z relaxes to δ, and the extinction rest carries the same admissible positive effort root as the interior branch: institutional memory sustains baseline commanded effort against zero realised harvest. Effort is an institutional deployment intensity, not a conserved stock; the core makes no closed-effort-energetics claim, and a materially closed application must add and donor-limit that support explicitly. □

2.3 The registered model family

The reference member of the family is **M3-B (boundary-exact gated)** — the core (1) itself, whose multiplicative saturation gate enforces $E \in [0, E_{\max}]$ exactly (Theorem 2.1). Its crossings, folds, and basins are the paper's reference records (Sections 5.1 and 8.3). Four further variants delimit the robustness of those results. No invariant set, equilibrium formula, local threshold, periodic branch, or admissibility result transfers between them without a separate argument.

- **M3-U (ungated).** The same ecology with the effort law without the outer gate; $E_{\max}$ is then a self-limitation scale only. The gate factor is the algebraic source of threshold relocation between the variants (Section 3.2).
- **M3-LC (two-channel stock law).** The logistic renewal is written as a gross birth–mortality decomposition $B(N) = S(N) + \kappa r N$, $M(N) = \kappa r N$ ($\kappa \ge 0$), and the pressure qEN is split by $\psi \in [0, 1]$:

$$\dot{N} = \max\{0,\ B(N) - (1-\psi)qEN\} - M(N) - \psi qEN. \tag{2}$$

Here ψqEN is a realised removal of standing stock and $(1-\psi)qEN$ is demographic suppression — a prevented inflow, not a material transfer out of the adult-stock compartment. Equation (2) is a phenomenological stock equation, not a closed mass ledger. Whenever the recruitment floor is inactive, (2) reduces exactly to $\dot{N} = S(N) - qEN$. The floor never binds at the interior equilibrium (the binding condition reduces to $-\psi S(N) > \kappa r N$, impossible), so the equilibrium, the Jacobian, the characteristic equation, and both Hopf points are independent of ψ and κ. Local equality does not imply excursion equality: the floor truncates recruitment on large excursions, and the two channels diverge there (Section 8).

- **M4-A (active-pool extension).** Logistic renewal is replaced by $R(N, A) = rN(1 - N/K)\,A/(A + A_0)$ with a dynamic active-support pool $\dot{A} = -B(N, A) + \omega_A(A^{\mathrm{eq}} - A)$ and a fully declared donor-limited gross draw B. The relaxation term makes this a reduced open-pool model unless its donor/receiver reservoir is included explicitly. Freezing A is not a justified fast-variable elimination at the baseline $\omega_A = 10^{-3}$ yr^{-1}.
- **MPF (primitive-flux core).** Living biomass X, detritus U, active material $A = \mathcal{M} - X - U$, primitive fluxes $g(X, A) = \mu X A/(K_A + A)$, $m(X) = dX + cX^2$, $h(X, E) = qEX$, signed memory $\dot{Z} = (-\dot{X} - Z)/\tau_m$, the detritus equation $\dot{U} = m(X) - \gamma_U U$ — the

detritus compartment receives the full mortality flux $m(X) = dX + cX^2$ and exports it at the specific rate $\gamma_U > 0$, yr^{-1}; this is the equation the invariance argument below uses — and the bounded effort law $\dot{E} = (1-E/E_{\max})[\eta EZ(t-\tau)/\Delta_{\mathrm{ref}} + \delta_0 - \eta E^2/E_{\max}]$. The core has a signed zero-equilibrium memory ($Z^* = 0$, a true constant baseline), detritus, and a different effort equilibrium; it is not (1). On the boundary $X + U = \mathcal{M}$ the active pool is $A = 0$ and $g(X, 0) = 0$, so $\frac{d}{dt}(X + U) = -qEX - \gamma_U U \leq 0$, which — with $g(0, A) = 0$, $m(0) = 0$, and the donor-limited flux assumptions — proves forward invariance of the simplex $\{X \geq 0,\ U \geq 0,\ X + U \leq \mathcal{M}\}$ for the ecological subsystem under admissible effort.

2.4 The four-state working core and its relation to (1)

The working four-state core restores the active abiotic pool as a state: $\dot{N} = R(N, A) - qEN$, $\dot{A} = -B(N, A) + \omega_A(A^{\mathrm{eq},W} - A)$ with $A^{\mathrm{eq},W} = A^{\mathrm{eq,intrinsic}} + \kappa_A K/\omega_A$, and the Z and E equations of (1) unchanged except that $S(N)$ is replaced by $R(N, A)$. At the baseline $\omega_A = 10^{-3}$ yr^{-1}, $\kappa_A = 0.05$ yr^{-1}, $A_0 = 0.01K$, $A^{\mathrm{eq,intrinsic}} = 0.5K$, so $A^{\mathrm{eq},W} = 5050$.

Two closure statements fix its status. First, in the ideal large-reservoir limit $\sigma_{\mathrm{geo}} = 1$ — where $\sigma_{\mathrm{geo}} \in (0, 1]$ is the geological-reservoir adequacy ratio, a dimensionless companion-ledger quantity measuring how far the geological reservoir sustains the working core's donor draw ($\sigma_{\mathrm{geo}} = 1$ the ideal limit, $1 - \sigma_{\mathrm{geo}}$ the scale of the finite-reservoir perturbation) — the specialised system satisfies the working-core equations exactly (an algebraic identity check), with detritus a driven auxiliary that does not feed back; for finite geological reservoirs the vector field is perturbed by $O(1-\sigma_{\mathrm{geo}})$ and U feeds back. Second, detritus slaving under a fixed intrinsic target is an $O(\varepsilon_U)$ singular-perturbation estimate on compact intervals — a standard Tikhonov/Fenichel-type argument (Hale and Verduyn Lunel, 1993, Ch. 9, Diekmann et al., 1995); ε_U is the fast-variable slaving parameter, of the order of the slow-to-fast timescale ratio r/γ_U, with γ_U the detritus export rate of the MPF core, Section 2.3. At the baseline $\gamma_U/r = 10$ detritus is an order of magnitude faster than the stock, so the fast-variable slaving parameter $r/\gamma_U \approx 0.1$ is small and the compact-interval estimate holds. It is only moderately small, however, so it does not control global periodic orbits.

At Candidate A the quasi-steady (QSS) core has a positive low-A equilibrium $(N^*, A^*) \approx (23.85, 0.159)$ and no high-A near-logistic exploited equilibrium (the high-stock branch gives $A^* \approx -137$, inadmissible). The QSS core is a distinct singular-limit object: a valid limit that is not dynamically connected to the high-A working equilibrium used for the reported thresholds, and the two objects are never merged. The working core is declared as an open projection: omitted turnover is routed to a diagnostic detritus/inert sink; the reduced (N, A, Z, E) trajectory is not mass-closed by itself; and its global periodic results are model-version-specific. The working equilibrium is a frozen-donor quasi-equilibrium sustained by geological support of order $\omega_A(A^{\mathrm{eq},W} - A^*) \approx 4.652$ stock units yr^{-1}, whose cumulative donor change is $4.652\,T/A^{\mathrm{geo}}$ — 1.2% on a century horizon at the lower geological ratio $A^{\mathrm{geo}}/A^* = 10^2$ and 0.12% at 10^3.

Remark (companion-interface numbers). *The quantities $A^{\mathrm{eq},W} = 5050$, the geological support ≈ 4.652 stock units yr^{-1}, and the century draws 1.2%/0.12% are interface values carried from the companion material-ledger paper — where the closed mass ledger and its quasi-steady slaving target are the subjects — for cross-paper consistency. They are not hypotheses of any theorem of this paper, and they do not make the working-core quasi-equilibrium a rest point of a closed ledger: the four-state working core is an open projection, sustained by continuing geological support, precisely as stated above. If this paper is read alone, the four-state core should be taken as an open working model, with the 4.652 accounting a remark citing the companion rather than a structural result.*

The finite-time connection between the four-state and three-state cores is the following bound. It justifies using (1) for local, near-equilibrium questions on institutional timescales — but not the large-amplitude cycle or its period, and not spectral quantities. A trajectory

bound does not transfer to Hopf thresholds, so the proximity of the four-state crossings to the three-state values (with τ_- 3.2% higher and τ_+ 0.2% lower than the three-state values, Section 8.4) is reported as a numerical observation, not derived from this bound.

Lemma 2.1 (Frozen-active-pool approximation)**.** *Let (N, Z, E) solve (1) and let $(\tilde{N}, \tilde{Z}, \tilde{E})$ solve the four-state core from the same initial data, with $A(t) \geq A_{\min} > 0$ on $[0, T]$. Measure state differences in the scaled norm*

$$\|\xi\|_* = \frac{|\Delta N|}{K} + \frac{|\Delta Z|}{rK} + \frac{|\Delta E|}{E_{\max}},$$

where $\Delta N = \tilde{N} - N$, $\Delta Z = \tilde{Z} - Z$, $\Delta E = \tilde{E} - E$ (stock, flux, and effort scales respectively). Assume

(H1) the vector field of (1) is Lipschitz on the admissible region.

Then

$$\sup_{t\in[0,T]} \|\xi(t)\|_* \leq C_T \frac{A_0}{A_{\min} + A_0}$$

with C_T a constant depending on the structural parameters and T. This is an inner approximation on $[0, T]$, not a Tikhonov reduction.

Proof. The four-state renewal differs from the saturated renewal by

$$R(N, A(t)) - S(N) = rN\left(1 - \frac{N}{K}\right)\left(\frac{A(t)}{A(t) + A_0} - 1\right) = -rN\left(1 - \frac{N}{K}\right)\frac{A_0}{A(t) + A_0},$$

so the mismatch is bounded pointwise by $\frac{rK}{4} \cdot \frac{A_0}{A_{\min}+A_0}$ (the renewal factor is at most $rK/4$ and the saturation deficit is at most $A_0/(A_{\min} + A_0)$). The Z and E equations of the two cores coincide in form, so their difference equations are driven only by ΔN and by the renewal mismatch $\sigma(t) = R(\tilde{N}, A(t)) - S(\tilde{N})$, with $|\sigma(t)| \leq \frac{rK}{4}\frac{A_0}{A_{\min}+A_0}$. The difference system is retarded (the E-row contains $\Delta Z(t - \tau)$), so the estimate is taken in the history norm: with $Y(t) = \sup_{s\in[-\tau,t]} \|\xi(s)\|_*$, Lipschitz continuity of the common vector field gives the retarded differential inequality

$$Y(t) \;\leq\; C_1 \int_0^t \left(Y(s) + \frac{A_0}{A_{\min} + A_0}\right) ds,$$

for the structural constant C_1 — the scaled-norm Lipschitz constant of the difference vector field, finite by the Lipschitz hypothesis of the statement. The retarded Gronwall estimate (Hale and Verduyn Lunel, 1993, Ch. 6) then yields $\sup_{s\in[-\tau,T]} \|\xi(s)\|_* \leq C_T A_0/(A_{\min} + A_0)$ on $[0, T]$, with $C_T = C_1 T e^{C_1 T}$ after absorbing constants; identical initial data give $\xi(0) = 0$. □

2.5 The protective channel and the two-channel interpolation

A protective institution is the quota-tracking law

$$\dot{E} = \left(1 - \frac{E}{E_{\max}}\right)\eta_p\big(E_{\text{cap}}(Z(t - \tau_p)) - E\big), \tag{3}$$

where E_{cap} is C^2, positive, strictly decreasing, with the calibration $E_{\text{cap}}(Z) = E_0 Z_{\text{ref}}/(Z_{\text{ref}} + Z)$ and $E_0 = E_A^*(Z_{\text{ref}} + \delta)/Z_{\text{ref}}$. The calibration places the unique interior rest of (3) on the stock–memory block of (1) at the Candidate A point $(N^*, Z^*, E^*) = (89.55188, \delta, 2.08962)$, so the stock–memory block is identical to (1)'s and only the effort law changes. At that rest $E_{\text{cap}}(\delta) = E^*$ and the linear gains are

$$C_E = -\left(1 - \frac{E^*}{E_{\max}}\right)\eta_p, \qquad C_Z = \left(1 - \frac{E^*}{E_{\max}}\right)\eta_p E'_{\text{cap}}(\delta),$$

which at $\eta_p = \eta_A = 0.914$ give $C_E = -0.850336$ and $C_Z = -1.661702$. Both signs are those of a restoring quota, not of scarcity mobilisation; the mobilising counterpart of (1) at the same point has $C_Z = +1.785$ and $C_E = -0.0595$ (Section 3.2). The channel-separation object of this paper is exactly this sign discipline.

The two-channel interpolation replaces the effort law by

$$\dot{E} = \left(1 - \frac{E}{E_{\max}}\right)\left[\chi_m F_m(E, Z(t-\tau_M)) + \chi_p \eta_p\big(E_{\text{cap}}(Z(t-\tau_p)) - E\big)\right], \tag{4}$$

with $\chi_m, \chi_p \geq 0$; the pure mobilising channel is $(\chi_m, \chi_p) = (1, 0)$, pure protection $(0, 1)$. The mobilising bracket is

$$F_m(E, Z) = \eta E\left(\frac{Z}{\Delta_{\text{ref}}} - \frac{E}{E_{\max}}\right) + \delta_0 \frac{Z}{Z_{\text{ref}} + Z},$$

the bracket of the mobilising law (1) written with its Z-argument displayed; τ_M is the mobilising deployment delay and τ_p the protective one. Throughout, the memory-filter timescale keeps its own symbol τ_m (used in the filter $\dot{Z} = (\Phi_k(\cdot) - Z)/\tau_m$); the two are distinct objects, distinguished by subscript wherever they co-occur.

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3. Equilibria and the Characteristic Equation

3.1 The interior equilibrium and the extinction face

At any interior equilibrium $qE^*N^* = S(N^*)$, the signal argument vanishes, the floor is inactive, and

$$Z^* = \Phi_k(0) = \delta,$$

independent of τ_m, k, and (N^*, E^*). Substituting into $\dot{E} = 0$ (away from $E = E_{\max}$) produces

$$-\frac{\eta}{E_{\max}}(E^*)^2 + \eta\frac{\delta}{\Delta_{\text{ref}}}E^* + \delta_0\frac{\delta}{Z_{\text{ref}} + \delta} = 0,$$

whose constant term is positive and quadratic coefficient negative, so there is exactly one positive root. Admissibility requires $0 < E^* < \min\{E_{\max}, r/q\}$, and then $N^* = K(1 - qE^*/r)$, positive if and only if $qE^* < r$. The equilibrium is independent of τ and k, which does not make its stability delay-independent. One certified value is used throughout: at Candidate A the positive quadratic root is $E^* = 2.08962$ and $N^* = K(1 - qE^*/r) = 89.55188$, with $Z^* = \delta \approx 0.0693$. (The four-state donor-limited equilibrium of Section 8.4, $N^* = 89.52562$, $A^* = 397.8665$, is a distinct object and is never used as the three-state rest.)

On the extinction face $N = 0$ the same zero raw signal gives $Z = \delta$, and the gated law has both the interior-effort extinction rest $(0, \delta, E^*)$ — when the interior root is admissible — and the boundary rest $(0, \delta, E_{\max})$ created by the multiplicative gate. The stock-direction eigenvalue at either branch is $r - qE$. The interior positive-stock branch exchanges stock-direction stability with the interior-effort extinction branch at the transcritical point $r = qE^*$, while the $E = E_{\max}$ boundary branch is classified separately and is not part of that exchange. The survival condition $r > qE^*$ is identical to $N^* > 0$; interior and extinction branches are not simultaneously stable.

3.2 The characteristic quasi-polynomial

With $x = N - N^*$, $z = Z - Z^*$, $e = E - E^*$, $\mathrm{sp}'_k(0) = 1/2$, and the floor inactive at $\delta > 0$, the linearisation of (1) is

$$\dot{x} = A_N x + A_E e, \qquad \dot{z} = B_N x + B_E e - \frac{1}{\tau_m} z, \qquad \dot{e} = C_E e + C_Z z(t-\tau),$$

with

$$A_N = r\left(1 - \frac{2N^*}{K}\right) - qE^*, \qquad A_E = -qN^*,$$
$$B_N = \frac{qE^* - S'(N^*)}{2\tau_m}, \qquad B_E = \frac{qN^*}{2\tau_m},$$
$$C_E = \left(1 - \frac{E^*}{E_{\max}}\right)\eta\left(\frac{\delta}{\Delta_{\mathrm{ref}}} - \frac{2E^*}{E_{\max}}\right), \quad C_Z = \left(1 - \frac{E^*}{E_{\max}}\right)\left[\frac{\eta E^*}{\Delta_{\mathrm{ref}}} + \frac{\delta_0 Z_{\mathrm{ref}}}{(Z_{\mathrm{ref}} + \delta)^2}\right].$$

The gate factors $(1 - E^*/E_{\max})$ distinguish the gated from the ungated variant and are the algebraic source of threshold relocation between the two. Substituting the modal ansatz $(x, z, e)e^{\lambda t}$ and expanding the 3×3 characteristic determinant along the third row gives the characteristic quasi-polynomial

$$P(\lambda) - C_Z L(\lambda) e^{-\lambda\tau} = 0, \qquad P(\lambda) = (\lambda - A_N)(\lambda + d)(\lambda - C_E), \qquad L(\lambda) = B_E(\lambda - A_N) + A_E B_N, \tag{5}$$

with $d = 1/\tau_m$. At the interior equilibrium the filter identities hold:

$$B_N = -\frac{A_N}{2\tau_m}, \qquad B_E = -\frac{A_E}{2\tau_m},$$

because the deficit signal is $qEN - S(N) = -A_N x - A_E e + o(\cdot)$ and $\mathrm{sp}'_k(0) = 1/2$. Consequently

$$L(\lambda) = B_E(\lambda - A_N) + A_E B_N = B_E\lambda - B_E A_N - A_E\frac{A_N}{2\tau_m} = B_E\lambda,$$

since $B_E A_N = A_E B_N$. This cancellation — the even-pairs algebra below — is the structural identity of the architecture.

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4. The Complete Hopf Cubic

The Hopf-frequency equation of the linearised system (5) takes a particularly clean form on this architecture: a cubic in $x = \omega^2$ for the modulus, plus a phase relation in τ. The cubic has at most three positive roots, so the system supports at most three Hopf-frequency families. The interior equilibrium's filter identity then collapses one structural term, and the resulting even-pairs algebra explains why, generically, two crossings are observed rather than one or three.

Theorem 4.1 (Cubic modulus condition and phase branches)**.** *Consider the linearisation (5). Assume*

(H1) $C_Z L(i\omega) \neq 0$ at every candidate imaginary root. This is automatic on the declared family: by the filter identity of Section 3.2, $L(i\omega) = B_E i\omega$, so $C_Z L(i\omega) = 0$ only at $\omega = 0$, which is excluded by $\omega > 0$; the degenerate case $C_Z B_E = 0$ is excluded by the standing assumptions. Hypothesis (H1) is stated for completeness, to make the phase balance (7) well defined; it imposes no additional restriction here.

Then $\lambda = i\omega$ ($\omega > 0$) is a characteristic root of (5) if and only if $x = \omega^2$ is a positive root of

$$H(x) = (x + A_N^2)(x + d^2)(x + C_E^2) - C_Z^2 \left[B_E^2 x + (A_E B_N - A_N B_E)^2\right] = 0, \tag{6}$$

and

$$\tau = \frac{-\arg\{P(i\omega)/(C_Z L(i\omega))\} + 2\pi k}{\omega} > 0, \qquad k \in \mathbb{Z}. \tag{7}$$

A cubic has at most three positive roots, so there are at most three Hopf-frequency families; higher branches recur within a family as $\tau_{n,0} + 2\pi k/\omega_n$ and are not additional frequencies. The certified branch pairs are separated — $\tau_{1,k} \neq \tau_{2,\ell}$ over the declared search range — so simultaneous double-Hopf collisions are excluded on the computed set (computed on the declared search range; not an algebraic consequence of the cubic). Simplicity and transversality are verified separately for each reported crossing.

Proof. At $\lambda = i\omega$, equation (5) reads $P(i\omega) = C_Z L(i\omega) e^{-i\omega\tau}$, so the moduli must match:

$$|P(i\omega)|^2 = C_Z^2 |L(i\omega)|^2. \tag{8}$$

Compute $|P(i\omega)|^2 = |(i\omega - A_N)(i\omega + d)(i\omega - C_E)|^2 = (\omega^2 + A_N^2)(\omega^2 + d^2)(\omega^2 + C_E^2)$, and $L(i\omega) = B_E i\omega + (A_E B_N - A_N B_E)$, whence $|L(i\omega)|^2 = B_E^2\omega^2 + (A_E B_N - A_N B_E)^2$. Substituting into (8) and writing $x = \omega^2$ gives exactly (6). Given a positive root x of (6), the phase condition reads $e^{-i\omega\tau} = P(i\omega)/(C_Z L(i\omega))$; since both sides have unit modulus on the root locus, τ is recovered as (7) up to the $2\pi k$ ambiguity. A cubic has at most three positive roots, giving at most three frequency families; for a fixed family ω_n the delays $\tau_{n,0} + 2\pi k/\omega_n$ recur with period $2\pi/\omega_n$ in τ, and are branches of the same frequency, not additional frequencies. Candidates qualify as Hopf crossings only after simplicity and transversality are verified separately; the cubic determines neither criticality nor global folds. □

Corollary 4.1 (Even pairs)**.** *At the interior equilibrium of (1), the cross term vanishes identically, $A_E B_N - A_N B_E \equiv 0$, so*

$$H(x) = (x + A_N^2)(x + d^2)(x + C_E^2) - C_Z^2 B_E^2 x.$$

Assume $A_N d C_E \neq 0$ (nonzero filter rate, memory rate, and equilibrium effort derivative). Then $H(0) = A_N^2 d^2 C_E^2 > 0$ and $H(x) \to +\infty$; the total algebraic multiplicity of the strictly positive roots of H is even, and generically H has zero or two distinct simple positive roots. The double-root boundary — the birth or annihilation of a Hopf-frequency pair — solves $H(x) = H'(x) = 0$.

Proof. The identities $B_N = -A_N/(2\tau_m)$, $B_E = -A_E/(2\tau_m)$ give $A_E B_N - A_N B_E = -A_E A_N/(2\tau_m) + A_N A_E/(2\tau_m) = 0$. With the cross term zero, H has the stated form, and $H(0) > 0$ follows from the nondegeneracy assumption. A polynomial with positive value at 0 and positive leading coefficient crosses the positive axis an even number of times counting multiplicity, which is the multiplicity claim; a single positive double root is the degenerate residual case and sits on the boundary $H = H' = 0$. □

For both Candidate A and Candidate B the cubic has exactly two positive roots on both the gated and the ungated variant; the certified local spectrum is stated in Section 5.1.

4.1 The scalar archetype

Two general statements frame the delay mathematics. First, the scalar Hayes result (Hayes, 1950; Hale and Verduyn Lunel, 1993, Ch. 3): for $\dot{x} = -ax - Bx(t - \tau)$ with $a > 0$, $B \geq 0$, the zero equilibrium is stable for all $\tau \geq 0$ if $B \leq a$; if $B > a$ it is stable for $\tau < \tau_{\text{crit}} = \arccos(-a/B)/\sqrt{B^2 - a^2}$ and unstable beyond, with *no restabilisation* as τ grows

— the crossing is destabilising. This is the scalar mechanism by which loop delay destroys stability that undelayed feedback sustains. Second, no delay conclusion is sign-free: the systems $\dot{x} = -ax - bx(t-\tau)$ and $\dot{x} = -ax + bx(t-\tau)$ $(a, b > 0)$ have different feedback signs and different stability properties. The named systems of this paper instantiate the two signs — the mobilising law of (1) carries $C_Z > 0$, the protective law (3) carries $C_Z < 0$ — and Sections 5–6 show that the local mathematics separates accordingly.

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5. The Mobilising Channel

The mobilising channel is the effort law of (1) with positive delayed gain on the deficit memory. Its behavioural reading is direct: when an institution sees a deficit signal, it deploys more extraction effort — a fleet expansion, a subsidy release, an open-access licence issuance — and the deployment is realised only after a delay. The mathematics of this section says two things. First, the undelayed gated version of such a law is already unstable. Second, an interval of intermediate delays — rather than the short-delay regime familiar from the scalar Hayes example — is the window in which the loop is stabilised by phase. The implication for governance is direct: on this architecture the institutional deployment delay is a phase filter that can either stabilise or destabilise depending on its value. It is not a one-directional hazard.

5.1 Local crossings and interval-certified delays

Two terms are used throughout, and both are defined here for the broad reader. A *Hopf crossing* is a parameter value at which a complex-conjugate pair of characteristic roots crosses the imaginary axis and oscillatory dynamics are born. *Interval-certified* means the crossing delay is enclosed by interval Newton iteration with outward rounding, so the enclosure is guaranteed to contain the exact root of the registered equations (Moore, 1979; Cloud, Moore, and Kearfott, 2009).

The complete cubic search (Theorem 4.1) with separately verified simplicity and transversality, independently checked by direct root tracking of the quasi-polynomial, gives the crossing pairs:

System	Candidate A τ_-/τ_+ (yr)	Candidate B τ_-/τ_+ (yr)
M3-U (ungated)	6.8814 / 132.3749	6.2136 / 76.2906
M3-B (gated)	3.67 / 150.36	5.5128 / 80.4245

System / candidate / crossing	Enclosure (yr)
M3-B (gated), A, τ_-	[3.6661490142739, 3.6661490142743]
M3-B (gated), A, τ_+	[150.3584773101408, 150.3584773101421]
M3-B (gated), B, τ_-	[5.5128407314433, 5.5128407314436]
M3-B (gated), B, τ_+	[80.4245267142270, 80.4245267142276]
M3-U (ungated), B, τ_-	[6.2135987340180, 6.2135987340183]
M3-U (ungated), B, τ_+	[76.2906356879512, 76.2906356879518]

The interval certificates are interval-Newton enclosures of the simple positive roots of H in $x = \omega^2$ (width $\leq 4 \times 10^{-17}$ in x) followed by branch-safe interval evaluation of the phase relation (7). The delay is the interval evaluation of the phase formula at a certified positive root of H, not a root of an argument formula. The registered interval pipeline uses outward rounding and interval transcendentals, checks simplicity and transversality signs — the lower crossing stabilising, $\mathrm{d}\,\mathrm{Re}\,\lambda/\mathrm{d}\tau < 0$; the upper crossing destabilising — and reproduces the displayed Candidate A intervals exactly on re-execution.

The certification tier is scoped precisely. These are certified enclosures of the *local spectrum* of the cubic — a strictly weaker statement than a certified Hopf bifurcation for retarded functional differential equations in the sense of Church and Lessard (2022) and Church and Queirolo (2024), whose radii-polynomial methods certify the bifurcation itself. No such full certificate is claimed here, and none is claimed for any global fold (Section 8, and the supplementary material). The interval machinery itself is standard (Moore, 1979; Cloud, Moore, and Kearfott, 2009). The pair is registered as gate G5 of the registration campaign — the gate log is deposited with the supplementary material, S9. The registered compute-core Hopf pair 3.666149 / 150.358477 yr — the base core (1)'s institutional-delay certificates, reproduced by the recovered compute core (Supplementary S9.5) — certifies this pair; it is a record of the compute core, not an object of the delayed-recruitment system of Section 9.

The undelayed gated mobilising law is already unstable ($\mathrm{Re}\,\lambda > 0$ at $\tau = 0$): the undelayed characteristic polynomial $P(\lambda) - C_Z B_E \lambda = (\lambda - A_N)(\lambda + d)(\lambda - C_E) - C_Z B_E \lambda$ in the symbols of Section 3.2 — the $-C_Z B_E \lambda$ term being exactly what the filter identity leaves of the delay coupling at $\tau = 0$ — at the declared Candidate A constants violates the Routh–Hurwitz condition. The linear coefficient is reduced by $C_Z B_E$, so the product of the first two coefficients falls below the constant term, which is the algebraic signature of the shortest-delay instability; and $P(\lambda) - C_Z B_E \lambda = 0$ is the undelayed case of (5). Institutional delay acts as a phase filter that opens the phase-stabilised window (τ_-, τ_+) and closes it again at τ_+. Delay-amplified instability refers to the upper crossing and the bistable windows, not to delay creating the short-delay instability. Enforcing the effort boundary relocates the local thresholds by approximately 47% (lower) and 14% (upper) at Candidate A without changing the equilibrium — thresholds do not transport between the gated and ungated variants, and the gate's threshold relocation is a registered comparison, not a calibration.

5.2 Lyapunov coefficients and criticality

The first Lyapunov coefficient at a Hopf point of the gated three-state core is the Hassard–Faria–Magalhães cubic (Hassard, Kazarinoff, and Wan, 1981; Faria and Magalhães, 1995) evaluated from the exact second and third derivatives of the vector field at equilibrium, under unit Hermitian normalisation of the right eigenvector v and its adjoint w, with the normalisation $w^* \Delta'(i\omega) v = 1$ (w^* the conjugate transpose; the eigenvector pair is denoted v, w so that q remains the catchability throughout):

$$\ell_1(\tau_-^{\mathrm{A}}) = +5.75 \times 10^{-5}, \qquad \ell_1(\tau_+^{\mathrm{A}}) = +3.55 \times 10^{-4},$$

both subcritical at gated Candidate A; the ungated Candidate B lower crossing is supercritical, $\ell_1(\tau_-^{\mathrm{B}}) = -9.84 \times 10^{-5}$ — hence no lower fold for that class — with $\ell_1(\tau_+^{\mathrm{B}}) = +2.19 \times 10^{-3}$. The subcritical small branch satisfies $\|N - N^*\| \sim C\sqrt{\tau - \tau_-}$ (slope 29.8 in amplitude-squared, $R^2 = 0.994$; the collocated orbit at $\tau = 3.700$ has residual $\sim 10^{-7}$ and escapes onto the large cycle by roundoff alone). These are numerical evaluations of the coefficient formulas at declared parameter points — computational results, stated as such.

Two status distinctions matter here. First, within the registered model family the criticality statements obtained from branch scaling — amplitude exponent 0.47 and surrogate cubic coefficient $\approx 3.9 \times 10^{-6}$ near the lower gated crossing — are inferred numerical classifications, not first Lyapunov coefficients from a centre-manifold calculation; the ℓ_1 values above are the computed coefficients for the gated core. Second, criticality is not invariant under the regularisation: the first Lyapunov coefficient contains $\mathrm{sp}_k''(0) = k/4$, so the reported $\ell_1(\tau_\pm)$ are at $k = 10$; Hopf points are invariant under $k \in \{5, 10, 20, 40\}$ at fixed δ — the sweep holds δ numerically fixed rather than maintaining the table relation $\delta = \log 2/k$, and this is the declared reading — and a sign change in ℓ_1 under that sweep would rearrange the lower window. The k-independence of equilibria and linearisations is a local statement only; fold locations depend on k, and no

k-uniform topology is claimed.

5.3 Hopf persistence under residual feedback

Remark 5.1 (Local Hopf persistence, conditional)**.** *Assume*

(H1) the five-state macro-reduction conjecture of the supplementary material — a slow-fast reduction whose fast-block uniform normal hyperbolicity is established as the proposition of S5, proved analytically on the parameter set $\alpha + \eta\beta < 1$, where α and β are the two coupling weights of the fast block's K–L Jacobian displayed in Supplementary S5, so that the inequality is that block's uniform-Hurwitz condition — with the sweep-quantified margin and the class-violation fraction recorded, and whose compact-memory-kernel condition for infinite-time persistence remains unverified,

(H2) the working-core projection of Section 2.4,

(H3) the working four-state (respectively three-state) characteristic equation has a simple pair $\pm i\omega$ at $\tau = \tau_\star \in \{\tau_-, \tau_+\}$ with $\mathrm{d}\,\mathrm{Re}\,\lambda/\mathrm{d}\tau \neq 0$ and no other imaginary eigenvalues,

(H4) the fast Jacobian remains uniformly Hurwitz at the joint equilibrium,

(H5) non-feedback mass compartments stay outside the delay loop.

Then: under the strict specialisation the core spectrum is a literal factor of the full characteristic function and the Hopf points persist exactly; under residual macroeconomic feedback of size ε the specialised system has a Hopf point $\tau_\star(\varepsilon) = \tau_\star + O(\varepsilon)$ in the ideal large-reservoir limit (add $O(1-\sigma_{\mathrm{geo}})$ for the finite reservoir, σ_{geo} being the geological-reservoir adequacy ratio of Section 2.4).

The conditionality is mathematical content, and it is stated as such: the statement is conditional on the reduction's residual hypotheses — fast-block Hurwitzness beyond the verified parameter set of S5, and the memory-kernel condition — and the global fold events of Section 8 lie explicitly outside its hypotheses.

Derivation (conditional, in full). Under the strict specialisation, the block-triangular identity granted by the reduction conjecture factors the full characteristic function as $\Delta_{\mathrm{full}}(\lambda,\tau) = \Delta_{\mathrm{core}}(\lambda,\tau)\,\Delta_{\mathrm{fast}}(\lambda,\tau)$ with Δ_{fast} uniformly Hurwitz on the declared set. The simple imaginary pair $\pm i\omega$ of the core is therefore a simple imaginary pair of the full function, and the implicit function theorem applied to $\Delta_{\mathrm{full}}(\lambda,\tau) = 0$ at $(\pm i\omega, \tau_\star)$ — where $\partial_\lambda \Delta_{\mathrm{full}} \neq 0$ by simplicity — preserves the crossing exactly. Under residual macroeconomic feedback of size ε, write $\Delta_{\mathrm{full}}(\lambda,\tau;\varepsilon) = \Delta_{\mathrm{core}}(\lambda,\tau) + \varepsilon\,\chi(\lambda,\tau;\varepsilon)$ with χ continuous. Near $(\omega_\star, \tau_\star, 0)$ the zero set of $F(\omega,\tau;\varepsilon) = (\mathrm{Re}\,\Delta_{\mathrm{full}}(i\omega,\tau;\varepsilon),\ \mathrm{Im}\,\Delta_{\mathrm{full}}(i\omega,\tau;\varepsilon))$ is a smooth curve through $(\omega_\star,\tau_\star)$ whenever $D_{(\omega,\tau)}F$ is nonsingular there. That nonsingularity is the Cauchy–Riemann restatement of the two hypotheses of the statement — simplicity of $\pm i\omega$ ($\partial_\lambda \Delta_{\mathrm{core}} \neq 0$) and transversality ($\mathrm{d}\,\mathrm{Re}\,\lambda/\mathrm{d}\tau \neq 0$) — so the implicit function theorem yields $\tau_\star(\varepsilon) = \tau_\star + O(\varepsilon)$. The finite-reservoir $O(1-\sigma_{\mathrm{geo}})$ term is the same argument with the perturbation scale replaced. The reduction conjecture itself remains the unproved hypothesis. □

5.4 The two-delay interpolation and the conditional mobilising-weight corollary

Linearising the interpolation (4) at the Candidate A point — an interior rest where both brackets vanish separately under the calibration of Section 2.5 — gives $\dot{e} = C_E e + C_m z(t-\tau_M) + C_p z(t-\tau_p)$, with $C_m = \chi_m C_Z^{\mathrm{mob}}$, $C_p = \chi_p C_Z^{\mathrm{prot}}$, and C_E the sum of the two gate-adjusted E-derivatives.

Proposition 5.1 (Two-delay characteristic identity)**.** *The characteristic function of the stock–memory block coupled to the interpolated effort law is*

$$P(\lambda) - L(\lambda)\big(C_m e^{-\lambda\tau_M} + C_p e^{-\lambda\tau_p}\big) = 0, \tag{9}$$

with P and L the polynomials of Section 3.2.

Proof. The variational system of the interpolation has the same stock–memory block as (1) and the effort row $\dot{e} = C_E e + C_m z(t-\tau_M) + C_p z(t-\tau_p)$. Expanding the characteristic determinant along the effort row, the minor associated with $z(t-\tau_M)$ is exactly $L(\lambda)e^{-\lambda\tau_M}$ — the cofactor of the memory row, the same cofactor that produced (5) — and likewise for τ_p. □

Proposition 5.2 (Weighted small gain)**.** *Assume*

(H1) the undelayed interpolation is stable (Hurwitz at $\tau_M = \tau_p = 0$),

(H2) the zero root is excluded ($P(0) - L(0)(C_m + C_p) \neq 0$).

If

$$\sup_{\omega>0} \frac{(|C_m| + |C_p|)\,|L(i\omega)|}{|(i\omega - A_N)(i\omega + d)(i\omega - C_E)|} < 1, \tag{10}$$

then (9) has no purely imaginary root for any $\tau_M, \tau_p \geq 0$, and — since for retarded equations finitely many roots lie to the right of any vertical line and the right-half-plane root count changes only through imaginary-axis crossings — the equilibrium is exponentially stable for every fixed pair of delays (pointwise in the delays; the decay rate need not be delay-uniform).

Proof. An imaginary root $\lambda = i\omega$ with $\omega > 0$ would satisfy $P(i\omega) = L(i\omega)(C_m e^{-i\omega\tau_M} + C_p e^{-i\omega\tau_p})$; taking moduli and using $|e^{-i\omega\tau}| = 1$,

$$|P(i\omega)| \leq |L(i\omega)|(|C_m| + |C_p|),$$

contradicting (10), which is the triangle inequality reversed strictly. The excluded case $\omega = 0$ is covered by the stated zero-root hypothesis; stability at the reference point is the stated undelayed-stability hypothesis; and the stability-switch principle (finitely many roots in the right half-plane, locally constant count off the imaginary axis; Hale and Verduyn Lunel, 1993, Ch. 11) transfers the reference stability to every delay pair. □

Corollary 5.1 (Mobilising weight, conditional)**.** *At Candidate A the pure mobilising loop gain exceeds 1 and the pure protective loop gain is 0.080. Assume*

(H1) the common equilibrium and all linear coefficients depend continuously on χ_m,

(H2) the characteristic denominator remains nonzero on the imaginary axis,

(H3) the protective endpoint has a strict gain margin.

Then there exists a conservative threshold radius $\chi_m^* \in (0,1)$ such that every interpolation with $\chi_m < \chi_m^*$ and $\chi_p = 1 - \chi_m$ satisfies (10): the small-gain exclusion remains valid on a nonzero protective-dominated weight interval. ***Claimed:*** *existence of a sufficient-condition radius.* ***Not claimed:*** *identification of the actual onset weight for Hopf frequencies — failure of (10) does not imply Hopf, and χ_m^* is a sufficient-condition radius, not a physical critical weight. A Hopf of the interpolated system therefore requires a sufficiently large mobilising weight; it cannot be produced by decreasing τ_p alone.*

The corollary holds exactly under the listed hypotheses. The interpolation family is not otherwise established to preserve a common equilibrium or a nonvanishing denominator, and no promotion of the corollary beyond them is made.

Proof (in full). Define the certificate map

$$F(\chi_m) = \sup_{\omega>0} \frac{(|C_m| + |C_p|)\,|L(i\omega)|}{|(i\omega - A_N)(i\omega + d)(i\omega - C_E)|}, \qquad C_m = \chi_m C_Z^{\mathrm{mob}}, \quad C_p = (1-\chi_m) C_Z^{\mathrm{prot}}.$$

Hypothesis (H2) keeps the denominator nonzero on the imaginary axis. Because $|L(i\omega)|/|(i\omega - A_N)(i\omega + d)(i\omega - C_E)| = O(\omega^{-2})$ as $\omega \to \infty$, and because the coefficients A_N, d, C_E are bounded uniformly on $[0,1]$ (hypothesis (H1), continuity on a compact set), the tail $\omega > \overline{\omega}$ contributes less than $F(0)/2$ for some $\overline{\omega}$ uniformly in χ_m. The supremum is

therefore a maximum over the fixed compact range $[\underline{\omega}, \overline{\omega}]$, on which the integrand is jointly continuous in (ω, χ_m), so F is continuous in χ_m (the maximum of a jointly continuous function over a fixed compact set is continuous). At the protective endpoint, $F(0)$ is the pure protective loop gain, $0.080 < 1$ (the stated value; hypothesis (H3) is its strict-margin form). Continuity supplies $\varepsilon > 0$ with $F(\chi_m) < 1$ for every $\chi_m < \varepsilon$; the conservative threshold radius is any $\chi_m^* \in (0, \min\{\varepsilon, 1\})$. For $\chi_m < \chi_m^*$ and $\chi_p = 1 - \chi_m$, (10) holds, which is Proposition 5.2's small-gain exclusion, valid for every pair (τ_M, τ_p). The final clause follows from the sufficient-condition status of the certificate: on the certified weight range no imaginary root can be produced by any τ_p, so a Hopf of the interpolated system requires $\chi_m \geq \chi_m^*$ — a sufficiently large mobilising weight — and cannot be produced by decreasing τ_p alone. □

For institutional design this says that, on the interpolated architecture, delay-induced cycles cannot be generated by accelerating the protective response alone; the mobilising weight must first cross a sufficient-condition threshold. This is a sufficient-condition statement, not an empirical calibration, and the actual onset of Hopf frequencies may lie below χ_m^*.

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6. The Protective Channel

The protective channel is the quota-tracking law of (3). Its behavioural reading for fisheries governance is direct: when an institution observes a deficit signal, effort is restored toward a cap rather than mobilised further; landing pressure is reduced as the signal worsens. The channel acts on the same stock–memory block as the mobilising channel, but with the sign of C_Z reversed and the gain modulus reorganised by the cap calibration. Its local mathematics is qualitatively different. There are no Hopf crossings at any delay at the calibrated point — a no-Hopf theorem — and the channel is stable under annual sample-and-hold review. The implication for institutional design is that, on this architecture, the calibrated quota-tracking law is the protective direction the loop should take, and the only sampled instability it exhibits is a discretisation artefact.

6.1 The quota-tracking law and its calibration

The protective law (3) with its calibration is stated in Section 2.5. One interpretation discipline travels with it. The effort variable may be an endogenous industry response, a legal quota-utilisation state, or an actual institutional control, and these interpretations are not interchangeable; the quota-tracking law is an institutional control law.

6.2 The no-Hopf theorem

Theorem 6.1 (Delay-independent local stability of the calibrated quota tracker)**.** *Consider the Candidate A stock–memory linearisation together with the protective gains of Section 2.5. The statement is pointwise in the declared coefficients — not a theorem for every decreasing cap function or every protective law. The modulus cubic*

$$H(x) = x^3 + c_2 x^2 + c_1 x + c_0$$

has

$$c_2 = A_N^2 + d^2 + C_E^2 = 0.76339, \qquad c_1 = 0.028946, \qquad c_0 = 9.278 \times 10^{-6},$$

all positive, with $c_2 c_1 - c_0 = 0.02209 > 0$. *Assume*
(H1) the undelayed Jacobian is Hurwitz,
(H2) the zero-root condition holds delay-independently, $P(0) \neq 0$.

Then the characteristic quasi-polynomial has no purely imaginary root for any delay $\tau_p \geq 0$, and the equilibrium is exponentially stable for every $\tau_p \geq 0$.

Proof. (a) *Exclude imaginary roots.* At $\lambda = i\omega$ the modulus balance of (5) reads $|P(i\omega)| = |C_Z||L(i\omega)|$; by Theorem 4.1 this is equivalent to $H(\omega^2) = 0$. For the protective channel, $C_E = -0.850336$ and $C_Z = -1.661702$ with the same $A_N = -0.0179$, $d = 0.2$, $B_E = 0.008955$, and the even-pairs cancellation gives

$$H(x) = (x + A_N^2)(x + d^2)(x + C_E^2) - C_Z^2 B_E^2 x = x^3 + c_2 x^2 + c_1 x + c_0$$

with $c_2 = A_N^2 + d^2 + C_E^2 = 0.76339$, $c_1 = A_N^2 d^2 + A_N^2 C_E^2 + d^2 C_E^2 - C_Z^2 B_E^2 = 0.028946$, $c_0 = A_N^2 d^2 C_E^2 = 9.278 \times 10^{-6}$. All three coefficients are strictly positive, so by Descartes' rule of signs $H(x) > 0$ for every $x \geq 0$ and no positive real root exists; no $\omega > 0$ solves the modulus balance. (The Routh–Hurwitz array of H, with $c_2 > 0$, $c_2 c_1 - c_0 = 0.02209 > 0$, $c_0 > 0$, is the strictly stronger statement that all roots of H lie in the left half-plane; it is reported for completeness, while the undelayed Jacobian's own stability array is the separate polynomial of step (b) below. A symbolic sufficient condition for the no-root property is $c_1 = A_N^2 d^2 + A_N^2 C_E^2 + d^2 C_E^2 - C_Z^2 B_E^2 \geq 0$, the only coefficient of H that can change sign.)

(b) *Delay-independent stability.* The undelayed linearisation is Hurwitz: its characteristic polynomial is $P(\lambda) - C_Z L(\lambda)$ with $L(\lambda) = B_E \lambda$ (even-pairs cancellation), i.e. $\lambda^3 + 1.0682\,\lambda^2 + c_1'\lambda + c_0'$ with $c_1' = 0.2038$ and $c_0' = 0.00305$; all Routh conditions are satisfied (the protective $C_E = -0.850$ dominates, so the λ^2-coefficient $-A_N + d - C_E = 1.0682$). Explicitly, the Routh array for $P(\lambda) - C_Z B_E \lambda$ has first column entries $1,\ 1.0682,\ c,\ c_0'$, where $c = (1.0682\,c_1' - c_0')/1.0682$ is the standard third first-column entry of a cubic's Routh array (the quotient $(a_2 a_1 - a_0)/a_2$ with $a_2 = 1.0682$, $a_1 = c_1'$, $a_0 = c_0'$), and both $c > 0$ and $c_0' > 0$. The zero root is excluded independently of delay: $P(0) - C_Z L(0) = P(0) \neq 0$ because $L(0) = 0$ and $P(0) = (-A_N)(d)(-C_E) = A_N d C_E \neq 0$. Since (a) shows the imaginary axis is never crossed for any $\tau_p \geq 0$, the stability-switch principle for retarded equations — finitely many characteristic roots lie to the right of any vertical line, and their count is locally constant while no root lies on the imaginary axis, so stability changes only through imaginary-axis crossings (Hale and Verduyn Lunel, 1993, Ch. 11) — transfers the undelayed Hurwitz property of (b) to every fixed delay: the equilibrium is exponentially stable for each $\tau_p \geq 0$, pointwise in τ_p, without a delay-uniform decay rate being claimed. □

The channel-separation reading is the theorem's interpretation: destabilisation by short delay is confined to the mobilising summand, and the protective channel has no periodic branch born from the equilibrium at any delay.

The same conclusion is visible in the loop-gain form. Define

$$\Gamma(\omega) = \left| \frac{C_Z L(i\omega)}{(i\omega - A_N)(i\omega + d)(i\omega - C_E)} \right|.$$

Then Γ is continuous on $[0, \infty)$, $\Gamma(0) = 0$ (since $L(0) = 0$), $\Gamma(\omega) \to 0$ as $\omega \to \infty$ — linear over cubic, $O(\omega^{-2})$: the numerator carries $L(i\omega) = B_E i\omega$ — and a numerically located maximum gives $\sup_\omega \Gamma(\omega) = 0.08011 < 1$, attained at $\omega \approx 0.0589$ yr^{-1} (frequencies in yr^{-1}, the unit of C_E, d, A_N; Section 2). The loop-gain exclusion argument of Section 10 then excludes every imaginary root for all delays directly. The maximum is a numerical location, stated as such; the analytic certificate is the Descartes/Routh–Hurwitz argument above.

6.3 The iso-gain sign flip

Proposition 6.1 (Iso-gain sign flip)**.** *Consider the gated Candidate A linearisation. Replacing C_Z by $-C_Z$, leaving every other coefficient unchanged:*

(i) leaves H identical (it depends on C_Z only through C_Z^2);

(ii) leaves the frequencies identical;

(iii) shifts each family's fundamental delay by π/ω on the branch that keeps it fundamental: the lower family ($\omega_1 \approx 0.02518$) moves up, $3.666 + \pi/\omega_1 = 128.374$ yr, and the upper family ($\omega_2 \approx 0.03944$) moves down, $150.358 - \pi/\omega_2 = 70.697$ yr, both remaining local Hopfs.

*The subscripts retain their original-family meaning — τ_- is the lower family's shifted delay, τ_+ the upper family's — so on the shifted axis the order is reversed: $\tau_+ < \tau_- < \tau_+^{\mathrm{unshifted}}$. The reversed-gain linearisation has loop gain $1.016 > 1$ and retains the factor $\eta E^*/\Delta_{\mathrm{ref}}$: it is **not** the quota law, whose genuine form changes the modulus as well as the sign. The proposition is the paper's sign-phase invariance statement: for a single delayed scalar feedback term $C_Z L(\lambda)e^{-\lambda\tau}$, replacing C_Z by $-C_Z$ leaves the imaginary-root frequency equation unchanged and shifts every admissible delay branch by an odd half-period — consequently, **feedback sign alone cannot create or eliminate Hopf-frequency families over the unrestricted delay axis**, and the protective no-Hopf property of Theorem 6.1 is a property of the quota law's modulus (its direct damping and delayed gain), not of its sign.*

This is the false-reversal identification hazard: a pure sign flip is not a protective institution, and attributing the reversed-gain crossings to a quota tracker would confuse two different effort laws.

Proof (in full). The imaginary-root condition of the gated Candidate A linearisation is $P(i\omega) - C_Z L(i\omega)e^{-i\omega\tau} = 0$ with the polynomials of (5). Separating modulus and phase,

$$|P(i\omega)| = |C_Z|\,|L(i\omega)|, \qquad \arg P(i\omega) - \arg\big(C_Z L(i\omega)\big) - \omega\tau \in 2\pi\mathbb{Z}.$$

The modulus equation depends on the effort coupling only through $|C_Z| = |-C_Z|$, so the frequency equation — the cubic in $x = \omega^2$ of Section 4 — and its certified positive roots are unchanged by the flip; the phase equation changes by exactly π, so every admissible delay branch is translated by an odd half-period, $(2k+1)\pi/\omega$. The certified fundamental pair and its frequencies are $(\tau_-, \tau_+) = (3.666149, 150.358477)$ yr and $(\omega_1, \omega_2) = (0.0251915, 0.0394366)$ (Section 5.1). The fundamental branch of each family is its smallest positive representative, which for the flipped gain is

$$\tau_-^{\mathrm{fl}} = \tau_- + \frac{\pi}{\omega_1} = 3.666149 + \frac{\pi}{0.0251915} = 128.374,$$
$$\tau_+^{\mathrm{fl}} = \tau_+ - \frac{\pi}{\omega_2} = 150.358477 - \frac{\pi}{0.0394366} = 70.697.$$

the lower family moving up and the upper family moving down, as stated. Both shifted crossings remain simple and transverse — the certified evaluation gives $\mathrm{d}\,\mathrm{Re}\,\lambda/\mathrm{d}\tau = -6.07\times10^{-6}$ at $(\omega_1, 128.374)$ and $+1.83 \times 10^{-5}$ at $(\omega_2, 70.697)$, both nonzero — so each remains a local Hopf point of the flipped system. The invariance of the frequency equation is the stated sign-phase invariance: over the unrestricted delay axis the families are translated, never created or destroyed, and the protective no-Hopf property of Theorem 6.1 is accordingly a property of the quota law's modulus — its direct damping and delayed gain — not of its sign. □

6.4 Sampled protection and the discretisation crossing

Proposition 6.2 (Protective Euler-reviewed zero-order-hold monodromy)**.** *Replace the protective delayed feedback by sample-and-hold of period T_r with one explicit Euler review step — an Euler-reviewed zero-order-hold controller, one particular sampled-data implementation among many (the standard reviewed controller of computer-controlled systems; Åström and Wittenmark, 1997), not sample-and-hold governance in general. Between reviews the variational system is $\dot{\xi} = A_{\mathrm{hold}}\xi$ with*

$$A_{\text{hold}} = \begin{pmatrix} A_N & 0 & A_E \\ B_N & -d & B_E \\ 0 & 0 & 0 \end{pmatrix},$$

(the effort is frozen between reviews), and at each review — at the end of the interval, on the flowed state (the flow-then-update convention of Section 7) — the effort update is the explicit Euler step $e_{k+1} = e_k + T_r\big(C_E e_k + C_Z z(t_{k+1}^-)\big)$*. The monodromy of one review interval is therefore*

$$M_p(T_r) = \begin{pmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & T_r C_Z & 1 + T_r C_E \end{pmatrix} \exp(A_{\text{hold}} T_r). \tag{11}$$

Proof. On one interval $[kT_r, (k+1)T_r)$ the frozen-effort system is linear with matrix A_{hold} (the effort row is zero), so the state at the end of the interval is $\exp(A_{\text{hold}}T_r)\xi_k$; the review then applies the affine update $\xi_{k+1} = \xi(t_{k+1}^-) + T_r(C_E e_{k+1}^- + C_Z z_{k+1}^-)\,\mathbf{e}_3$, i.e. the matrix factor displayed in (11). The sampled equilibrium is exponentially stable if and only if every eigenvalue of $M_p(T_r)$ lies in the open unit disc. □*

At the protective gains, $\rho(M_p(1)) = 0.9838 < 1$: annual review of the quota-tracking channel is linearly stable at Candidate A. On the grid $T_r \in [0.2, 20]$, the spectral radius is strictly below one on $[0.2, 2.306)$ and strictly above one on $(2.306, 20]$, so the Euler hold map crosses $\rho = 1$ at $T_r \approx 2.306$ (a grid resolution).

That crossing is carried by the discretisation, not by the governance system. Specifically, it comes from the interaction of the Euler-update factor $(1+T_rC_E)$ with the hold flow $\exp(A_{\text{hold}}T_r)$ — the C_E row of the update restoring the coupling that the hold flow's zero third row lacks — and not from the scalar condition $1 + T_rC_E = 0$ alone, which (with $C_E = -0.850$) would place a threshold near $T_r \approx 1.176$, not at 2.306. So the crossing is a property of the Euler factor *combined with* the hold dynamics. It also sits within about 2% of the explicit-Euler stability limit of the scalar effort equation, $T_r < 2/|C_E| = 2.352$ yr (with $C_E = -0.850336$; the limit at which the scalar factor $1 + T_rC_E$ itself reaches -1), which makes the discretisation origin of the crossing self-evident. It is **not** a Hopf point of the protective delay equation, whose continuous spectrum Theorem 6.1 excludes at every delay. The finding is the mathematical statement: the $T_r = 2.306$ crossing belongs to the discretisation. The separating computation confirms it: under the exact held-measurement update (Proposition 7.1) the protective monodromy has spectral radius below one at every tested review interval — 0.9838 at $T_r = 1$ yr and at most 0.9967 on $[0.2, 300]$ yr — so the crossing is carried entirely by the explicit-Euler command step, and the protective channel's sampled stability is not cadence-limited. The operator-specific scope is recorded: the mobilising hold map is unstable at $T_r = 1$ because the undelayed mobilising Jacobian is already unstable, and the protective map does not inherit that instability (Section 7).

6.5 Channel-specific pacing

Corollary 6.1 (Channel-specific pacing)**.** *Three pacing statements hold.*

(i) For the mobilising bracket the equilibrium is linearly unstable for $0 < \tau < \tau_-$*.*

(ii) For the protective law at Candidate A the equilibrium is linearly stable for every $\tau_p \geq 0$*.*

(iii) For the two-channel system, in every connected parameter region where the undelayed interpolation is stable, the zero root is excluded, and the strict weighted small-gain bound (10) holds, stability is independent of both delays — any delay-induced stability loss must occur outside that certified region.

Proof. The first clause is Section 5.1 (undelayed instability with a stabilising lower crossing at τ_-). The second is Theorem 6.1. The third is Proposition 5.2 together with Corollary 5.1:

wherever (10) holds, no imaginary root exists for any (τ_M, τ_p), so instability — if present — must originate in parameter regions where the mobilising weight is large, and is independent of the protective delay there. The synthesis inherits Corollary 5.1's interpolation hypotheses wherever its clause applies. □*

The policy-scope reading, instantiated: faster protective governance is not the hazard — the mobilising sign is.

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7. The Review Interval as Control

Periodic review is the form that institutional delay takes in modern fisheries governance: assessment is conducted on a fixed cadence (annual, biennial), and the assessment result is acted on with a discrete update. The review interval T_r — the **review cadence** — is the length of time between successive assessments, the interval at which the institution re-decides. On this architecture the review interval is a spectral design parameter that acts in opposite directions on the two channels. Periodic review is a *hybrid* system — a frozen-effort continuous hold interleaved with a discrete effort update at the review instants — and is not the delay equation of Sections 2–6 sampled at $\tau = T_r$. Its stability diagram is therefore not a re-plot of the continuous-delay τ-axis, and "lengthen the interval" is not a relabelling of "decrease the delay". The protective channel is stable at annual review under both the Euler and the exact update; its apparent 2.3-yr instability threshold is a discretisation artefact. The mobilising channel is unstable at annual review and restabilises as the review interval is lengthened — but its restabilising crossing is a complex unit-circle crossing of the sampled-data map, not a Hopf of the continuous-delay equation. The implication for institutional design is that the review interval is a controller knob, not a governance recommendation. An interval that stabilises the equilibrium may still allow severe inter-review depletion, and inter-review tube safety is an additional criterion not certified here. Throughout this section the *monodromy* is the one-period linear return map of the reviewed loop — the product of the held flow and the review update — whose spectral radius decides the linear stability of the sampled equilibrium (radius below one: stable; above one: unstable).

Theorem 7.1 (Sampled-data monodromy of the mobilising channel)**.** *Linearise the gated three-state core about the interior equilibrium and replace the delayed feedback by sample-and-hold of period T_r with one Euler review step. Between reviews the variational system is $\dot{\xi} = A_{\mathrm{hold}}\xi$ with A_{hold} as in Proposition 6.2. The monodromy of one review interval is*

$$M(T_r) = \begin{pmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & T_r C_Z & 1 + T_r C_E \end{pmatrix} \exp(A_{\mathrm{hold}} T_r), \tag{12}$$

now with the mobilising gains $C_E = -0.0595$, $C_Z = +1.785$. Assume the flow-then-update timing convention (the pre-review state $\xi_k^- = \xi(t_k^-)$ flows for one period under the held effort, and the review applies the Euler update to the flowed state, giving the post-review state $\xi_{k+1}^- = M(T_r)\xi_k^-$ with M as displayed — the update factor acts on the end-of-period assessment; the reversed ordering $\exp(A_{\mathrm{hold}}T_r)R(T_r)$ is a different hybrid system, not used here, the two orderings having closely related linear stabilities (proof); the nonlinear map differs, so the convention remains fixed throughout).

Then: (i) the sampled equilibrium is exponentially stable if and only if every eigenvalue of $M(T_r)$ lies in the open unit disc; (ii) a complex unit-circle crossing — the linear spectral signature of a Neimark–Sacker bifurcation (transverse crossing, nonresonance, and the nonlinear first Lyapunov coefficient of the map are not verified here, and the bifurcation name is reserved for that verified case) — occurs at those T_r where $M(T_r)$ has a simple pair $e^{\pm i\theta}$, $\theta \notin \{0, \pi\}$,

with the remaining spectrum inside the disc; (iii) $(M(T_r) - I)/T_r$ *converges to the continuous undelayed Jacobian as* $T_r \to 0$*. The statement concerns this sample-and-hold/Euler review scheme, not the continuous-delay equation.*

Proof. The monodromy formula is Proposition 6.2's derivation with the mobilising gains; the stability criterion is the discrete-time spectral radius condition; the unit-circle characterisation is the standard eigenvalue-crossing criterion for maps. The consistency limit follows from $\exp(A_{\text{hold}}T_r) = I + A_{\text{hold}}T_r + O(T_r^2)$ and the update factor $I + T_r R_0$, where $R_0 = \mathbf{e}_3\big(C_Z\mathbf{e}_2^\top + C_E\mathbf{e}_3^\top\big)$ is the rank-one effort-update row, whose sum with $A_{\text{hold}}T_r$ is $I + T_r(A_{\text{hold}} + R_0) + O(T_r^2)$; writing $J_{\text{cont}} = A_{\text{hold}} + R_0$ for the full undelayed Jacobian of the continuous linearisation, the limit is $(M(T_r) - I)/T_r \to J_{\text{cont}}$ — R_0 is precisely what restores the C_E, C_Z entries that vanish in the third row of A_{hold}, which is why the consistency limit is not read off from A_{hold} alone. For the ordering comparison: $R_0 \exp(A_{\text{hold}}T_r)$ and $\exp(A_{\text{hold}}T_r)R_0$ have identical spectra whenever R_0 is invertible (they are then similar matrices), so the linear stability classification transfers between the two orderings in that case; the nonlinear map differs, and the convention stays fixed. □

What Theorem 7.1 says. The theorem turns review-cadence stability into an eigenvalue test: compute the one-period return map $M(T_r)$ of the reviewed loop, and the sampled equilibrium is stable exactly when every eigenvalue of that map lies inside the unit disc. The rest of the section runs this test for the two channels.

On the gated Candidate A hold map the Euler scheme reports: annual review unstable, $\rho(M(1)) = 1.00055$ (the undelayed linearisation being already unstable, Section 5.1), with the sampled equilibrium's restabilising complex unit-circle crossing at $T_r^{\text{UC}} = 47.536$ yr and a multiplier reaching -1 at $T_r^{(-1)} = 79.143$ yr, and the reported stable interval $(47.536, 79.143)$ yr. These quantitative Euler values do **not** survive the exact held-measurement update; they are artefacts of the explicit-Euler command step and are recorded as such in Remark 7.1, with the exact record reported in Proposition 7.1 (unit-circle crossing at $T_r = 6.50$ yr; the Euler-reported crossings give exact spectral radius 0.786 at 47.536 yr and 0.597 at 79.143 yr). The load-bearing feature is therefore the *cadence* statement — lengthening the review interval restabilises a system that annual review destabilises — not the Euler map's particular thresholds. The computation is the zero set of $\det(M(T_r) - e^{i\theta}I) = 0$ on the declared map: for a slow stock under this review operator, **the review interval is a local spectral design parameter**.

> **Governance warning.** Local spectral stability is not governance: a long review interval may locally stabilise the equilibrium while allowing severe transient depletion between reviews, and inter-review tube safety — the stock floor held along the whole interval under the declared disturbance class — is the additional criterion a governance recommendation would require, which is not certified here.

The contrast with the protective channel — stable at annual review and at every tested interval under the exact update — is complete: the two sign channels respond to the review interval in opposite directions, and the protective channel's only sampled instability is the Euler command step's.

Remark 7.1 (Euler command-step artefact)**.** *The Euler map's reported restabilisation at* $T_r^{\text{UC}} = 47.536$ *yr with stable interval* $(47.536, 79.143)$ *yr is not a property of periodic review; Proposition 7.1 shows those crossings do not survive under the exact held-measurement update (exact spectral radius* 0.786 *at* 47.536 *yr and* 0.597 *at* 79.143 *yr, both below one) and that the exact monodromy has a single unit-circle crossing on* $[0.2, 300]$ *yr — the restabilising complex pair at* $T_r = 6.50$ *yr. The Euler thresholds are artefacts of the explicit command step, which is why they are labelled here and not reported as crossings of the review operator. The exact map and its single crossing are the quantities reported in Proposition 7.1.*

The design conclusion is not, however, an artefact of the update: as $T_r \to 0$ every consistent scheme inherits the undelayed linearisation (Theorem 7.1(iii)), and as $T_r \to \infty$ the exact hold tends to a rank-one operator whose eigenvalue is the DC gain, proportional to $L(0) = 0$ by the filter identity, so the stock–effort coupling is lost and the equilibrium is stable for long enough review intervals. Each consistent scheme therefore restabilises at some T_r; only the location is scheme-dependent, quantified in the closing scheme-dependence remark of this section, which reports the tight 6.50–6.73 *yr band across the three consistent schemes. Scheme-dependence is the* discretisation *axis and it is narrow; it is not to be conflated with* parameter *sensitivity, which is wide and is stated where that remark closes.*

Proposition 7.1 (Exact held-measurement monodromy)**.** *Keep the sample-and-hold architecture, the frozen-effort hold flow $\dot{\xi} = A_{\text{hold}}\xi$, and the flow-then-update timing convention of Proposition 6.2 and Theorem 7.1, and replace the explicit-Euler command step by the exact held-measurement update: the effort law $\dot{e} = C_E e + C_Z z$ is integrated exactly over the review interval with the measurement held at the flowed end-of-period value, giving the update*

$$e_{k+1} = e^{C_E T_r} e_k + \frac{e^{C_E T_r} - 1}{C_E}\, C_Z\, z(t_{k+1}^-).$$

The monodromy of one review interval is therefore

$$M_{\text{ex}}(T_r) = \begin{pmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & \dfrac{e^{C_E T_r} - 1}{C_E}\, C_Z & e^{C_E T_r} \end{pmatrix} \exp(A_{\text{hold}} T_r), \tag{13}$$

which satisfies the same consistency limit as the Euler map, $(M_{\text{ex}}(T_r) - I)/T_r \to J_{\text{cont}}$ as $T_r \to 0$ (the update factor is the exact exponential of the effort gain, of which $1 + T_r C_E$ is the first-order Taylor). Then, at the gated Candidate A point:

(i) Protective channel ($C_E = -0.850336$, $C_Z = -1.661702$). $\rho(M_{\text{ex}}(1)) = 0.9838$, and $\rho < 1$ at every tested review interval on $[0.2, 300]$ yr, with maximum 0.9967 *attained at the shortest interval tested. The protective hold map is stable at every review interval tested; the Euler crossing at* 2.306 *yr is carried entirely by the explicit-Euler command step and vanishes under the exact update.*

(ii) Mobilising channel ($C_E = -0.0595$, $C_Z = +1.785$). $\rho(M_{\text{ex}}(1)) = 1.00035$: the annual-review instability survives the exact update. The Euler-reported crossings do not: at 47.536 *yr and* 79.143 *yr the exact spectral radius is* 0.786 *and* 0.597*. On $[0.2, 300]$ yr the exact monodromy has exactly one unit-circle crossing — the restabilising complex pair at*

$$T_r^{\text{UC,ex}} = 6.50 \text{ yr},$$

with crossing eigenvalues $0.9846 \pm 0.1746\,i$ ($|\lambda| = 1$), third eigenvalue 0.1647 *inside the disc, a non-resonant argument ($2\pi/\theta \approx 35.8$), and transverse direction ($\rho - 1$ changes sign). The restabilisation is a property of review cadence, not of the command step. Holding the measurement at the start-of-period value instead of the flowed end-of-period value does not move the reported crossings at the reported precision. The Neimark–Sacker reading keeps the discipline of Theorem 7.1: the nonlinear conditions are not verified.*

Proof. The update is the exact solution of the scalar effort equation with z held constant; the monodromy factor follows from the flow-then-update convention, under which the update acts on the flowed state, whose effort coordinate is unchanged by the frozen hold. The consistency limit follows exactly as in Theorem 7.1 with $e^{C_E T_r} = 1 + C_E T_r + O(T_r^2)$ and $(e^{C_E T_r} - 1)/C_E = T_r + O(T_r^2)$. The channel statements are the computed spectral-radius records of M_{ex} on the declared grids, re-executable from the registered model constants; the same computation reproduces the

registered Euler numbers of Proposition 6.2 and Theorem 7.1 before the exact map is evaluated, and both update conventions (start- and end-of-period measurement) agree at the reported precision. □

What Proposition 7.1 says. Under the exact update the protective channel stays inside the unit circle at every tested review interval, while the mobilising channel crosses it exactly once, restabilising at a review interval of about 6.5 yr; the Euler-reported half-century thresholds are artefacts of the command step, not of the cadence.

Remark (scheme-dependence of the sampled-data map). Neither M nor M_{ex} is the physically natural zero-order hold of the delay equation, in which the delayed signal z is held at its sampled value z_k while the effort dynamics $\dot{e} = C_E e + C_Z z_k$ runs continuously alongside the stock response. That scheme has monodromy, in closed form,

$$M_{\mathrm{ZOH}}(T_r) = e^{A_{\mathrm{Z}} T_r} + \Big(\int_0^{T_r} e^{A_{\mathrm{Z}} s}\, ds\Big) C_Z\, \mathbf{e}_3 \mathbf{e}_2^{\top},$$

with A_{Z} the full undelayed Jacobian with the C_Z entry removed. The two maps analysed above freeze effort throughout the flow and integrate the effort law impulsively at the review instant; they are one choice among several consistent schemes, so the phrases "the Euler-reported crossings are artefacts of the command step" and "the restabilisation is a property of review cadence" are properly read as scheme-dependent claims, not as properties of periodic review per se. The *qualitative* conclusion is not scheme-dependent, and it is provable without computing the location: as $T_r \to 0$ every consistent scheme inherits the undelayed linearisation ($(M - I)/T_r \to J_{\mathrm{cont}}$, Theorem 7.1(iii)); and as $T_r \to \infty$ the exact hold tends to a rank-one operator whose eigenvalue is the DC gain, which is proportional to $L(\lambda)$ evaluated at $\lambda = 0$ — and $L(0) = B_E \cdot 0 = 0$ by the filter identity of Section 3.2, so the map loses the stock–effort coupling and the equilibrium is stable for long enough review intervals. Hence every consistent periodic-review scheme restabilises at *some* T_r; only the location is scheme-dependent. The scheme-dependence is in fact small for this architecture. Re-executing the monodromy on the registered Candidate A constants: the exact held-measurement map has its mobilising complex crossing at $T_r = 6.5013$ yr; the same map with the measurement held at the *start*-of-period value gives $T_r = 6.5013$ yr (so the paper's clause that start-of-period measurement does not move the reported crossings holds); and the physically native zero-order hold M_{ZOH} gives $T_r = 6.7279$ yr (crossing eigenvalues $0.9855 \pm 0.1699\, i$), within about 4% of the exact value. All three consistent schemes therefore restabilise in a tight 6.50–6.73 yr band: the crossing is not an artefact of the discretisation convention, which is the question this remark was set up to answer. **That is a different question from whether the crossing is identifiable, and on that question the answer is no.** Scheme agreement constrains the *discretisation* axis; it says nothing about the *parameter* axis. Re-running the monodromy of this section under joint parameter perturbation (Part II, Section 3.4.1, using the same Candidate A reconstruction) sweeps the crossing from 0.87 yr to 10.67 yr under a half-percent joint perturbation of the six influential parameters, and under a one-percent perturbation the crossing disappears altogether — the loop stable at every tested interval — in twenty of sixty-four corners. Perturbing q, K or $E_{\max}$ downward by 2% individually also removes it. The margin that decides this is $\rho(1) - 1 = 3.5 \times 10^{-4}$. The 6.50 yr figure is accordingly a property of one declared parameter vector: reproducible across discretisation schemes, fragile across parameters, and not to be read as a certified invariant of periodic review. The protective channel is stable under M_{ZOH} as well (no crossing on $[0.2, 200]$ yr; $\rho(1) = 0.9928$), consistent with the protective no-Hopf claim. The rank-one limit is verified numerically: $\rho(M_{\mathrm{ZOH}})$ falls to 0.242, 0.0165, and below 10^{-6} at $T_r = 200$, 500, 2000 yr, with the map's numerical rank dropping to 2 at 500 yr and 1 at 2000 yr — confirming the DC-gain ($L(0) = 0$) loss of coupling.

The consolidated sampled-data record (channel × scheme × review interval). Every entry below is a record already stated in this section (Propositions 6.2 and 7.1, Theorem

7.1, and the scheme-dependence remark), collected in one table; the mobilising native-ZOH row's annual radius is not a recorded quantity of this paper.

Channel	Monodromy scheme	ρ at $T_r = 1$ yr	Unit-circle crossings on the tested grids
Protective	Euler (11)	0.9838	$\rho = 1$ at $T_r \approx 2.306$ yr (grid resolution; Euler command-step artefact, within 2% of the scalar Euler limit $2/\lvert C_E\rvert = 2.352$ yr)
Protective	Exact held-measurement (13)	0.9838	none on $[0.2, 300]$ yr; maximum 0.9967 at $T_r = 0.2$ yr
Protective	Native ZOH (M_{ZOH})	0.9928	none on $[0.2, 200]$ yr
Mobilising	Euler (12)	1.00055	complex pair at 47.536 yr and -1 at 79.143 yr (command-step artefacts; exact-map radii there 0.786 and 0.597)
Mobilising	Exact held-measurement (13)	1.00035	one complex pair at $T_r = 6.50$ yr ($0.9846 \pm 0.1746\,i$; third eigenvalue 0.1647)
Mobilising	Native ZOH (M_{ZOH})	not recorded	one complex pair at $T_r = 6.7279$ yr ($0.9855 \pm 0.1699\,i$)

Two readings travel with the table. First, the annual-review mobilising growth rate is small in absolute terms: $\rho - 1 = 5.5 \times 10^{-4}$ per review cycle (Euler) and 3.5×10^{-4} (exact) — e-folding times of roughly 1800 and 2900 review cycles respectively — so the annual-review instability is the $\tau = 0$ instability of Section 5.1 seen through a one-year hold that supplies too little phase, not a fast divergence. Second, the protective annual radius is the slow stock mode: the uncoupled modes of the protective hold-and-update map at annual cadence sit at $e^{A_N} \approx 0.982$ (stock; $A_N = -0.0179$), $e^{-d} \approx 0.819$ (memory; $d = 0.2$), and $1 + C_E \approx 0.150$ (effort; $C_E = -0.850336$), and the recorded $\rho(M_p(1)) = \rho(M_{\mathrm{ex}}(1)) = 0.9838$ — the same value under the two update schemes — is that stock mode, only weakly coupled by the controller. The protective maximum 0.9967 at $T_r = 0.2$ yr likewise matches the stock mode's $e^{A_N T_r} \approx 0.996$. The protective sampled-data record therefore says that the slow stock mode decays under review, not that the controller stabilises an otherwise unstable loop.

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8. Global Numerics at Declared Certification Levels

All results in this section are numerical results at their declared status: computed outputs tied to registered equations, parameter sets, history classes, methods, tolerances, and finite domains, with basins restricted to the histories actually tested. The fold-status discipline governs every global statement. The events below are numerical continuation, multiplier, basin, and turning-region results; the lower fold of Section 8.2 and the second fold of the same section are both enclosed as Moore–Spence candidates in finite collocation systems (Krawczyk enclosures for both; a discrete nondegeneracy record for the upper candidate). Off-grid residual bounds and a lift to the continuous delay equation have not been established; no continuous-delay fold

proof is claimed. Rigorous saddle-node-of-periodic-orbit results for delay equations exist for specific classes (Beretka and Vas, 2020); none is claimed for the events below. The implication for fisheries governance is that, on the declared architecture, the global attractor topology is a five-regime structure with two Krawczyk-enclosed finite-dimensional turning candidates and a distinct basin-transition interval; the corresponding continuous-delay fold and saddle-node classifications remain open.

8.1 Methods

Hopf points are located by root tracking of the characteristic equation and independently by the cubic (Theorem 4.1); folds by Keller pseudo-arclength continuation of the periodic orbit with 1.5–5 $\times 10^6$ yr persistence tests per step and far-from-equilibrium cross-checks (collocation-continuation numerics in the DDE-bifurcation tradition: Engelborghs, Luzyanina, and Roose, 2002; background: Guckenheimer and Holmes, 1983; Kuznetsov, 2004). Fixed-initial-condition bisection mislocates folds by more than 20 yr under the critical slowing down near each Hopf (linear rates 10^{-4}–10^{-5} yr^{-1}) and is not used.

8.2 The attractor topology and the lower boundary events

For the effort-bounded three-state core at Candidate A the lower boundary is carried by the pre-registered five-regime campaign (plan frozen before any run; executed once; pre-registered but not yet independently replicated; records deposited with the supplementary material, see Data availability). The record resolves the lower boundary into a **single fold at $\tau_f = 5.5872362$ yr of one S-shaped branch**: the small arm born at the subcritical Hopf $\tau_- = 3.666$ and the stable large arm are two arms of the same branch that meet at one apparent turning point. For the finite collocation map, a Krawczyk inclusion encloses a unique Moore–Spence zero of the $m = 64$ system in $[5.587236198689, 5.587236198691]$, the three-order solves ($m = 64/96/128$) agree to 2.69×10^{-11}, and both arms' dominant multipliers — real at every record, imaginary part identically zero to 10^{-6} — cross $+1$ at the fold: the campaign's recorded crossing proximity is 1.1×10^{-7} in τ, the finite-collocation location enclosure is the Krawczyk box above, and the multiplier records are corroboration at their own, coarser, precision (small arm: 1.0192 at $\tau = 5.584$ falling to 0.9942 at 5.587; large arm: 0.2040 at $\tau = 4.0$ rising to 0.9692 at 5.5815 — τ recorded to three decimals, multipliers to four, so the tabulated points bracket the crossing without resolving it at the 10^{-7} proximity, which is the campaign's record, not an interpolation of the table). The two cycles coexisting at $\tau = 5.575$ are genuinely distinct (amplitudes 24.91 and 19.94; periods 322.60 and 308.16 yr) but are arms of one branch, not two families with two folds. The registered record places the basin-bisection endpoint 0.011 yr below the fold. The saddle-node-of-periodic-orbits classification remains **provisional** — the certification is of the discrete collocation system, the continuum off-grid residual stage is still open, and the crisis alternative survives as a reading of the archived record — and no Neimark–Sacker or torus event is involved in either reading (the multiplier is real throughout). The registered records reproduce the gate-floor facts near $\tau = 5.55$ ($E \leq 9.21$, $N \geq 68.6$): the termination is not a gate singularity, not a graze of $N = 0$, and not a homoclinic connection over the computed segment (the period is 324 yr and decreasing there; a homoclinic limit farther along the branch is not excluded by a finite segment).

The attractor sequence is the five-regime topology on the explored fundamental-delay range — $0 < \tau \lesssim 160$ yr, before the first recurrent branch of the lower frequency family (≈ 253 yr, Theorem 4.1), so the classification is a statement about the declared domain (Figure 1), not the entire positive delay axis: (i) $0 < \tau < \tau_-$, the equilibrium unstable with a single large-amplitude cycle the only attractor found from the declared tested histories. (ii) $\tau_- < \tau < \tau_f$, a stable focus coexisting with the large cycle (lower bistable window, width ≈ 1.92 yr, closed at the fold $\tau_f = 5.5872362$ yr). (iii) $\tau_f < \tau < \tau_{f2}$, settling to the equilibrium from the tested histories

(finite searches support, but cannot prove, basin monostability; the second-fold branch records show an unstable upper arm existing mathematically only above the second fold — over regime (iv)'s $64.4 < \tau < 150.4$ yr, as Section 8.3 registers — without generic basin). (iv) $\tau_{f2} < \tau < \tau_+$, mathematical bistability: the second S-branch's stable returning arm coexists with the still-stable equilibrium from the Krawczyk-enclosed second turning candidate $\tau_{f2} = 64.4023272$ yr, but the registered basin records find the stable arm generically unreachable near the fold — the capture onset in $[148.6, 149.5]$ yr is a basin boundary inside this window, not a fold. (v) $\tau > \tau_+$ within the explored range, the equilibrium unstable and the captured family the only attractor found from the declared tested histories. The registered basin archive resolves the lower-window asymmetry at $\tau = 5.575$: the depleted history (H2) is captured by the cycle while the large-stock (H1) and near-equilibrium (H3) histories settle. A basin diagnostic records that the H2 capture at $\tau = 5.6$ is a saddle-node ghost transient (the same run settles at four times the horizon), consistent with the collocation fold at 5.5872362 yr. In the ungated family (M3-U) the upper-window attractor is a period-1 limit cycle (at $\tau = 131.8$ yr ungated: Poincaré spread $< 5 \times 10^{-3}$, envelope constant to 0.005% over 2×10^6 yr, fundamental period ≈ 135.6 yr, all Floquet multipliers inside the unit circle with dominant nontrivial pair modulus ≈ 0.81; an independent method-of-steps RK45 reproduces the envelope). In the gated core the registered record's upper-window captured object is the $E \approx E_{\max}$ face cycle (below).

Regime	Delay range	Attractor record
(i)	$0 < \tau < \tau_-$	Equilibrium unstable; a single large-amplitude cycle the only attractor found from the declared tested histories.
(ii)	$\tau_- < \tau < \tau_f$	A stable focus coexisting with the large cycle — the lower bistable window, width ≈ 1.92 yr, closed at the fold $\tau_f = 5.5872362$ yr.
(iii)	$\tau_f < \tau < \tau_{f2}$	Settling to the equilibrium from the tested histories; finite searches support, but cannot prove, basin monostability.
(iv)	$\tau_{f2} < \tau < \tau_+$	Mathematical bistability: the second S-branch's stable returning arm coexists with the still-stable equilibrium; the capture onset in $[148.6, 149.5]$ yr is a basin boundary inside this window, not a fold.
(v)	$\tau > \tau_+$	Within the explored range, the equilibrium unstable and the captured family the only attractor found from the declared tested histories.

At the upper boundary the registered second-fold record (the pre-registered second-fold campaign; same deposit and status discipline as the five-regime campaign) resolves the boundary into a **finite-collocation second turning candidate at** $\tau_{f2} \in [64.402327203368, 64.402327203372]$ **yr**: the upper branch, carried downward from the converged cycle at $\tau_+ - 0.05$, neither terminates at $\tau = 130$ nor fails collocation anywhere — 183 accepted natural steps at residuals $\leq 3.7 \times 10^{-12}$ from $\tau = 150.308$ to the turning-point stall at 64.402327895 — and turns at the interval-Krawczyk-enclosed Moore–Spence zero (unique in the box, G' nonsingular throughout, both nondegeneracy constants excluding zero; three-order

itted records, both campaigns (2026-09-02, first runs, no independent reruns yet; all data from the deposited p4_five_regime_campaign/ and a025_second_

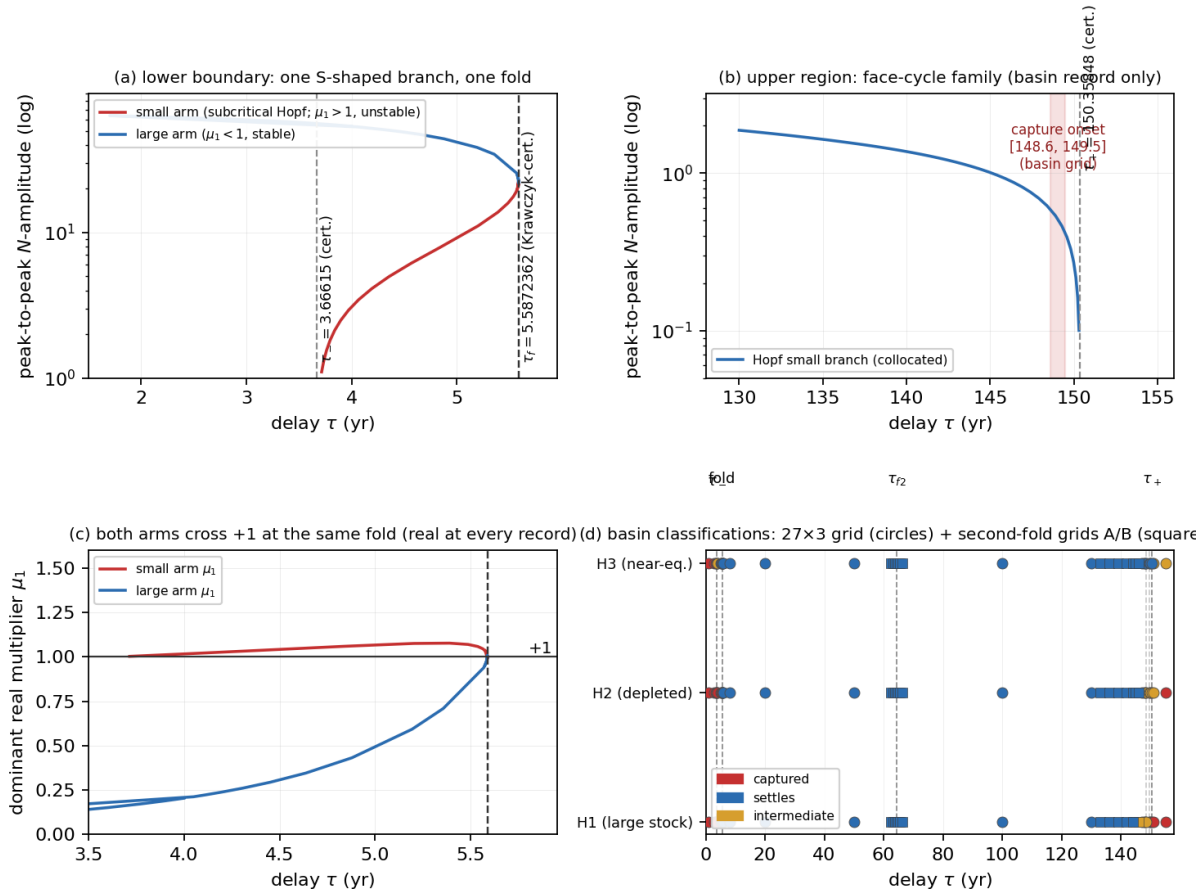


Figure 1: The five-regime attractor topology from the two pre-registered continuation records (five-regime campaign and second-fold campaign; every panel drawn from the deposited campaign CSVs, see Data availability). (a) The lower S-branch: the subcritical-Hopf small arm (red, $\mu_1 > 1$) and the stable large arm (blue, $\mu_1 < 1$) meet at the Krawczyk-enclosed finite-dimensional turning candidate $\tau_f = 5.5872362$ yr. (b) The upper S-branch: the unstable arm from the Hopf at τ_+ folds near the collocation-enclosed $\tau_{f2} = 64.4023272$ yr and returns as a stable arm (inset: its real multiplier through the fold), with the basin capture onset $[148.6, 149.5]$ yr — a basin boundary, not a fold — and the certified $\tau_+ = 150.35848$ yr. (c) The lower branch's dominant real multipliers crossing $+1$ at the same fold. (d) Basin classifications (captured / settles / intermediate) on both registered grids (27×3 plus the second-fold grids A/B, dt = 0.02): the depleted history is captured at $\tau = 5.575$; every tested history settles from 5.6 to 146 (H1 intermediate at 145–146); capture returns only inside $[148.6, 149.5]$ yr.

agreement 5.53×10^{-7} at $m = 64/96/128$). The branch is a second S: the unstable subcritical arm from the Hopf at τ_+ (real multiplier > 1 at every record, peaking at 1.27745 at $\tau = 74.902$) meets the fold at amplitude ≈ 10.5, and sixteen pseudo-arclength points past the turn record a **stable** returning arm ($\mu_1 = 0.9993$ falling monotonically to 0.9435 at $\tau = 64.438$). **The stable arm is not generically reachable near the fold**: the campaign's basin grids find every tested history settling around τ_{f2} (21 runs at $\pm 0.5/\pm 1/\pm 2$; dt-halving at 64.4 unchanged) and throughout the 133–146 gap (21 runs; H1 intermediate at 145–146, critical slowing toward the capture window). The capture onset $[148.6, 149.5]$ yr is therefore a **basin boundary** inside the mathematically bistable window (τ_{f2}, τ_+) — not the stable family's birth (that is τ_{f2}) and not the equilibrium's stability boundary (that is the certified τ_+). The captured object there remains the $E \approx E_{\max}$ face cycle (peak-to-peak $N \approx 44$; $E \in [0.35, 29.91]$ on the basin seeds), which no campaign could Fourier-collocate at $m = 64$ or $m = 128$ (two recorded collocation failures): the connection from the stable arm's last record ($\tau = 64.438$) to the face-cycle family is the natural conjecture, unverified over the 64.438–148.6 gap (a follow-up campaign would be required). The Hopf small branch is reproduced at both ends of the amplitude window (0.100 at $\tau = 150.31$ and 1.874 at $\tau = 130$). The face-riding cycle is reached from admissible histories in the upper window: the admissible topology of regimes (iv)–(v) is carried by the face cycle.

8.3 The registered numerical families

M3-U. Candidate A crossings 6.8814 / 132.3749 yr (Candidate B 6.2136 / 76.2906 yr); persistence boundaries near 7.355 and 131.24 yr; lower bistable window ≈ 0.47 yr, upper ≈ 1.1 yr; basin-dependent capture at $\tau = 131.8$ yr; the upper-window attractor a stable period-one cycle with all nontrivial computed Floquet multipliers inside the unit circle and the near-lower-Hopf small orbit unstable (dominant multiplier > 1). The saddle-node-of-periodic-orbits classification of either persistence boundary remains conjectural unless branch collision and nondegeneracy are demonstrated. Sensitivity sweeps over $\eta \in [0.5, 3.0]$ find zero or two positive cubic roots; the two-crossing window in r is bounded ($\approx (0.008, 0.022)$ yr^{-1} at $\eta = 0.914$, extending to ≈ 0.061 yr^{-1} at $\eta = 3$) — model sensitivity results, not empirical calibration. A Droop nutrient–quota variant of the same core — an external nutrient pool and an internal quota with quota-limited growth $\mu(q) = r(1 - q_{\min}/q)$, a growth-coupled pool, and the interior equilibrium preserved — leaves the r-window unchanged, with window upper edge ≤ 0.023 yr^{-1} at $\eta = 0.914$ and no Hopf crossing at any $r \geq 0.2$ yr^{-1}: a growth-coupled pool cannot be slow at large r (the quota self-relaxes at exactly r, and the nutrient pool's relaxation grows with the throughput; the only r-independent slow pool is the working core's ω_A-type exchange, which narrows the window).

M3-B. Candidate A crossings 3.67 / 150.36 yr (Candidate B 5.5128 / 80.4245 yr; no gated-Candidate-B global branch or fold result is registered). The lower global boundary is the single fold of one S-shaped branch at $\tau_f = 5.5872362$ yr (Krawczyk-enclosed in the finite collocation system; both arms cross $+1$ there); the upper boundary is the finite-collocation second turning candidate near $\tau_{f2} \in [64.402327203368, 64.402327203372]$ yr (interval Krawczyk on the $m = 64$ system; three-order agreement 5.53×10^{-7}), the upper branch an S whose stable returning arm is not generically reachable near the fold — the capture onset $[148.6, 149.5]$ yr (basin grid) is a basin boundary inside the bistable window (τ_{f2}, τ_+), and the fold type (saddle-node of periodic orbits) remains undetermined. Multiple far-from-equilibrium seeds failed to reach a large cycle throughout the tested interior range $5.57 < \tau < 148.3$ yr — finite searches that support, but cannot prove, interior monostability (basin-scoped: the second-fold branch records show an unstable upper arm existing mathematically on $64.4 < \tau < 150.4$ yr without generic basin, so the statement concerns basins, not branch existence). The inferred subcritical signatures (amplitude exponent 0.47; surrogate cubic coefficient $\approx 3.9 \times 10^{-6}$) are inferred numerical classifications, not centre-manifold Lyapunov coefficients; the two arms belong to one branch meeting at a single fold, so a single square-root collision law applies in principle but is not verified (no continuum certificate); k-independence is local only. Ungated-Candidate-B global values are registered: the

three-state ungated Candidate B has no lower fold (its lower crossing is supercritical, Section 5.2) and an upper fold of the large cycle at 76.075 yr against $\tau_+ = 76.2906$ yr — a narrow upper bistable window $(76.075, 76.29)$ — while the four-state ungated Candidate B carries the corresponding Hopf crossings at 6.25 and 76.33 yr with its upper fold not pinned; these are registered numerical values, their classification as saddle-nodes of periodic orbits remains open, and no gated-Candidate-B fold location follows from them.

M3-LC. Upper persistence boundary ≈ 132.0 yr at $\psi = 1$ versus ≈ 132.5 yr at $\psi = 0$ (raw outputs 131.998/132.499; separation 0.501 yr), with an inter-locator discrepancy of ≈ 0.8 yr against an independent locator's 131.24 yr taken as the localisation-uncertainty scale; the defensible conclusion is an order-of-one-year shift, not an exact percentage. At $\tau = 115$ yr, $\kappa = 0.5$, long-lived transients of order 10^4–10^5 yr reach minimum $N \approx 33$ under pure stock culling and $N \approx 10$ under pure recruitment suppression: the two channels of harvest pressure — removing standing stock versus suppressing recruitment — differ in the depth of their transient excursions, though the local bifurcation structure cannot see them. In the fixed-demand experiment ($D = 0.7 > S_{\max} = 0.5$, $N(0) = 50$), stock culling is stopped at its first hitting time of $N = 0$ (equivalently implemented with an explicit donor limiter; the unconstrained vector field is not continued into negative stock), placing pure culling at zero near time 158, while pure recruitment suppression approaches zero asymptotically and reaches $N < 1$ near time 430 — a first-hitting-time result. Local equivalence does not imply excursion equivalence; assigning ψ to a field system requires age-, stage-, or replenishment-specific evidence.

Four-state working core. Ungated Candidate A donor-limited equilibrium is

$$(N^*, A^*, E^*) = (89.5256, 397.8665, 2.0896)$$

with crossings 6.982022 / 132.272044 yr and persistence boundaries near 7.374 and 130.77 yr (upper bracketed $[130.770, 130.771]$); gated counterpart crossings ≈ 3.7849 / 150.12 yr with cycle periods $\approx$ 360–380 yr (lower regime) and 150–160 yr (upper regime). The turnover stability boundary at $\tau = 0$ is $\omega_A^* \approx 0.001316$ (the sweep's recorded value is 0.001316298; quoted at sweep resolution, not enclosure precision; gated ≈ 0.001330), located by a 1798-point equilibrium sweep, sixty sub-threshold simulations (ten ω_A values, six delays in $[0, 300]$ yr, all converging to the equilibrium), and sixty continuation points above it producing a continuous Hopf pair — finite sweeps supporting sub-threshold stability across the tested delays, not a theorem for all parameter values. The reduced open-pool caveat stands: freezing A is not a justified fast-variable elimination at the baseline $\omega_A = 10^{-3}$ yr^{-1}. The certified three-state second fold at 64.4023 yr falls inside the four-state's upper fold bracket $[64.25, 64.5]$ yr (persistent orbit of amplitude ≈ 11 at 64.5, continuation failure by 64.25): reproduced to within the bracket width — the bracket spans 0.25 yr, or 0.4% of the fold value, so the version-robustness of the upper fold location is a bracket-resolution statement — while the Hopf pair stays within a few percent and the basin reachability of the stable arm is version-specific; the characteristic-pinned detail is in Section 8.4.

MPF (primitive-flux core). At the illustrative baseline

$$(\mathcal{M}, \mu, K_A, d, c, \gamma_U) = (100, 0.340, 24.5, 0.072, 0.00995, 0.388),$$
$$(q, E_{\max}, \eta, \delta_0, \Delta_{\text{ref}}, \tau_m) = (0.0384, 35.8, 2.23, 0.0118, 2.29, 5.13).$$

the equilibrium is $(X^*, U^*, E^*) \approx (16.68, 10.23, 0.435)$, with no local Hopf crossing for $0 \le \tau \le 500$ (characteristic-root and argument-principle counts) and small perturbations decaying at all directly tested delays. The apparent onset near $\tau = 33.4$–33.6 is a long-lived decaying transient (return within $\approx 2 \times 10^4$ time units throughout the tested 33.4–34.8 interval); for $\tau \gtrsim 35$ a slow-fast oscillation persists for some tested memory histories and not others — basin-selective global dynamics, not a local Hopf or a classified periodic-orbit fold. The absence of a baseline Hopf is parametric: Hopf roots first appear at $\eta_{\text{crit}} \approx 2.337$; above τ_+ at $\eta = 10$ the attractor

is diagnosed as irregular slow-fast intermittency (inter-excursion-interval coefficient of variation 1.58, return-map anticorrelation $r = -0.47$; the underlying global bifurcation mechanism is unresolved). The full η_{crit} sweep with the interleaving pair values and the pair-birth structure, the intermittency diagnostics, and the sigmoid-gated effort screen (more than 300 parameterisations without finding a genuine delay-induced Hopf — a numerical negative result over the sampled domain, not a structural impossibility theorem) are deposited in the supplement (S11), with Section 10.4 carrying the screen's loop-gain reading. The MPF regime is neither the M3-B regime nor a transfer of its threshold values.

8.4 The four-state working core

The characteristic matrix $\Delta(\lambda) = \lambda I - A_0 - A_\tau e^{-\lambda\tau}$ of the gated working core at the frozen-donor equilibrium $(N, A, Z, E) = (89.52562, 397.8665, \log 2/10, 2.08962)$ has simple imaginary roots at

$$\tau_- = 3.78487 \text{ yr (period } 250.44 \text{ yr)}, \qquad \tau_+ = 150.12175 \text{ yr (period } 159.13 \text{ yr)},$$

with $|\det \Delta(i\omega, \tau)| < 10^{-18}$ — the characteristic-pinned pair, with τ_- 3.2% higher and τ_+ 0.2% lower than the three-state values — the directly computed spectral sensitivity of the crossings to the frozen-A perturbation, 12.7 times the parameter ratio $A_0/A^* \approx 2.5 \times 10^{-3}$, and not a consequence of the trajectory bound of Lemma 2.1, which bounds state deviations on $[0, T]$, not eigenvalue locations. Fourier collocation produces a small periodic orbit of residual $\sim 10^{-13}$ immediately below τ_+ (amplitude 0.090 at $\tau = 150.082$). Continuation of the large-amplitude cycle (converged orbit state and delay history carried between steps, 10^4–10^5 yr per step) locates the two global folds at

$$\tau^{(4)}_{\text{term},L} \approx 5.63 \text{ yr}, \qquad \tau^{(4)}_{\text{fold},R} \approx 64.4 \text{ yr},$$

the lower bracketed by a persistent periodic orbit at $\tau = 5.62$ (peak-to-peak N-amplitude ≈ 23) and failure of continuation by 5.64, the upper by a persistent periodic orbit at $\tau = 64.5$ (amplitude ≈ 11) and failure of continuation by 64.25. The resulting topology is a narrow lower bistable window $(3.78, 5.63)$ yr, a wide monostable interval $(5.63, 64.4)$ yr, and a wide upper bistable window $(64.4, 150.1)$ yr in which the large cycle coexists with the stable equilibrium but, from generic far-from-equilibrium histories, is reached only for τ above roughly 75–100 yr (depleted-stock histories only near $\tau \gtrsim 135$). Cycle periods run from ≈ 371 yr at $\tau = 4.5$ to ≈ 320 yr at the lower fold, and from ≈ 156 yr near τ_+ to ≈ 73 yr at the upper fold. Whether either fold is a saddle-node of periodic orbits is not established. A Neimark–Sacker bifurcation of a periodic orbit changes its stability and may create a torus but does not by itself terminate the branch, so branch termination, stability loss, and basin-capture loss are reported separately, and a torus-mediated reading would explain simulation-based loss of attraction without explaining branch nonexistence. A second bifurcation parameter invisible to the three-state core appears: at $\tau = 0$ the four-state equilibrium is unstable for $\omega_A > \omega_A^* \approx 0.001316$ and delay-independently stable below it; above it, on the ungated system, both Hopf points and both folds track smoothly, τ_- falling from ≈ 17.5 to ≈ 6.9 yr with the monostable interval remaining $\approx$ 120–260 yr; the gated $\tau_- \approx 3.78$ is the baseline (ω_A, κ_A) value, not a uniform constant. The coexistence of the characteristic-pinned pair with the sub-threshold delay-independent stability is not a contradiction, but the two objects must be labelled distinctly. The characteristic-pinned pair ($\tau_- = 3.78487$, $\tau_+ = 150.12175$ yr) is the spectrum of the frozen-donor characteristic matrix $\Delta(\lambda) = \lambda I - A_0 - A_\tau e^{-\lambda\tau}$ with the donor pool A held fixed at its equilibrium value — the standard quasi-polynomial spectrum of the reduced 4×4 system written with A pinned. The ω_A^* boundary (≈ 0.001316) is a different object: the dynamic-A turnover sweep at $\tau = 0$ of the full system (2), in which A is itself a state. Call the first the *frozen-donor characteristic pair* and the second the *dynamic-A stability boundary*; they are two different operators, and reporting

both as "the four-state working core" invites exactly the confusion the distinction resolves. The frozen-donor pair is only a spectral statement about the quasi-polynomial with A frozen; it is not the spectrum of the dynamic-A system, whose $\tau = 0$ equilibrium is delay-independently stable below ω_A^*. Oscillation periods throughout $\tau < \tau_-$ are 250–390 yr, essentially independent of τ: the frequency is pinned by r and τ_m, not by institutional delay, and uniformly raising r toward fish-like values destroys the oscillatory regime rather than compressing the period into a decade.

The frozen-donor quasi-equilibrium discipline governs all of these numbers: the working point requires continuing geological support (≈ 4.652 stock units per year — a companion-interface quantity, Remark of Section 2.4) and is not a rest point of the closed mass ledger; it is incompatible with the formal quasi-steady slaving target of Section 2.4; and the working-core thresholds are $\sigma_{\text{geo}} = 1$ properties.

8.5 Parameter windows

Within the parameter subregion where the Candidate A two-Hopf pair exists, one-at-a-time variation across the reported ranges preserves the pair and the lower-boundary structure. Outside the bounded r-window the system is delay-independently stable or lacks a positive equilibrium. τ_- lies in 3.7–7 yr across the two effort laws and both candidates, and in 4–25 yr across the full (r, η) rectangle; τ_+ is primarily controlled by the biological time $1/r$ but also depends on the effort-response chart — Candidates A and B share $r = 0.02$ and have $\tau_+ \approx 132$–150 yr and ≈ 76–80 yr respectively across the two effort laws (A: 132.37 ungated, 150.36 gated; B: 76.29 ungated, 80.42 gated). Reported literature ranges of intrinsic growth rates are wider than the instability window; the quoted $r \in [0.005, 0.4]$ yr^{-1} range is not independently sourced in this paper, so the identification warning is scoped to the paper's own computed two-crossing window, $r \in (0.008, 0.06)$ yr^{-1} (Section 8.3) — against any plausible empirical spread of r, most assessed stocks fall outside the window. The dimensionless groups that fix N^*/K and $r\tau_\pm$ once effort is scaled by $E_{\max}$ — and the complementary fact that the separate effort scale is not identifiable from (N, Z) alone — are identification statements treated in the companion analyses of sampled governance and assessment architectures (under review).

8.6 The scaffold companion: registered and verified records

The three records that entered the archive as unreproducible "reproduction targets" are the completed, verified results of a distinct research object — the **scaffold** model of the companion flow-balance framework (a separate manuscript) — whose generating code and result files have been recovered and are reproduced here verbatim. The scaffold is **not** the model of Sections 2–6 and 9; it is a stage/age-structured Ricker-cascade core with a material block, capital stock, and gated effort (states X_A, X_J, P, U, A, K_C, E — seven implemented nonlinear states, of which the appendix prose calls the classical scaffold "nine states" and lists ten symbols), and its delay-feedback loop is the companion's, not (1). Status: these are verified, machine-precision-computed records, not open directions. The three records, each reproduced by the verbatim scripts:

(1) Elevated-forcing cod-class Hopf. The scaffold_item3_refine scan over the economic parameters (η, ζ, K_0, q) finds, for the cod-class life history ($g = 5$, $P_0 = 2$, $N_c = 60$, $d_A = 0.2$, $d_J = 0.6$), a genuine Hopf at $\tau^* = 43.29$ with period 263.4 time units (crossing period ≈ 263; $E^* = 0.245$, $\bar{X}_A^* = 54.2$, $g_0 = 1.000$) at the elevated-but-plausible economic setting $(\eta, \zeta, K_0, q) = (5.0, 0.8, 0.03, 0.01)$ — maxgap +0.1631. The reported values $\tau^* \approx 43$ and period ≈ 263. In the code symbol convention the savings parameter is ζ (the manuscript's ς), and the fuel-efficiency parameter is K_0; at $\eta = 5$, $\varsigma = 0.8$, $K_0 = 0.03$, $q = 0.01$ are confirmed.

(2) Life-history anchoring at default economics. The scaffold_item_3_anchor run over the three anchored classes gives delay-independent stability at the default economic set-

tings: anchovy-class ($g = 1$, $\bar{X}_A^* \approx 27.7$), sprat-class ($g = 2$, ≈ 34.6), cod-class ($g = 5$, ≈ 54.8) all satisfy the interior survival condition with a negative modulus gap (-3×10^{-4} to -10^{-3}) and no crossing over the tested delay search — confirming no crossing at default economics. (P_0 values that violated the Step-2 survival condition produced no equilibrium — a self-consistency check.)

(3) Broad crossing search. The scaffold_hopf_search + items_1-2 scans over the biological parameters give: among the strong-candidate genuine Hopfs (periods $\approx$ 900–11,000 time units on the broad search, and $\approx$ 3,800–35,000 on the six classified crossings), every classified crossing has $\mathrm{d}\,\mathrm{Re}\,\lambda/\mathrm{d}\tau < 0$ (stabilising), with crossing rates 10^{-7}–10^{-9}, and **no clean two-crossing τ_-/τ_+ window** — the stable windows after the stabilising crossing are fragmented and narrow, so the scaffolding's delay-induced structure is stabilising-only, not the paper's two-crossing (τ_-, τ_+) organisation. This is a genuine structural difference between the scaffold and the paper's three-state core, and it is the section's actual content.

State-count flag. The scaffold's appendix prose states "nine states" but lists ten symbols (the mass-conserved block $(\bar{X}_A, X_J, P, U, A, G)$ plus V_N, Z, K_C, E); the implemented code carries seven nonlinear states (G fixed at its mass-balance value, V_N fixed at 0 at the interior equilibrium, Z folded into the characteristic equation via the $(1 + \tau_m\lambda)$ memory factor). The count should be stated as seven implemented nonlinear states (equivalently ten conceptual states, nine per the appendix label) in one place, not inconsistently across the prose.

Status. These are registered, verified numerical records of the scaffold companion, reproduced here from the recovered scripts; they are reproduced on the declared instances and carry no saddle-node or collocation-grade certificate. They are included because they delimit the scaffold's delay-induced instability structure, which differs from the paper's — a difference worth recording, not as open directions but as the verified companion record it is.

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9. The Delayed-Recruitment (Maturation-Delay) System

The two channels of Sections 5 and 6 act on the scalar gated core (1), whose recruitment is instantaneous logistic inflow. This section registers the applied analogue in which the same governance structure — a delayed feedback law carrying institutional memory, deployed effort, and a gated response — acts on a stock whose own delay is the maturation lag from spawning to recruitment. The computational machinery is the recovered stage-analysis code of the companion empirical study, archived verbatim and re-executed without modification (the recovery audit is deposited with the supplementary material); the records below reproduce the recorded stage-analysis results at every anchor, and the section is the registration counterpart of the reproduction targets of Section 8.6.

9.1 The registered system

The working system has the three states of the core — stock N, deficit memory Z, deployed effort E — with the stock compartment replaced by the Gurney–Blythe–Nisbet delayed-recruitment plant (the class already cited in the manuscript for its laboratory evidence): recruitment into the stock at regeneration rate r, at the cohort $N(t-g)$ of the spawning event one maturation delay g earlier,

$$\dot{N} = r\,N(t-g)\big(1 - N(t-g)/K\big) - q\,E\,N,$$

so the perceived deficit is extraction minus the currently maturing regeneration,

$$d = q\,E\,N - r\,N(t-g)\big(1 - N(t-g)/K\big),$$

and the filter and effort law are exactly those of (1),

$$\dot{Z} = \frac{1}{\tau_m}\big[\Phi_k(\ell) - Z\big],$$

$$\dot{E} = \left(1 - \frac{E}{E_{\max}}\right)\left[\eta\, E\Big(\frac{Z(t-\tau)}{\Delta_{\mathrm{ref}}} - \frac{E}{E_{\max}}\Big) + \delta_0 \frac{Z(t-\tau)}{Z_{\mathrm{ref}} + Z(t-\tau)}\right].$$

Here g is the **maturation delay** (the recruit-to-mature lag), entering the stock equation and the deficit signal; τ is the **institutional delay**, entering **only** the effort equation (as in the base core); and η is the **effort-response coefficient**. The multiplicative gate is $(1 - E/E_{\max})$, the same as in (1), and is *not* denoted g here (the recovered code and its results file fix these labels: g is the maturation delay, η the effort-response coefficient). The two switches are the section's analogue of the two channels of Sections 5 and 6: at $g = 0$ the maturation delay is switched off and the system reduces **exactly** to the base scalar core; at $\tau = 0$ the institutional delay is switched off (the institutional channel still acting, undelayed). Because at steady state $N(t-g) = N^*$, the equilibrium is identical to the base core's ($Z^* = \delta$, E^* from the same quadratic, $N^* = K(1 - qE^*/r)$), giving a clean comparison with the base windows.

9.2 Validation at zero maturation delay

At $g = 0$ (maturation delay off) the two-delay crossing criterion — the delayed-recruitment counterpart of the stability-switch analysis of Section 5.1 — returns the recorded base windows exactly: recruitment-rate windows $r \in (0.00796, 0.02191)$ at $\eta = 0.914$ and $r \in (0.00676, 0.06028)$ at $\eta = 3.0$, both reproduced by the verbatim re-run with zero deviation. These are the registered validation rows for the two-delay machinery: the delayed-recruitment crossing structure exists, at zero maturation lag, only on a bounded window of recruitment rates, and the window widens with the effort response η.

9.3 Fine-map institutional bands

With the institutional channel on, the $\tau = 0$-stable two-crossing structure occupies four recruitment-rate bands at $\eta = 0.914$, on the locus $r\,g \approx 1.5$–1.6: $g = 1$: $r \in (1.565, 1.585)$; $g = 2$: $r \in (0.77, 0.81)$; $g = 3$: $r \in (0.50, 0.55)$; $g = 5$: $r \in (0.28, 0.33)$. The re-run locates the same bands with refined edges (1.564–1.586, 0.767–0.805, 0.492–0.547, 0.260–0.332), within the recorded fine-map grid resolution of ± 0.03; the band cells carry two delay crossings each on the wide characteristic mesh (rightmost root ≈ -0.06). A longer maturation delay moves the band to lower recruitment rates on that product locus — $g = 7$ near $r \approx 0.17$, $g = 10$ near $r \approx 0.08$ — so the stronger and longer the maturation lag, the slower the stock for which it carries the same two-crossing structure. The band τ-windows at the band centres lie within documented governance lags ($g = 1$: $(1.6, 3.5)$ yr; $g = 2$: $(2.6, 7.8)$ yr; $g = 5$: $(9.9, 20.3)$ yr) with periods ~ 4, ~ 8, and ~ 17 yr respectively. The $g = 2$ window lies inside the documented 2–13 yr governance-lag distribution (Hocherman, Trop, and Ghermandi, 2025) with a decadal (~ 8 yr) period, which is the sharpest falsifiable empirical avenue: fast-maturing small pelagics under continuous harvest control rules.

A mesh-range caveat is registered with the fine map. The map's characteristic scan is a bounded mesh, and an isolated apparent $\tau = 0$-stable crossing cell at $g = 5$, $\eta = 3.0$, $r \approx 1.54$–1.60 fails on the wide mesh (true rightmost root $+0.244$, unstable — the cell is a cohort regime of the delayed plant); the recorded absence of institutional $\tau = 0$-stable cells at $\eta = 3.0$ is confirmed, and band cells are claimed stable only where the wide mesh confirms them. The bands and their widths are parameter-sensitive and are characterised at finite parameter resolution — no collocation-grade classification and no τ-converged Hopf-locus mapping is claimed.

9.4 Nonlinear ground truth

The verbatim nonlinear integrators reproduce the recorded cycle rows. The slow-stock class ($r = 0.02$, $g = 5$, $\tau = 0$) carries the registered 358.8-yr biological cohort cycle (re-run 358.7 yr; stock swinging $\approx$ 28.5–95.4, E saturating at $E_{\max}$), reinforcing the centuries-long-period untestability conclusion — the slow-r system is not even quietly stable once recruitment is lagged, so the institutional-delay question becomes moot at slow r. The fast class ($r = 0.5$, $g = 5$, $\tau = 0$) carries the $\sim$ 20-yr cohort cycle (re-run 20.1 yr). The institutional cycle at ($r = 0.3$, $g = 5$, $\tau = 10$) has recorded period 16.96 yr with amplitude 8.66 (re-run 16.95 yr; the amplitude is tail-window dependent, re-run 9.36 under the recovered integrators' tail convention), and at $\tau = 21$ the same configuration is stable — so the **institutional-delay** window in which the delayed-recruitment channel carries its cycle is bracketed ($\tau \approx 10$ carries the $\sim$ 17-yr cycle; $\tau \approx 21$ is stable), the stage analogue of the two families' fundamental-window structure of Section 5.1. The recorded $\eta = 3.0$ band-centre rows close the section: ($r = 1.57$, $g = 1$, $\tau = 2.5$) gives period 4.0 yr, amplitude 3.5 (recorded; re-run 4.00/3.49), and ($r = 0.8$, $g = 2$, $\tau = 5.5$) gives period 8.04 yr, amplitude 12.4 (recorded; re-run 8.05/12.35), both τ-converged.

9.5 The distinct two-stage companion

The recovered batch also contains a separately-provenanced object — the two-stage (adult/juvenile) harvest model with states X_A, X_J, Z, E, where the harvest channel is adult take versus juvenile take. That model is the subject of a separate companion article and is **not** the delayed-recruitment plant of this section: there g is the stage *duration* (a demographic residence timescale) and the only genuine delay is the institutional τ, whereas here g is the maturation *delay*. The two objects' crossing records must not be conflated. Its archived computation (adult-take Hopf at $\tau^* \approx 52.07$ yr, period 269.4 yr, on the $g = 5$ class; 16.76/121.80 yr on $g = 1$; 31.66/200.88 yr on $g = 2$; none for juvenile take) is reproduced byte-identical from the recovered `stage_core.py`, and is recorded in the supplementary material as that companion's table, not as a result of the present section.

9.6 Registration and status

The records of this section are certified by the registration campaign (its gate log is deposited with the supplementary material, S9; all gates pass): the $g = 0$ validation rows (G2), the fine-map bands with refined edges, the wide-mesh verification of the recorded cells and the mesh-range caveat (G3), and the nonlinear ground truth (G4). The fine-map table is deposited with the supplementary material (S9). The records are a reproduction on the recovered, declared instance — a continuous-time characteristic-root scan with RK4 verification (the DDE analogue of a multiplier scan) — not a proof for all parameters; no discrete sampled-review stage-map Floquet computation was ever built (the recovered readme states so explicitly), and no saddle-node, Neimark–Sacker, or collocation-grade classification is claimed for the stage records.

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10. The Loop-Gain Family

The two channels of Sections 5 and 6 look different on the stock–memory block, but they share a common linearised feedback identity. The unifying object is the loop-gain family: an ecological resolvent supplying admissible frequencies, an institutional gain that must close the loop, and a delay supplying phase. The gain condition is delay-independent; the phase condition rotates through match and past it. Non-monotone delay-stability is therefore possible and structurally natural in multi-frequency loop architectures. The implication for institutional design is that

the two channels are two instances of one feedback identity, and the channel-separation theorem is a property of the institutional gain — its modulus and the phase of the direct damping — not of the sign alone.

10.1 The general feedback identity

The vector ledger and the reduced cores share a linearised feedback identity, not one nonlinear system. Writing the ledger in primitive fluxes — assimilation g, mortality m, decomposition d_U, harvest h — the signed depletion signal $\ell = h-(g-m) = -\dot{X}$ is an identity, and liquidation is $[\ell]_+$. Let $\boldsymbol{\xi}$ be the ecological perturbation, $\dot{\boldsymbol{\xi}} = \mathbf{J}\boldsymbol{\xi} + \mathbf{b}_E e$, with memory a gain-$\gamma_m$ filter of $-\mathbf{c}^\top\dot{\boldsymbol{\xi}}$. The linearised loop of every core in this paper is

$$\lambda - C_E - C_Z e^{-\lambda\tau}\,\frac{-\gamma_m\lambda\,\mathbf{c}^\top(\lambda\mathbf{I}-\mathbf{J})^{-1}\mathbf{b}_E}{1+\tau_m\lambda} = 0. \tag{14}$$

The three-state core is the 1×1 case ($\mathbf{J} = A_N$, $\mathbf{b}_E = A_E$, $\gamma_m = 1/2$); the primitive-flux (stoichiometric) core is the 2×2 stock–detritus block with $\gamma_m = 1$ (its signed memory $\ell = -\dot{X}$ is unregularised, which places it outside the $Z \geq 0$ invariance statement of Section 2.2). Both reductions are symbolic identities: the three systems share this characteristic identity but differ in their nonlinear objects — the upper fold is version-robust at bracket resolution (≈ 64.4 yr: finite-collocation second turning candidate 64.4023; four-state bracket $[64.25, 64.5]$, whose 0.4% width sets that resolution). The Hopf pair stays within a few percent, the equilibrium memory differs ($Z^* = \delta > 0$ versus 0), and the primitive-flux core has no baseline Hopf ($\eta_{\text{crit}} \approx 2.34$). The three ingredients are an ecological resolvent supplying admissible frequencies, an institutional gain $C_Z\gamma_m$ that must close the loop, and a delay supplying phase; the gain condition is τ-independent, the phase condition rotates through match and past it, and non-monotone delay-stability is possible and structurally natural in multi-frequency loop architectures (an open-set transversality argument for genericity is not supplied, and no generic claim is made).

10.2 The loop-gain exclusion theorem

Theorem 10.1 (Loop-gain exclusion of delay-induced Hopf)**.** *Write the general feedback equation as $\lambda - C_E - C_Z e^{-\lambda\tau}G(\lambda) = 0$. Assume*

(H1) $\sup_{\omega\in\mathbb{R}} \frac{|C_Z|\,|G(i\omega)|}{|i\omega - C_E|} < 1$,

(H2) the undelayed system is Hurwitz,

(H3) the zero root is excluded for every delay ($-C_E - C_Z G(0) \neq 0$; in the three-state architecture $G(0) = 0$ and $C_E < 0$ supplies this).

Then there is no purely imaginary characteristic root for any $\tau \geq 0$, hence no delay-induced Hopf, and the equilibrium is exponentially stable for each fixed $\tau \geq 0$.

Proof. An imaginary root $\lambda = i\omega$ would give $i\omega - C_E = C_Z e^{-i\omega\tau}G(i\omega)$, hence $|i\omega - C_E| = |C_Z||G(i\omega)|$, contradicting the strict supremum (H1); the zero-root and reference-stability hypotheses (H2)–(H3) close the switch argument exactly as in Theorem 6.1. □

The companion Nyquist reading: a Hopf root exists if and only if the loop transfer $L(i\omega,\tau) = C_Z G(\lambda)e^{-\lambda\tau}/(\lambda - C_E)$ equals 1 for some $\omega > 0$, with the small-gain bound $|L| < 1$ as the uniform special case. A separate sufficient certificate in the same family is the Halanay-type delay-independent stability theorem, stated for systems of the form $\dot{x}(t) = A_0 x(t) + A_1 x(t-\tau)$ in a declared norm: if for some $\alpha_0 > \beta_0 \geq 0$ the logarithmic matrix measure satisfies $\mu_*(A_0) \leq -\alpha_0$ and $\|A_1\|_* \leq \beta_0$, then the zero solution is exponentially stable for every fixed $\tau \geq 0$, with decay rate the unique $\nu > 0$ solving $\nu = \alpha_0 - \beta_0 e^{\nu\tau}$ (Halanay, 1966; the decay rate is denoted ν here so that η remains the effort-response coefficient of Section 2 throughout). It is not a consequence of the scalar loop-gain formulation and is stated as its own theorem, not as "the" general certificate. Two qualifications travel with it: the decay rate depends on τ and tends to

zero as the delay grows (the stability is pointwise in τ, not delay-uniform), and the condition is sufficient, not necessary — its failure removes the certificate without proving instability; in nonlinear applications the theorem is local after a declared linearisation unless a global incremental bound is proved. The named-system theorems of Sections 5.4 and 6.2 are instances of this certificate family at declared linearisations.

10.3 The logistic identification lemma

Lemma 10.1 (Logistic identification)**.** *In the primitive-flux core take $A \gg K_A$ and set $\mu - d = r > 0$, $c = r/K$. Then, uniformly on a declared domain where X is bounded and A is bounded away from zero,*

$$g - m = rX\left(1 - \frac{X}{K}\right) + O\left(\mu X \frac{K_A}{A}\right),$$

with the remainder carrying flux units. The identification requires the mortality identification: $K_A \to 0$ without $\mu - d = r$ and $c = r/K$ yields $g \to \mu X$, which is not logistic.

Proof. Write $g - m = \mu X \frac{A}{K_A + A} - dX - cX^2$; for $A \gg K_A$, $\frac{A}{K_A+A} = 1 - \frac{K_A}{A} + O((K_A/A)^2)$, so $g - m = (\mu - d)X - cX^2 + \mu X\, O(K_A/A)$. With $\mu - d = r$ and $c = r/K$ this is $rX(1 - X/K) + O(K_A/A)$. Taking $K_A \to 0$ alone gives $\frac{A}{K_A+A} \to 1$, hence $g - m \to (\mu - d)X - cX^2$, which is μX only if $d = c = 0$; the logistic identification is a joint statement about mortality and saturation, not a limit of the assimilation flux alone. The claim concerns the linearised feedback identity, not the nonlinear objects, which also differ in the equilibrium memory level, the effort baseline, and the closed form of E^*. □

10.4 The saturating-gate negative screen

Replacing the autocatalytic factor $\eta E Z_\tau / \Delta_{\mathrm{ref}}$ by a saturating gate $\sigma(Z_\tau/Z_0)$ whose linearisation at equilibrium decreases in deployed effort and contains no factor $\eta E^*/\Delta_{\mathrm{ref}}$, a search of more than 300 randomised parameterisations found no genuine imaginary-axis root of the general feedback equation (14) — the screen record, with the method stack of Newton eigenvalue tracking, joint modulus minimisation, and nonlinear integration, and the nonexistence reading, is deposited in the supplement, S11, which also carries the MPF sweep and intermittency records. This is a numerical nonexistence report on a compact searched set, not an analytic exclusion on the parameter space, and it does not establish that autocatalysis is necessary for a Hopf pair — other non-autocatalytic feedback laws could supply sufficient loop gain and phase. The gated law (1) is not of this class — its C_Z contains both the gate factor and the autocatalytic factor, and it admits the Hopf pair. In the searched parameter set, removing the autocatalytic factor reduced the gain enough that no imaginary-axis root was detected.

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11. Discussion

11.1 The two channels as a design distinction

The sign separation is the paper's structural finding. On the identical stock–memory block the two rules carry opposite mathematics:

- **The mobilising law ($C_Z > 0$)** has an interval-enclosed imaginary-root crossing pair and a numerically resolved five-regime attractor topology.
- **The protective quota-tracking law ($C_Z < 0$)** carries the no-Hopf theorem — stability at every delay — and stability under annual review.

The distinction is not an artefact of the calibration: the iso-gain sign flip (Proposition 6.1) shows that a sign flip alone, without the genuine quota law's modulus change, retains the Hopf pair at shifted delays and a loop gain above one — so the sign alone does not decide, and the protective channel's no-Hopf property is a property of the quota law's combined direct damping and delayed gain. On this architecture, institutions whose calibrated effort law combines weak direct damping with sufficient delayed mobilising gain are exposed to delay-induced cycles; institutions tracking a quota with the calibrated damping are not. The channel-separation reading answers the question posed by the behavioural literature (Moxnes, 1998): the hazard is not slow governance as such, but slow governance of the calibrated mobilising law — weak direct damping with sufficient delayed gain, of which the sign is one ingredient among three (modulus, phase placement, and digital implementation).

11.2 The review interval as control

Under periodic review the two channels respond in opposite directions to the same instrument. The protective channel is stable at annual review and — under the exact held-measurement update (Proposition 7.1) — at every tested review interval. Its only sampled instability is the explicit-Euler command step's near $T_r \approx 2.3$ yr — an artefact to be recognised, not a design threshold; for the calibrated local continuous-delay model, protective delay does not generate a Hopf instability, which is the precise form of the claim, not a general immunity of quota-tracking institutions. The mobilising channel is unstable at annual review and restabilises as the review interval is lengthened, through its complex unit-circle crossing at $T_r = 6.50$ yr under the exact held-measurement update *at the declared parameter vector*. The location is reproducible across discretisation schemes and fragile across parameters (the scheme-dependence remark of Section 7), so it is reported as a property of this instantiation rather than as a threshold. The practical reading: for a slow stock governed by a mobilising effort law, frequent assessment and rapid response is destabilising, and within this operator the only computed spectral control is the review interval itself — a governance parameter, not an ecological one.

> **Management caution.** Local spectral stabilisation is not governance: an interval that stabilises the equilibrium may still admit severe inter-review depletion, so the statement stops at the spectral level (Section 7).

The result echoes, in a single-species institutional setting, the sampled-data principle that the sampling period is a controller design variable (Åström and Wittenmark, 1997), and it supplies the missing dynamical content to the observation that institutional timing structures resource governance (Ostrom, 1990). Where harvester forecasting can dampen the cycles induced by delayed knowledge of the harvested population state (Adamson and Hilker, 2020), the review interval is the design variable analysed here.

11.3 Generality: what carries beyond the analysed class

Three coordinates of the institutional loop are independent of the ecology they govern: the sign structure and modulus of the response law, the deployment delay, and the review cadence. The theorems of this paper fix what those coordinates do within the declared stock–memory–effort class — the delay window and its two certified crossings, the no-Hopf property of the calibrated quota law, the opposite cadence responses of the two rules — and the class itself is stock-agnostic: a harvested renewable stock under institutional control, of which fisheries are the motivating instance, with the liquidation channels of Section 1.1 (culling the standing stock, suppressing recruitment, decoupled consumption) marking how far the same loop reaches beyond a single-sector reading. What transfers to other periodically reviewed renewable-resource regimes is therefore the mechanism's form, not the numerical thresholds: the thresholds are theorems about the declared parameterisation, and any field application must re-identify the institutional

coefficients — the gains, delays, and cadences of the governance loop — with the same standing as the biological ones.

For management readers the paper's technical vocabulary has a direct translation:

Technical term	Management reading
Mobilising law ($C_Z > 0$)	Effort-escalation rule: decline triggers more deployment
Protective quota-tracking law ($C_Z < 0$)	Catch/quota rule: harvest restored toward a cap
Hopf crossing	Onset or loss of oscillatory stock dynamics
Neimark–Sacker-type crossing	Quasi-periodic instability emerging under periodic review
Monodromy spectral radius	Stability margin of one review cycle
Review interval T_r	Assessment/management review cycle length
Sample-and-hold review	The harvest control rule in force between assessments — a total allowable catch (TAC) or effort decision set at each review and held fixed until the next
Subcritical Hopf crossing	A tipping point for cycles: the boundary where an oscillatory attractor is already present, so the cycle regime arrives with coexistence rather than a clean switch
Inter-assessment stability margin	The monodromy spectral radius in plain words — how far one review cycle's contraction of small deviations sits below one

Two cautions travel with the translation:

- **Discretisation artefacts are a general warning for management modelling, not a curiosity of this system.** In both channels an explicit first-order update manufactured thresholds (near 2.3 yr for the protective channel, the half-century restabilisation for the mobilising channel) that the exact update of the same reviewed loop does not carry, so a discrete-time management model can report stability boundaries that belong to its discretisation rather than to the institution.
- **Local spectral stabilisation is not governance.** An interval that stabilises the equilibrium may still admit severe inter-review depletion (Section 7), and the capture boundaries of Section 8 are basin statements, not thresholds a manager can read off a spectrum.

The design reading that survives both cautions is the paper's central one: the response rule's class is a design choice, the review interval is a control variable with opposite effects under the two rules, and the institutional coefficients are dynamical parameters to be identified, bounded, and designed against. The documented institutional timelines collected in the next subsection ground the scales of those coordinates in the management record.

11.4 Documented institutional timelines

The two timing coordinates of the analysed loop — the deployment delay and the review cadence — have documented values at management scale, and the northern cod (NAFO 2J3KL) record is the canonical instance. Assessment of that stock proceeded on an annual cycle while the estimated spawning-stock biomass fell from about 735 kt in the 1991 assessment to about 31 kt in the 1994 assessment (DFO, 2016); the directed-fishery moratorium was announced on 2 July 1992, and the renewed commercial fishery on 26 June 2024 — thirty-two years later (DFO, 2016, 2024). The collapse itself has been attributed to overexploitation under sustained harvest pressure (Hutchings and Myers, 1994), with the assessment-and-response chronology dissected as a management failure (Walters and Maguire, 1996).

For this paper the record does two things. It documents that institutional response delays of years, and management-cycle decisions spanning decades, are real objects at the scale the analysed windows occupy — the mobilising channel's delay window between crossings near 3.7 and 150 yr, and the stage-analysis bands of Section 9.3, whose $g = 2$ τ-window lies inside the documented 2–13 yr governance-lag distribution (Hocherman, Trop, and Ghermandi, 2025). And it supplies the canonical instance of the inter-review depletion that the local spectral statements of Section 7 explicitly do not certify: a stock falling by a factor of more than twenty between annual assessments while the institutional response remained incremental.

The review cadence, likewise, is an actively managed design parameter in operating fisheries rather than a theoretical construct. Annual assessment and annual catch-setting are the historical norm for major stocks, and management agencies are actively weighing moves to multi-year assessment cycles; the assessment-frequency literature finds that lengthening the interval changes fishery performance — less frequent assessment reduced relative yield and increased the risk of stock depletion and interannual variation in yield in simulation (Li, Bence, and Brenden, 2016), with the same caution echoed for a large coastal shark fishery under multi-year assessment (Peterson et al., 2022). That literature treats the review interval as a management-design variable with risk consequences — the empirical counterpart of this paper's reading of T_r as a local spectral design parameter — and its direction of caution matches the theory's: the mobilising channel restabilises only for review intervals above about 6.5 yr at the declared parameter vector — a figure that is not identifiable and should be read as illustrative — but local spectral stabilisation is not governance, and inter-review depletion is the uncertified cost that both the theorems and the assessment-frequency record place at the centre.

What the documented timelines ground is the scale of the two timing coordinates, not the coefficients of any theorem. No stock, effort law, or delay value in this paper is calibrated to a named fishery, and the northern cod record is a timescale grounding and a caution, not a calibration: the institutional coefficients — the gains, delays, and cadences of the governance loop — remain to be identified from field data before any threshold of the declared class is read against a fishery. A structured search across more than thirty resource systems for an unconfounded instance of the delayed institutional feedback mechanism — a responsive rule, an independently dateable lag, no dominant competing driver, and unit-level data — found no eligible case; that search and its null result are the subject of the companion sampled-governance paper, and they mark the empirical programme named in the conclusion as prospective. The documented record therefore supplies what a sustainability or management readership requires of a theory paper — real governance timelines at the analysed scales — without converting any theorem into a claim about a named institution.

11.5 Certification levels

The paper's results carry three distinct certification levels, and the distinction is maintained claim by claim:

1. **Proved theorems.** The cubic modulus condition, the even-pairs algebra, the no-Hopf theorem, the monodromy formulas, and the loop-gain exclusions.
2. **Interval certificates.** The Hopf delay enclosures are interval-Newton certificates of the local spectrum of the cubic — re-execution-verified with outward rounding — but are not certificates of the bifurcation in the sense of Church and Lessard (2022); upgrading them to that standard is a stated open task. The 13-digit enclosures certify roots for the exact registered decimal parameter vector; they do not represent parameter-estimation uncertainty, and any policy reading of the thresholds must add a sensitivity layer over the declared parameter windows (Section 8.5).
3. **Declared-status numerical results.** The global folds, the attractor classification, and the Lyapunov coefficients are numerical results at declared status, with the lower boundary

events left as a provisional saddle-node against a crisis reading, with Krawczyk enclosures registered for both finite-dimensional turning candidates (the lower candidate's Moore–Spence/Krawczyk enclosure, and the second fold's interval-Krawczyk certificate on the $m = 64$ system with three-order agreement 5.53×10^{-7}; the continuum off-grid residual stage and the continuous-delay lift remain open for both), documented component by component in the supplementary material.

Rigorous saddle-node results for delay equations exist for specific classes (Beretka and Vas, 2020); closing the gap for this system is a second stated open task.

11.6 Relation to the early-warning literature

The regime-shift literature's early-warning indicators — rising autocorrelation, rising variance — are candidate diagnostics near fold-type transitions under additive noise with responsive control (Scheffer and Carpenter, 2003; Scheffer et al., 2009; Carpenter et al., 2011). Four indicators are candidate diagnostics for the fold events of Section 8: critical slowing down (the lag-1 autocorrelation of the stock rises near a fold); rising variance (rolling variance increases under additive noise near a bifurcation); epistemic divergence (disagreement among independent estimators, innovation residuals, or a widening set-membership assessment tube — the true state is not observed, so the raw gap $|\hat{N} - N|$ between an estimated and the true stock (written in the paper's stock symbol, since $S(N)$ is the yield) is not itself an operational statistic, and the observable substitutes are listed instead); and policy inertia (physical stress increasing without a corresponding control adjustment). The caveat is load-bearing: these are not universal early-warning signals — rising autocorrelation and variance are tied to specific local bifurcations under specific noise and timescale assumptions, while the capture onset of Section 8.2 is a basin boundary — the finite-collocation turning candidate lies 84 yr lower in τ — and the lower fold's crisis alternative remains live, so an indicator tuned to a fold threshold would be watching the wrong event, and each indicator must be tied to a hypothesised mechanism before it is used; they do not follow from the local machinery of this paper, and each is a separate falsifiable empirical-statistical claim. In the bistable windows of Section 8.2 the relevant diagnostic is not a pre-collapse trend but the history-dependence of capture — the registered basin record's reversal at $\tau = 5.575$ (the depleted history captured, the large-stock history settling) is the load-bearing asymmetry — which the variance/autocorrelation pair cannot see.

11.7 The ecological reading

(i) **The exposed life histories are doubly selected.** With the maturation delay switched on, the two-crossing structure survives only on the product locus $r\,g \approx 1.5$–1.6 (Section 9.3): longer maturation delays move the band to slower stocks, down to $g = 10$ near $r \approx 0.08$. Selection acts a second time through the institutional delay. The band τ-windows grow with the maturation lag — $g = 1$: $(1.6, 3.5)$ yr; $g = 2$: $(2.6, 7.8)$ yr; $g = 5$: $(9.9, 20.3)$ yr — and the $g = 2$ window is the one that lies inside the documented 2–13 yr governance-lag distribution (Hocherman, Trop, and Ghermandi, 2025). Fast-maturing small pelagics are therefore the predicted exposed class, not an illustrative example. At the slow end the question inverts: with recruitment lagged, the slow-r system carries a 358.8-yr cohort cycle, so the institutional-delay question becomes moot there (Section 9.4).

(ii) **Ecological delay and institutional delay sit in different compartments and produce different cycles.** The maturation delay enters the stock equation and the deficit signal; the institutional delay enters only the effort equation (Section 9.1). Their cycles differ accordingly: cohort cycles whose periods track the life history (~ 20 yr at $r = 0.5$, $g = 5$; 358.8 yr at $r = 0.02$, $g = 5$) versus loop cycles whose period tracks the deployed

response (16.96 yr at $r = 0.3$, $g = 5$, $\tau = 10$). With both delays present the institutional window survives on top of the ecological one — $\tau \approx 10$ carries the $\sim$ 17-yr cycle; $\tau \approx 21$ is stable — the stage analogue of the fundamental window of Section 5.1 (Section 9.4).

(iii) **The form of extraction is ecological even where the local mathematics is not.** The two harvest channels of M3-LC — removing standing stock versus suppressing recruitment — leave the equilibrium, the Jacobian, the characteristic equation, and both Hopf points identical (Section 2.3). The collapse dynamics still separate them: at $\tau = 115$ yr, $\kappa = 0.5$, long-lived transients reach minimum $N \approx 33$ under pure stock culling and $N \approx 10$ under pure recruitment suppression; under fixed demand, culling first hits $N = 0$ near time 158 while suppression approaches zero asymptotically, reaching $N < 1$ near time 430 (Section 8.3). Local spectra cannot see the extraction form; the excursion record can.

(iv) **Growth-coupled ecology cannot widen the window.** The Droop nutrient–quota variant leaves the r-window unchanged — upper edge ≤ 0.023 yr^{-1} at $\eta = 0.914$, and no Hopf crossing at any $r \geq 0.2$ yr^{-1} — because a growth-coupled pool self-relaxes at exactly r and cannot supply the slow companion variable (Section 8.3). The only r-independent slow pool in the registered family is the working core's ω_A-type exchange: decoupled storage, not growth-coupled ecology, is what can slow the loop.

Each reading consolidates records registered in Sections 2.3, 9, and 8.3; none adds a computation, an archetype, or a calibration, and the limitations of Section 11.8 apply to all four unchanged.

11.8 Limitations

(i) The model class is a mathematical parameterisation: no stock, effort law, or delay value is calibrated to a named fishery, and the institutional coefficients have not been identified from field data; the results are theorems about the declared class, with the parameter windows of Section 8.5 delimiting the regime of the two-crossing structure. The documented governance timelines of Section 11.4 ground the scales of the delay and review coordinates; they are not a calibration, and no institutional coefficient is identified from them. Identification is its own layer: different gain–delay–memory triples can realise similar observed harvest trajectories, so the institutional coordinates are latent structural parameters — separable only through exogenous variation or direct process measurement, not from the observed trajectory alone — and the functional form of the response law is identified, if at all, separately from its parameters. (ii) The delay is a single discrete lag — a reduced compression of the observation, assessment, decision, deployment, and compliance stages that a multi-stage institution carries separately; the documented cod chronology of Section 11.4 (the annual assessment cycle, the 1992 moratorium decision, the 2024 reopening) is a record of such stages at management scale, and what the single lag absorbs is their composite timing, not any one stage. Distributed delays and variable-time institutional lags require separate analysis. (iii) The sample-and-hold results concern the declared Euler review scheme and its exact held-measurement counterpart (Theorem 7.1, Proposition 7.1); other discretisations have different monodromies, and the continuous-delay and periodic-review recommendations are not interchangeable. (iv) The global classification of Section 8 is provisional where stated: the lower boundary's saddle-node classification, the upper boundary's type, and interior monostability are open. (v) The Hopf-persistence statements are conditional on the unproved reduction conjecture (Section 5.3). (vi) The scalar core of Sections 2–6 and the global numerics of Section 8 concern a scalar-logistic ecology; stage structure enters in the delayed-recruitment (maturation-delay) generalisation of Section 9, and spatial structure, multispecies interactions, and the distinct two-stage

adult/juvenile harvest model (the separate companion of Section 9.5) remain outside the analysed class, with the delayed-recruitment literature (Gurney, Blythe, and Nisbet, 1980; Costantino et al., 1995) engaged as background.

11.9 Open problems

Two genuine next theorems are stated with their declared gaps, with the first Lyapunov coefficient of the exact-hold map's Neimark–Sacker crossing as the matching open computation: the Church–Lessard upgrade of the Hopf certificates (Section 11.5's first stated open task, taken together with the continuum off-grid residual stage of both folds), and persistence of a transverse fold of periodic orbits under small typed coupling (requires a verified fold baseline, spectral separation, and regularity of the infinite-dimensional Poincaré map). The further programme directions carried by the open-problem register — an RFDE/hybrid transition-persistence analogue, an n-patch super-equilibrium criterion, a variable-time delayed-hybrid information kernel, a restricted delay-separation principle for modularly identified governance loops, and an exergy-limited controller class with its declared conjecture and non-universality caveat — are recorded in the supplement's open-problem register (S6) and are not elaborated here: none connects to a theorem proved in this paper. A third stated open task is parameter-box certification: the interval enclosures of Section 5.1 certify the registered decimal parameter vector, and the one-at-a-time windows of Section 8.5 are not boxes — extending the interval Newton stage from the point vector to declared parameter boxes is the upgrade that would carry the threshold statements from arithmetic precision to the policy robustness that Section 11.5 reserves to a sensitivity layer.

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12. Conclusion

Where the delay sits determines what it does. Delays in the ecological dynamics of harvested stocks are the classical subject of delayed-logistic and delayed-recruitment analysis. The delay analysed here sits in the institutional loop that converts observed decline into deployed effort, and its mathematics separates sharply across the two calibrated response laws.

The mobilising channel — effort responds to decline by deploying more effort — carries a subcritical Hopf pair with interval-certified fundamental crossings of two positive frequency families, and a five-regime attractor topology on the explored fundamental-delay range with two Krawczyk-enclosed turning candidates in finite collocation maps (Section 8.2), and with a numerically observed basin-transition interval at 148.6–149.5 yr. Off-grid residual validation and a continuous-delay fold proof remain open. Under periodic-review governance, the mobilising channel is unstable at the annual scale and restabilises only for review intervals above about 6.5 yr at the declared parameter vector — reproducible across discretisation schemes, fragile across parameters, and therefore illustrative rather than prescriptive — via a Neimark–Sacker-type crossing of the exact held-measurement monodromy; the half-century crossing (47.5 yr) reported by a first-order (Euler) discretisation is a numerical artefact — a local spectral statement, not a governance recommendation. The protective channel — effort is restored toward a quota — carries a no-Hopf theorem valid at every delay at the calibrated point, stability under annual review, and an instability threshold that is provably an artefact of the discretisation.

The unifying object is the loop-gain family: an ecological resolvent, an institutional gain, and a delay supplying phase, with the gain condition delay-independent and the phase condition rotating through match and past it. For the design of extractive governance the conclusion is specific. The laws differ not merely in sign but in modulus, damping, phase placement, and digital implementation. The calibrated mobilising law is the hazard on this architecture. The

review interval is a control variable with opposite effects in the two channels. And the institutional coefficients — the gains and delays of the governance loop — are dynamical parameters with the same standing as the biological ones, to be identified, bounded, and designed against.

Beyond the declared class, the coordinates themselves generalise: any periodically reviewed renewable-resource institution — whatever the stock, the sector, or the rule's legal form — admits the same three design coordinates analysed here (the response law's class, the deployment delay, and the review cadence), and the theorems fix their local effects in one analysable instance of that family. The mechanism is available wherever an institution answers an observed decline with a lagged rule. The empirical programme that follows is the identification of these coordinates in field institutions.

Bridge between operators. Part I varies an institutional response lag τ inside a continuous feedback law; Part II varies the review interval T_r of a sample-and-hold map. Their spectral boundaries are not estimates of one shared delay, so each result retains its own model and evidential scope. The local spectral comparisons use the characteristic equation and sampled monodromy, not a fold theorem for periodic orbits. A continuous-delay fold theorem would strengthen the global-attractor interpretation, but is not a premise of the operator-contrast result.

Part II — The sampled channel: governance delay as a review clock

Study scope and principal findings

Environmental governance runs on clocks: institutions observe resource systems at discrete times and hold the resulting controls until the next review. Formal models routinely substitute this sampled architecture with a continuous delay or a single annual step. We show the substitution matters. Modelling periodic review as sample-and-hold governance — a loop that observes and updates only at review times — we establish two exact properties, forward invariance of the sampled state space and rapid-review consistency over finite horizons only, and compute stability review-interval by review-interval, demonstrated on fisheries. Stability boundaries move or vanish when the operator is changed: at an illustrative baseline the exact map crosses once near a 6.5-year interval while one-step approximations report artefact crossings, and the protective channel is stable at every tested interval. That crossing is reported as a band, not a threshold: under a half-percent joint perturbation of the model parameters it ranges from under one year to nearly eleven, and under a one-percent perturbation the instability verdict itself fails in a third of cases (Section 3.4.1), so the robust content of the paper is the operator contrast and the protective-channel result rather than the location of the crossing. A multiplicity-controlled screen of 42 fish stocks finds no robust institutional cycles and a 32-system cross-sector case search no unconfounded oscillator, showing periodicity alone cannot diagnose governance feedback; the northern cod record splits into a formulation-dependent crash and an unresolved post-collapse identification problem. For the specified model and tested intervals, the timing and form of policy revision affect the computed local stability verdict; the numerical crossing is not a threshold for other institutions or resource systems.

Evidential status. Every central claim carries its exact status — exact, conditional, archived, provisional, or prospective — in the statement inventory (Supplementary S1); the main text states each result once with its record pointer, and Section 4.7 collects the limitations.

1 Introduction

Fisheries governance is periodic. Surveys are scheduled, assessments are produced at fixed cadences, advice is delivered at review points, and the resulting catch or effort controls are held in force until the next review. The review interval — the time between successive opportunities to revise the command — is sometimes a single year, sometimes a multi-year plan, sometimes a moratorium of indefinite length (Punt and Donovan, 2007; DFO, 2016). The formal literature treats this institutional clock in two ways, neither of which matches the operating object. Population-dynamic theory models institutional response either as a continuous-time lag, that is, a delayed signal entering an evolving control law, in the lineage of Gurney, Blythe, and Nisbet (1980); or as an annual discrete-time decision appended to a surplus-production model, in which the between-decision dynamics are compressed into a single annual step. Real institutions do neither. They observe and assess at discrete times, choose a rule-based command, and then hold or interpolate that command until the next decision opportunity. This paper calls that architecture *sampled governance* — a governance loop in which state observation and control update occur at discrete review times, not continuously. It is the same architecture the title and abstract call *sample-and-hold governance*: one object under two names — *sampled governance* names the institutional loop, *sample-and-hold* its control-theoretic update law — and the article body uses *sample-and-hold* (with *sampled* for its maps and loops) throughout.

Two failure modes follow from this mismatch. The first is *architecture substitution*. Replacing the sample-and-hold architecture by a continuous delay, or by an annual difference equation, can move or delete stability boundaries. Conclusions about a governance system then depend on an operator that the institution does not implement. The second is *causal promotion*. Spectral peaks, cohort cycles, and crisis-driven monitoring records circulate as evidence for institutional-feedback mechanisms when they carry no information about the controller's sign, cadence, or timing at all. The marine-science literature already contains the disciplinary template for resisting this promotion. The skeptical re-analysis of the Russell Cycle (McManus, Licandro, and Coombs, 2016) showed that an 88-year zooplankton series long treated as a cycle failed to show a resolvable true periodicity. This article extends that discipline from climate-forced cycles to governance-feedback claims.

The contribution is fivefold. First, a sample-and-hold model of periodic review is specified, in which the governance clock is decomposed into seven time objects (observation interval, assessment interval, review interval, decision lag, deployment lag, ecological response lag, and memory timescale) and the extractive controller is an explicit discretisation with specified comparator classes. Second, two properties of the sampled process are established exactly: forward invariance of the sampled state space, and the precise scope of the rapid-review limit — a finite-horizon numerical-consistency statement that implies nothing about stability at positive review intervals. Third, the review-interval spectrum is computed under the discipline that the sample-and-hold map and the continuous-delay equation are *different operators*. A stability window located on one map does not transfer to the other, and the paper states which operator carries which statement. Fourth, an empirical layer is reported: a multiplicity-controlled spectral screen of 42 annually assessed stocks, a power analysis of that screen, and a structured case search across more than thirty resource systems. Fifth, the northern cod case (NAFO 2J3KL) is carried through the resulting falsification discipline. These five are retrospective; the paper's constructive content is the prospective programme — five prospective designs specified as preregistration targets — that could convert or refute the mechanism.

The model class examined here is the extractive controller: a rule that raises extraction effort in response to a perceived decline. This is the demand-side form of the overshoot loop. The response to a perceived shortfall falls on the productive stock itself rather than on its regeneration. Extraction is raised precisely because the base has declined, which accelerates the decline it is reacting to. Its protective counterpart (effort reduced in response to decline), fixed multi-year plans, and hybrid rules serve as comparators.

The extractive rule is not proposed as desirable management. It is a diagnostic representation of perverse pressure: when a perceived shortfall threatens short-term employment or revenue, the political-economic response — subsidies that sustain effort, pressure to hold catches up — can push extraction upward as the base declines. Capacity-enhancing subsidies are the documented form of this pressure (Sumaila et al., 2019). The protective controller is the policy-relevant comparator; the pair exists to show how controller sign interacts with review timing, not to recommend the extractive rule.

Conclusions drawn inside the extractive class are not generalised to governance as a whole, and the screen below asks whether such a channel is empirically resolvable at all. The implication for institutional design is direct: an annual-review extractive rule and a multi-year protective plan are not interchangeable instruments, and a sampled-data analysis must state which operator a given stability claim is computed on.

The architecture is not unique to fisheries. National climate pledges under the Paris Agreement are communicated every five years and informed by five-yearly global stocktakes (UNFCCC, 2015, Articles 4.9 and 14): held commands under periodic review (Figure 1A). Failures then take a common form across sectors: the governance clock (the review interval), the ecological clock (the recovery rate), and the environmental clock (exogenous cycles such as ENSO, Section 3.7) fall out of phase. The timing and form of policy revision — the review interval together with the revision rule — is the decision clock. This article develops the fisheries case in full; Section 4.8 returns to the cross-sector reading.

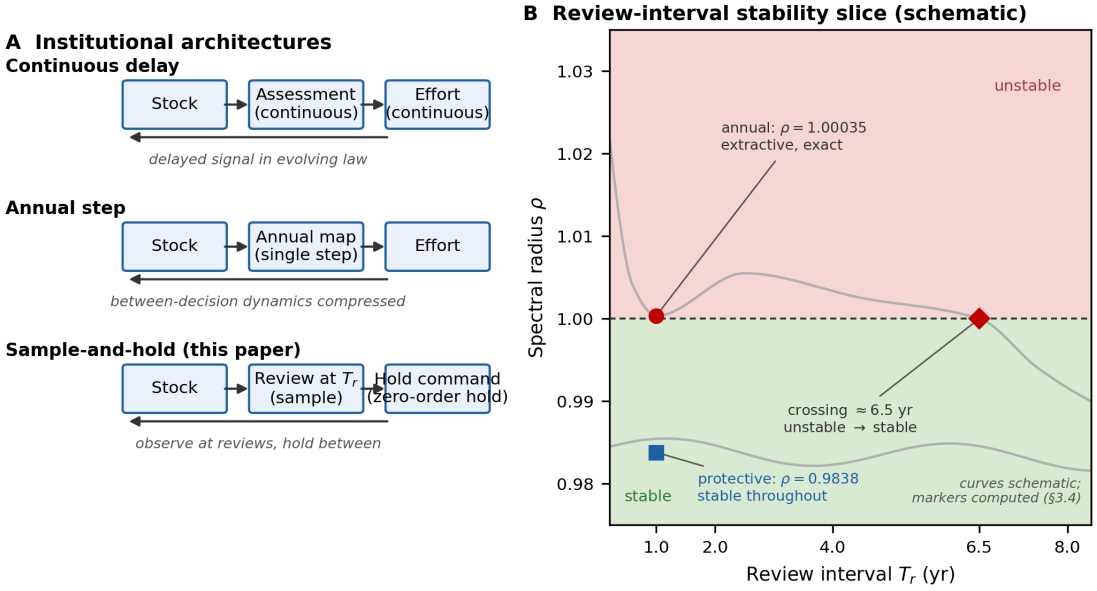


Figure 2: The decision clock. **A:** Three institutional architectures: continuous-delay feedback, the annual-step compression, and sample-and-hold governance, which observes at review times and holds the command between them. **B:** Review-interval stability slice (schematic): the extractive exact update is marginally unstable at annual review ($\rho = 1.00035$) and crosses once near a 6.5-year interval (unstable → stable), while the protective update is stable throughout ($\rho = 0.9838$ at annual review). Curves are schematic; markers show computed values (Section 3.4).

Related work

Sampled-data control. That a controller which samples, computes and holds is a different dynamical object from one that acts continuously is foundational in control engineering, not a refinement of it: the sampled-data theory of Chen and Francis (1995) and the digital-control treatment of Åström and Wittenmark (1997) are built on the distinction, and the framework of

Nešić and Teel (2004) exists precisely because stability of a continuous design does not transfer automatically to its sampled implementation. The zero-order hold appears there as an operator with spectral properties of its own. Little of that machinery is routinely imported into resource-governance modelling. The two exact properties established below — forward invariance of the sampled state space (Section 3.1) and rapid-review consistency over finite horizons only (Section 3.2) — are the governance analogues of facts that are standard in that literature and, to our knowledge, are not stated in this one.

Periodic review as practice, and the interval as a design variable. Real institutions do review periodically and hold. Management procedures are evaluated by closed-loop simulation and then applied on multi-year cycles (Butterworth and Punt, 1999; Butterworth, 2007; Punt et al., 2016), with review intervals of a few years the norm and the interval itself a choice the institution makes rather than a constant of nature. Practice is therefore sampled by construction, while the modelling literature reaches for the continuous-time delay or the single annual step. This paper treats the review interval as the design lever it already is in practice.

Delay, and the substitution this paper questions. Where lag is modelled in fisheries it is usually continuous. Shertzer and Prager (2007) simulate a spectrum of fisheries and find that management delay requires deeper catch reductions sustained for more years to recover benchmark status, and causes collapse most often under depensatory recruitment or density-dependent catchability. Brown et al. (2012) find that delays beyond about five years raise collapse probability under environmental decline, and that the long-run cost of delay exceeds the short-term benefit of overfishing. Adamson and Hilker (2020) model delayed knowledge of the stock and show it can induce harvester–resource cycles. Karlsson and Gilek (2020) treat delay as a governance pathology to be managed. Each of these is a statement about a *continuous* lag. The question asked here is prior to theirs: whether substituting a continuous lag for a sampled review is admissible at all. Sections 3.3 and 4.1 show that it is not — response regions and stability boundaries move or vanish when the operator is changed — so the delay literature and this paper concern different operators, and a result about one is not automatically a result about the other.

Position. The contribution is therefore not the observation that institutions act periodically, which is old, nor that delay is costly, which is well established. It is that the sampled operator has properties the continuous surrogate does not; that two of those properties can be stated exactly and proved; and that the review interval consequently carries a stability signature that can be computed rather than negotiated.

2 Material and methods

2.1 The sample-and-hold model of periodic review

The model treats periodic review as a sample-and-hold control architecture: the command set at a review instant is held fixed until the next review, and only the between-review ecological dynamics are continuous in time. Seven time objects are kept distinct, and no two are collapsed into a single "governance lag" unless the empirical record cannot resolve them and the aggregation rule is stated (Table 1).

For fisheries readers, the control-theoretic vocabulary maps onto management-procedure language: the controller is the harvest control rule; the command is the catch or effort limit it sets; the review interval is the advice cycle; the hold is the plan period between revisions; and the operator — the decision architecture on which a stability claim is computed — is the management procedure itself. Assessment error and implementation error enter where observation meets advice and where decisions meet deployment (Table 1).

Table 1. The governance-time ontology.

Object	Definition
Observation interval T_{obs}	Time between raw measurements
Assessment interval T_{assess}	Time between formal state estimates
Review interval T_r	Time between opportunities to change the command
Decision lag τ_{dec}	Assessment completion to formal decision
Deployment lag τ_{dep}	Decision to implemented change in extraction pressure
Ecological response lag τ_{eco}	Implementation to detectable ecological response
Memory timescale τ_m	Relaxation time of a filtered institutional signal; not a discrete delay

observation → assessment → advice → decision → implementation → ecological response

The chain is the sequence Table 1 resolves into distinct objects; collapsing any two stages without a stated aggregation rule is the measurement-level form of architecture substitution.

Reviews can be annual while deployment is delayed; observations can be frequent while decisions are legally fixed for several years; and stock and realised-effort series alone may identify only a combined closed-loop phase shift rather than the separate τ_{dec} and τ_{dep}. Separating them empirically requires dated observation releases, assessment products, commands, and implementation records. Alternatively, a proved structural-identifiability argument for the specified observation model can serve. Two further conventions apply. The review opportunity is not the same as the response sign: the extractive command studied here raises effort in response to a perceived decline, and the protective controller enters only as a comparator. The fixed-τ form is also an idealisation. Real institutions confronting visible scarcity typically change their instrument set — buyouts, engineered transfers, new infrastructure — rather than hold a single response law with constant delay. Any estimated τ therefore summarises a changing control architecture, not a structural constant.

Box 1. Management regimes as sample-and-hold architectures.

Four institutional regimes in the review–hold vocabulary.

Regime	Review	Instrument	Hold	Emergency clause
Annual TAC fishery	1 yr	TAC (quota)	1 yr	in-year adjustment where provided
Multiannual plan	3–5 yr	fishing-mortality rule	multi-year	trigger review
Moratorium	indefinite	closure	until review	reopening rule (Section 3.8)
Data-limited system	irregular	effort controls	variable	crisis response

Notation. Throughout, N denotes the resource stock, Z the institutional deficit signal, and E the extraction effort. The review Poincaré map — the discrete-time map obtained by sampling the continuous flow at review instants — is written $\mathcal{P}_{T_r}$; its Jacobian at the fixed point X^*, the monodromy matrix of the sampled loop, is $D\mathcal{P}_{T_r}(X^*)$ (Section 2.3's multiplier condition is written on it); the projection onto the admissible effort interval $[0, E_{\max}]$ is $\Pi_{[0,E_{\max}]}$. The signal map Φ is a nonnegative functional of the deficit, with memory timescale $\tau_m > 0$; its softplus realisation is Φ_k, with shift $\delta = \log 2/10 \approx 0.069$ and sharpness k (the shift is the equilibrium memory level, distinct from and unrelated to the effort-law gain δ_0, as the display below states). The effort law is F_B. The assessment operator is $\mathcal{A}_n$ and the observation operator is $\mathcal{O}$. In the control sections $S(\cdot)$ is surplus production — $S(N)$ on the logistic plant (defined

after equation (2)) and $S(A)$ on the stage plant (Section 3.4) — while g (Section 3.3), C_E and C_Z (Section 3.4), and the four-state loop (Section 3.7) are defined where they are used. In the northern cod case (Sections 2.7 and 3.8, Table 2) S is instead spawning stock biomass (reported as SSB in Table 2), $\mathfrak{s}$ the unstable threshold, K the unexploited carrying capacity, $C(t)$ removals, M instantaneous natural mortality (with extra mortality M_x in Lemma 2.2), and F fishing mortality. The two scopes of S never share an equation, and no other symbol serves two sorts: the monodromy is written $D\mathcal{P}_{T_r}(X^*)$ throughout, and Lemma 2.2's production maximum is written $f_{\max}$, leaving M to natural mortality alone. Symbols are tabulated with their definition sites in Table 4 of Appendix A.

Let review times be $t_n = nT_r$. Between reviews, extraction effort is held at E_n and the resource stock follows the logistic process under held effort:

$$\dot{N}(t) = rN(t)\left(1 - \frac{N(t)}{K}\right) - qE_nN(t), \qquad t \in [t_n, t_{n+1}), \tag{1}$$

while the institutional signal Z is the filtered nonnegative deficit

$$\dot{Z}(t) = \frac{\Phi\big(qE_nN(t) - S(N(t))\big) - Z(t)}{\tau_m}, \qquad \tau_m > 0, \tag{2}$$

with $S(N) = rN(1 - N/K)$ the surplus production and $\Phi \geq 0$ a nonnegative signal map with memory timescale τ_m. The assessment available at review n is

$$\widehat{Z}_n = \mathcal{A}_n(\{Y_j : t_j \leq t_n - \tau_{\mathrm{dec}}\}), \qquad Y_j = \mathcal{O}\big(N(t_j)\big) + \varepsilon_j, \tag{3}$$

which separates the latent process N, the observations Y, and the assessments $\widehat{Z}$. Measurement and assessment errors need not be independent or Gaussian, but the admissible error support is restricted to $\widehat{Z}_n \geq 0$. The multiplicative-error robustness experiment of Section 3.3 satisfies this by construction. In the executed calibration all parameters entering the effort law are positive (r, K, q, $E_{\max}$, Δ_{ref}, Z_{ref}, $\tau_m > 0$); because $Z_{\mathrm{ref}} > 0$ and $\widehat{Z}_n \geq 0$, the rational term is well defined on the admissible state space. The effort law's rational term is undefined at $\widehat{Z}_n = -Z_{\mathrm{ref}}$ and changes sign beyond it, and projection of the final command does not repair an undefined update. In the baseline deterministic record the assessment operator is exact and contemporaneous, $\widehat{Z}_n = Z(t_n^-)$: each command uses the just-flowed end-of-period state, indexed by the review that reads it ($\widehat{Z}_{n+1}$ in (4)); the exact update holds this same flowed value. The review map is the projected forward-Euler step

$$E_{n+1}^{\mathrm{cmd}} = \Pi_{[0,E_{\max}]}\left\{E_n + T_rF_B(E_n, \widehat{Z}_{n+1})\right\}, \tag{4}$$

— one specific reviewed controller among many (direct command rules, exact integration under held assessment, incremental rules with fixed step). Because the increment scales with T_r, a sweep in T_r changes both the hold duration and the accumulated gain per review. The separating computation is the exact held-assessment update (the exact update), reported in Section 3.4 with attribution to the companion delay study (Abaee, 2026). The effort law is

$$F_B(E, Z) = \left(1 - \frac{E}{E_{\max}}\right)\left[\eta E\left(\frac{Z}{\Delta_{\mathrm{ref}}} - \frac{E}{E_{\max}}\right) + \delta_0\frac{Z}{Z_{\mathrm{ref}} + Z}\right],$$

The protective comparator is the quota-tracking law

$$F_B^{\mathrm{prot}}(E, Z) = \left(1 - \frac{E}{E_{\max}}\right)\eta_p\big(E_{\mathrm{cap}}(Z) - E\big), \qquad E_{\mathrm{cap}}(Z) = E_0\frac{Z_{\mathrm{ref}}}{Z_{\mathrm{ref}} + Z},$$

with $\eta_p = \eta$ and $E_0 = E^*(Z_{\mathrm{ref}} + \delta)/Z_{\mathrm{ref}}$, calibrated so its unique interior rest coincides with the extractive fixed point; its linearised gains there are $C_E = -0.850336$, $C_Z = -1.661702$ (Abaee, 2026, equation (3)), reviewed through the same projection $\Pi_{[0,E_{\max}]}$. At $E = 0$ the law

commands $\eta_p E_{\mathrm{cap}}(Z) > 0$, so zero effort is not absorbing; saturation at $E_{\max}$ is through the gate and the projection.

together with the shifted, floored softplus signal map

$$\Phi_k(s) = \max\left\{0,\ \frac{1}{k}\log\bigl(1+e^{ks}\bigr) - \frac{\log 2}{k} + \delta\right\}.$$

with shift $\delta = \log 2/10 \approx 0.069$, the equilibrium memory level ($Z^* = \Phi(0) = \delta$; distinct from and unrelated to the effort-law gain δ_0) and sharpness k — the value used in the computed records, $k = 10$, is stated where it is used in Section 3.4 and collected in Table 3 of Appendix A. The multiplier records of Section 3.4 presuppose a fixed point interior to the nonsmooth regions of Φ_k and $\Pi_{[0,E_{\max}]}$: $\Phi_k(0) = \delta$ gives the equilibrium deficit $s^* = 0$ uniquely, with softplus slope $s p_k'(0) = 1/2$, and $E^* = 2.09$ lies in the interior $(0, E_{\max})$; the coordinates $(89.55188, \delta, 2.08962)$ are listed in Table 3 of Appendix A.

Equations (1)–(4) with $\Phi = \Phi_k$ constitute the extractive controller studied throughout. It is an explicit controller discretisation, not a generic harvest-control rule and not a model of protective quota reduction. Three comparators are distinctly specified: the protective controller (the quota-tracking law above, the same machinery with the response to decline entering with the opposite sign), the fixed-plan controller (no state-responsive update between scheduled resets), and the hybrid controller (annual-change limits, emergency triggers, or legal overrides). The comparators are specified for the prospective management-strategy evaluation (Section 4.5), not presented as completed experiments. A fixed plan and a state-responsive plan with the same nominal review period are not the same intervention. The model-level run of the comparator evaluation on the logistic hold-map core — the plant of the closed-form operator results — is executed and reported in Section 3.4 at the stated evidential status (deterministic, seed-fixed, on the specified architecture only). The prospective real-system design of Section 4.5 remains unexecuted.

2.2 Registration conventions

The deterministic record fixes a one-step flow-then-update convention, stated in pre-review/post-review terms. On $[t_n, t_{n+1})$ effort is held at E_n. At the review instant t_{n+1} the contemporaneous, exact assessment $\widehat{Z}_{n+1} = Z(t_{n+1}^-)$ of the flowed state is read ($\tau_{\mathrm{dec}} = \tau_{\mathrm{dep}} = 0$), and equation (4) computes the command E_{n+1} held on the following interval. The pre-review state is $X_n^- = (N(t_n^-), Z(t_n^-), E_{n-1})$ and the map used is the pre-review Poincaré map $X_{n+1}^- = \mathcal{P}_{T_r}(X_n^-)$. The reversed ordering (command first, flow second) is a different hybrid system and is not used. No decision or deployment queue is present, and multiplicative assessment error enters only the designated robustness experiment. A positive τ_{dep} requires an explicitly specified hold or interpolation rule.

The computational status of the response-region records is stated with the same care. Classification used long-horizon trajectories, tail amplitudes, multiple histories, and integration-step refinement. It did not use Poincaré-map multipliers, monodromy matrices, or Floquet multipliers. The reported bands are therefore finite-grid, trajectory-classified response regions, and no Neimark–Sacker, flip, or other multiplier-crossing classification is claimed. Until the solver configuration and initial histories are attached (Appendix A), the stage-output values carry provisional status. For the logistic hold-map core the multiplier-based boundary calculation is reported in Section 3.4. For the stage-structured map the boundary record is supplied by the labelled reconstruction of Section 3.4 — a newly specified, pre-registered object whose full specification (flow/update ordering, information pattern, derivative construction, multiplier trajectories, nonlinear trajectories, solver configuration) is given there. The original stage record remains unattached, and its output values keep provisional status. The computational requirements of this section and of Sections 2.4, 2.5, 3.3, and 3.4 — solver configurations, initial histories, stock identifiers, the eligibility table, the null-calibration record, simulation code and

seeds, and the reconstruction's plan, code, and output tables — are consolidated in Appendix A, which also fixes the pre-registration convention.

2.3 The review-map operators

Three objects are in play: the continuous-delay equation, the logistic hold map, and the stage-structured review map. The continuous delay equation and the logistic sample-and-hold map are the same feedback loop under two delay operators. The stage-structured map is a different ecological plant (delayed-recruitment stage structure against scalar logistic surplus production) with a different state dimension; it is not an operator substitution on the logistic plant. A valid architecture comparison holds the ecological plant, controller linearisation, equilibrium, and parameter vector fixed while changing only the timing operator. The 3–4 yr windows and the ≈ 6.5 yr exact-update crossing below are therefore reported as a comparison across plants and operators, which does not isolate the operator effect; the one-plant operator contrast in Section 3.4 isolates the operator for the logistic plant. Changing the operator can move or delete a crossing, and that relocation is a property of the review map's spectrum, not a refutation of the loop mechanism. The discipline throughout is that every stability statement names its operator, and statements computed on the hold map, on the stage-structured review map, and on the continuous-delay equation do not transfer to one another. The operative condition is $\det(D\mathcal{P}_{T_r}(X^*) - e^{i\theta} I) = 0$ on the map to which the statement refers, with $D\mathcal{P}_{T_r}(X^*)$ the Jacobian of the review map at its fixed point (the monodromy matrix of the sampled loop).

For a deterministic sampled system, the inter-review flow and the review update combine into a Poincaré map $X_{n+1} = \mathcal{P}_{T_r}(X_n)$, where X_n includes the ecological state, memory, held command, and any explicitly modelled queue. A fixed point is locally asymptotically stable if and only if every eigenvalue of $D\mathcal{P}_{T_r}(X^*)$ lies inside the unit disk. A verified crossing through the unit circle as the fixed parameter T_r changes is a sampled-data parameter bifurcation. It is not rate-induced tipping, which requires a nonautonomous parameter path and a loss of tracking as the ramp speed changes (Ashwin, Wieczorek, Vitolo, and Cox, 2012).

2.4 The cross-sectional spectral screen

The screen's input layer is a fixed 42-stock cohort of the RAM Legacy Stock Assessment Database v4.66 (Ricard, Minto, Jensen, and Baum, 2012): the annual-review designation (applied from general regime knowledge, not per-stock primary verification) covers the full 58-stock panel, and 42 stocks pass the $n \geq 20$ valid-point filter (series lengths 20–77 yr). The release records stock and assessment metadata and series such as biomass, fishing mortality or exploitation rate, total allowable catch, catch advice, and effort. It does not encode controller sign, the decision rule, or dated decision and deployment queues. The annual-review designation is therefore a criterion applied in this analysis, not a database classification of a stock as extractive or protective.

For each eligible biomass (spawning-stock biomass) series, the analysis removes a linear trend, computes a Lomb–Scargle periodogram (Lomb, 1976; Scargle, 1982), normalises it to unit total power, and integrates power in prespecified bands: target bands A (2.5–5 yr, anchovy class) and B (5–9 yr, sprat class), bracketing the observable-specific dominant biomass peaks near 4 and 8 yr of the archived stage-map diagnostics (provisional records, Section 3.3), with contextual bands C (9–14 yr, cohort/recruitment) and D (14–30 yr, trend/regime). No multiplier-crossing verdict depends on the archived diagnostics. The tested statistic is band-integrated power, compared with a per-series AR(1) red-noise null (coefficient by lag-1 autocorrelation of the linearly detrended series; 200 replicates, seed 7, detrending inside each replicate; Lomb-Scargle on 1,500 frequencies from $1/(2 \cdot \text{span})$ to 0.5 yr^{-1}). Empirical p-values over the 84 target-band cells (42 stocks by bands A and B) are adjusted by the Benjamini–Hochberg false-discovery-rate procedure (Benjamini and Hochberg, 1995), which controls the false discovery

rate, not the familywise error rate. Dependence across stocks (shared climate forcing, shared assessment methods) is acknowledged (Benjamini and Yekutieli, 2001). The reported zero count is the BH-adjusted result. A peak is classified as robust only if it survives the null comparison and the multiplicity adjustment. The cohort table, screen routine, and multiplicity reconstruction (`ram_target_stocks.csv`, `ram_crosssection.py`, `verify_bh.py`) are deposited with the article. The screen is a restricted result: its weight is limited to the refutation stated in Section 4.2, and its null verdict survives the eleven-variant sensitivity battery of Supplementary S9.1.

2.5 Power analysis

Power experiments inject the model-generated effort signal into white noise (multiplicative log-normal, fractional σ) and apply the same band-power statistic. Formally the reported power is $P(\text{reject} \mid \text{injected signal})$ at the 5% level over the design grid (the conventional power-analysis framing of Cohen, 1988). Power is estimated on 100–200 yr synthetic records across noise scales. The demonstration routine and grid driver with the executed configuration and seeds (`power_demo.py`, `power_driver_v2.py`) are deposited with the article. The executed grid configuration is Supplementary S13.1.

2.6 The structured case search

The case search is a curated inventory, not a systematic review. It considered more than thirty systems spanning fisheries, aquaculture, groundwater, surface water, rangeland, wildlife harvest, forestry, and produced-capital markets, under four criteria:

(i) a responsive institutional feedback rather than a one-time cap or ban;
(ii) an independently dateable response or implementation lag;
(iii) no major environmental, biological, or structural driver empirically supported as generating the same outcome, where the comparative rule is that a candidate institutional mechanism must outperform pre-registered alternatives on held-out prediction, phase ordering, or intervention response (since "capable of generating" is not a falsifiable exclusion in open ecological systems);
(iv) individual-resource or individual-station data rather than an aggregate series.

Behind criterion (iii) is an explicit list of the alternative mechanisms that can mimic or replace an institutional cycle: stage/maturation delays and cohort resonance; recruitment suppression versus stock culling; support-pool/nutrient limitation; pollution/waste feedback; predator–prey or climatic forcing; and direct abiotic mining and slow-store exhaustion. The rule is that a candidate institutional mechanism must be compared with these alternatives rather than identified from periodicity alone. No system in the searched set meets all four criteria; the case-screening table (32 systems) and query log are recorded in Supplementary S4.

2.7 The northern cod case: constitutive model and evidence

The northern cod case (NAFO 2J3KL) is treated as a bounded empirical object. Its illustrative constitutive model is the strong-Allee surplus equation:

$$\frac{dS}{dt} = rS\left(1 - \frac{S}{K}\right)\frac{S - \mathfrak{s}}{K - \mathfrak{s}} - C(t), \tag{5}$$

with S the spawning stock biomass, r the intrinsic growth rate, K the unexploited carrying capacity, $\mathfrak{s}$ the unstable threshold, and $C(t)$ the removals. The Schaefer model is the formal limit of this family ($\mathfrak{s} \to -\infty$) in which the factor $(S-\mathfrak{s})/(K-\mathfrak{s})$ is replaced by 1. No value of $\mathfrak{s}$ makes the displayed factor identically 1 (it tends to 1 only in the limit $\mathfrak{s} \to -\infty$), so the Schaefer comparison is a change of growth law, not a parameter specialisation. The constitutive equation

is illustrative, not a fit — and the obstruction class is the one-dimensional autonomous equation with fixed parameters and fixed removals. The displayed equation is autonomous exactly when removals are fixed.

Remark 2.1 (Phase-line obstruction). Let $f : \mathbb{R} \to \mathbb{R}$ be locally Lipschitz, and let $x(\cdot)$ be a nonconstant solution of the scalar autonomous ODE $\dot{x} = f(x)$. Then x is strictly monotone; it never attains an equilibrium value and cannot cross an equilibrium in either direction. Consequently, an exact path that repeatedly rises and falls across a common interval is incompatible with any single fixed scalar autonomous model.

Proof. If x is nonconstant on an interval and $x(t_1) = x(t_2)$ with $t_1 < t_2$, then by Rolle's theorem there exists $\tau \in (t_1, t_2)$ with $\dot{x}(\tau) = 0$, i.e. $f(x(\tau)) = 0$. By uniqueness the solution through $(\tau, x(\tau))$ is the constant equilibrium solution, a contradiction. Hence every solution is strictly monotone or constant. If a trajectory met an equilibrium value x^* at any time, it would be constant thereafter and, by uniqueness applied backwards, before; a nonconstant solution therefore never attains an equilibrium value and cannot cross one in either direction. A path that rises and falls across a common interval attains some level twice with opposite directions, contradicting monotonicity. □

Lemma 2.2 (Extra-loss threshold shift, conditional).

Statement. Assume $0 < \mathfrak{s} < K$. (i) Constant loss. In the constitutive model with an additional constant removal $C > 0$, the production function is replaced by $f_C(S) = f(S) - C$; the positive equilibria of the modified function are the solutions of $f(S) = C$, the smaller one (the effective threshold) lies to the right of $\mathfrak{s}$, and at the production maximum the two positive equilibria coalesce and beyond it disappear — all conditional on the solutions existing, i.e. on C being below the production maximum. For $C = 0$ the smaller positive equilibrium is $\mathfrak{s}$; the strict rightward shift holds for $C > 0$.

(ii) Proportional mortality. With extra mortality $M_x > 0$ the modified production is $f_M(S) = f(S) - M_x S$; positive equilibria solve $f(S)/S = M_x$, $S = 0$ remains an equilibrium (unlike case (i), where $\dot{S}|_0 = -C < 0$ drives the model below zero unless the removal is constrained near zero or the state equation is modified), and the smaller modified positive root lies strictly above $\mathfrak{s}$ (in $(\mathfrak{s}, S_h)$, where S_h is the vertex of the per-capita curve $h(S) = f(S)/S$).

Proof. The constitutive production $f(S) = rS(1 - S/K)(S - \mathfrak{s})/(K - \mathfrak{s})$ vanishes exactly at 0, $\mathfrak{s}$, K; it is negative on $(0, \mathfrak{s}) \cup (K, \infty)$, positive on $(\mathfrak{s}, K)$, and its derivative vanishes exactly at

$$S_\pm = \frac{K + \mathfrak{s} \pm \sqrt{K^2 - \mathfrak{s}K + \mathfrak{s}^2}}{3}, \qquad 0 < S_- < \mathfrak{s} < S_+ < K,$$

so f has exactly one critical point in $(\mathfrak{s}, K)$ — the unique maximum $S^* = S_+$ with value $f_{\max} = f(S^*) > 0$ — and f is strictly increasing on $(\mathfrak{s}, S^*)$ and strictly decreasing on (S^*, K).

(i) The equilibria of $f_C(S) = f(S) - C$ solve $f(S) = C$, so none lie outside $(\mathfrak{s}, K)$, where $f \le 0 < C$. For $0 < C < f_{\max}$, the intermediate value theorem on $[\mathfrak{s}, S^*]$ ($f(\mathfrak{s}) = 0 < C < f_{\max} = f(S^*)$) and on $[S^*, K]$ ($f(K) = 0 < C < f_{\max}$) supplies one root in each interval, and strict monotonicity makes each root unique. The smaller root satisfies $S_1(C) \in (\mathfrak{s}, S^*)$ — the effective threshold lies strictly to the right of $\mathfrak{s}$. As $C \uparrow f_{\max}$ the two roots approach S^* from either side (coalescence), and for $C > f_{\max}$ the equation $f(S) = C$ has no solution: the two positive equilibria have disappeared.

(ii) Write $f_M(S) = f(S) - M_x S = S\,(h(S) - M_x)$ with $h(S) = f(S)/S = \frac{r}{K - \mathfrak{s}}(1 - S/K)(S - \mathfrak{s})$, a quadratic with roots $\mathfrak{s}$ and K, negative on $(0, \mathfrak{s}) \cup (K, \infty)$, and unique maximum $H = h(S_h)$ at the vertex $S_h = (\mathfrak{s} + K)/2$. Then $f_M(0) = 0$: zero remains an equilibrium in every case. The positive equilibria solve $h(S) = M_x$; for $0 < M_x < H$ the same two-interval argument gives exactly two roots, the smaller in $(\mathfrak{s}, S_h)$ — the threshold shift — with

coalescence at $M_x = H$ and disappearance for $M_x > H$, where $f_M(S) = S(h(S) - M_x) < 0$ on $(0, \infty)$.

In case (i), by contrast, $f_C(0) = -C < 0$, so $\dot{S}|_0 = -C$ drives the state below zero unless the removal is constrained near zero or the state equation is modified — as claimed. □

The case evidence is the assessment record (the recovery-potential and sequential stock assessments: DFO, 2011, 2016, 2022, 2024). The crash window (1991–1995) and the subsequent windows are documented by the assessment table of DFO CSAS SAR 2016/026 (Table A2), produced by the Northern Cod Assessment Model (NCAM; Cadigan, 2016), together with the dated governance events (the moratorium announcement of 2 July 1992; the reopening of 26 June 2024 with a total allowable catch of 18 kt). The survival column $\exp(-M)$ is a transformation of the reported instantaneous mortality estimate, not an independently observed survival series.

3 Results

3.1 Forward invariance of the sampled process

A first well-posedness check on the sampled controller is that the state space on which the sample-and-hold dynamics act — nonnegative stock, nonnegative memory signal, admissible effort — is preserved by the review map. This check comes first because every later result relies on the state remaining inside this set between reviews.

Proposition 3.1 (Forward invariance of the sampled process). Assume:

(H0) $r, K, q, E_{\max}, \tau_m, Z_{\mathrm{ref}}, \Delta_{\mathrm{ref}} > 0$.

(H1) $N(0) \geq 0$, $Z(0) \geq 0$, $E_0 \in [0, E_{\max}]$.

(H2) The signal map Φ is non-negative.

(H3) Every effort command is projected by $\Pi_{[0,E_{\max}]}$.

(H4) Any between-review deployment interpolation remains in the convex interval joining consecutive commands. The main model takes $E(t) = E_n$, which satisfies (H4) trivially; the proposition is stated for the hold-or-interpolate class.

Then $N(t) \geq 0$, $Z(t) \geq 0$, and $E(t) \in [0, E_{\max}]$ for every time at which the sampled solution exists.

Proof. Proceed by induction over review intervals. Suppose that at a review time t_n one has $N(t_n) \geq 0$, $Z(t_n) \geq 0$, and that the command to be held or interpolated lies in $[0, E_{\max}]$. Projection places the next command in the same closed interval; a hold retains an endpoint of that interval, and any convex interpolation $E(t) = (1 - \theta(t))E_n + \theta(t)E_{n+1}$ with $0 \leq \theta(t) \leq 1$ also remains in $[0, E_{\max}]$. On $[t_n, t_{n+1}]$, Eq. (1) gives

$$N(t) = N(t_n) \exp\left\{ \int_{t_n}^{t} \left[r\left(1 - \frac{N(s)}{K}\right) - qE(s) \right] ds \right\} > 0$$

for a positive initial value, while if $N(t_n) = 0$, uniqueness gives the identically zero solution on the interval; N cannot become negative. Set $\nu(t) = \Phi(qE(t)N(t) - S(N(t))) \geq 0$. Variation of constants in Eq. (2) with $\tau_m > 0$ gives

$$Z(t) = e^{-(t-t_n)/\tau_m} Z(t_n) + \frac{1}{\tau_m} \int_{t_n}^{t} e^{-(t-s)/\tau_m} \nu(s)\, ds \;\geq 0.$$

Boundedness from above holds automatically on the stock equation and is used in the induction: on $[t_n, t_{n+1})$ the stock solves the logistic equation under held effort, and $\dot{N} \leq 0$ at $N = K$ (the harvest term is non-negative), so $N(t) \leq \max\{N(t_n), K\}$ throughout the interval. All three inequalities therefore hold through t_{n+1}; they hold at $t_0 = 0$ by hypothesis, and induction proves them on every review interval for which the sampled solution exists. □

Positivity, boundedness, persistence above a threshold, and viability are distinct properties; Proposition 3.1 establishes positivity and effort admissibility only. The institutional implication

is modest but essential: the projected Euler step does not push the state outside its specified admissible set on a single review interval, and any later stability or instability claim is therefore a statement about dynamics inside that set, not about escape from it.

3.2 The rapid-review limit and what it does not establish

Define $u = (N, Z, E)$ and the continuous no-delay system $\dot{u} = G(u)$ assembled from Eqs. (1)–(2) with the effort law evaluated continuously. Suppose G is continuously differentiable with bounded derivative on a compact neighbourhood containing the compared trajectories on $[0, T]$, assessments equal the contemporaneous state, no additional decision or deployment queue is present, projection is inactive, and the sampled and continuous systems share the initial state. The frozen-effort flow followed by the explicit effort step then has a one-step defect of order $O(T_r^2)$ relative to the exact flow of G. The usual discrete Gronwall estimate yields an $O(T_r)$ review-time error and uniform convergence on every fixed finite horizon as $T_r \to 0$ (the sampled-data approximation framework of Nešić and Teel, 2004).

This is a numerical-approximation property, not a finite-review stability theorem. It gives no uniform-in-time estimate, does not preserve stability automatically, and does not control any trajectory-classified response region reported below. It excludes active projection, delayed or erroneous assessment, a deployment queue, and stochastic updates. In particular, it neither identifies a finite T_r with a continuous discrete delay nor implies that sufficiently small positive T_r is stable when the continuous target is unstable. The converse caution is equally important. Under the additional assumptions of hyperbolicity of the continuous equilibrium and C^1-consistency of the sampled map with projection inactive locally, the multipliers satisfy $\mu_j(T_r) = 1 + T_r\lambda_j(A) + O(T_r^2)$ (Stuart and Humphries, 1996), so local stability persists for all sufficiently small positive review intervals provided every continuous eigenvalue satisfies $\Re\lambda_j < 0$; if any $\Re\lambda_j > 0$, the sampled equilibrium inherits instability for sufficiently small T_r. The finite-horizon result therefore neither proves nor refutes stability transfer; the transfer hypotheses are what any such claim would have to supply. The institutional reading is that "rapid review" is not a stability guarantee — a governance loop that reviews frequently approximates a continuous controller only on finite horizons, and the question of whether short review intervals stabilise or destabilise the equilibrium is decided by the spectrum of the map, not by the rapid-review limit.

3.3 Response regions of the delayed-recruitment review maps

The delayed-recruitment records, classified from long-horizon trajectories on the stage-structured review map, locate exploratory response regions by stock class (the delayed-recruitment oscillation literature of Gurney, Blythe, and Nisbet, 1980, supplies the classical reference point). These are archived records whose generating computation was never attached (Section 2.2): they are reported below at provisional status, and the pre-registered stage-map reconstruction of Section 3.4 does **not** reproduce the multi-year windows on the anchovy- or sprat-class parameterisation of that plant (reporting convergence there, with a separate long-horizon band that has no archived counterpart). The regions below are therefore archived, unreproduced records, not results of the reconstructed object:

- **Anchovy-class:** persistent tail oscillation near $T_r \approx 3$–4 yr, with a weak response at $T_r = 2$ yr; annual-review trajectories converge over the tested grids for every tested effort-response value.
- **Sprat-class:** persistent tail oscillation near $T_r \approx 6$–12 yr.
- **Cod-class:** trajectories converge to equilibrium for every tested $T_r \in [1, 20]$ yr, although the corresponding continuous-delay calculation has an oscillatory interval — a convergence-over-tested-grid record, not a stability theorem for every history.

- **Slow-stock class:** for $r \in (0.01, 0.05)$ yr^{-1}, oscillation over part of the tested grid below approximately 20–30 yr and convergence at longer review intervals, with transition brackets between approximately 30 and 50 yr depending on r. This one-sided pattern does not contradict rapid-review consistency: the continuous no-delay target is itself unstable over part of the slow-r regime, so small-T_r trajectories may approximate an unstable target on finite horizons. The record is conditional on the delayed-recruitment variant and is not a general claim that slower review stabilises governance; the dominant timescales are centuries (the archived slow-stock cohort cycle: $P \approx 250{-}360$ yr), beyond the length needed to resolve multiple cycles in most institutional records.

The corresponding continuous-delay calculations on the delayed-recruitment parameterisation — g is its maturation delay, and the response regions are located through the product rg — locate response regions near $rg \approx 1.5$–1.6: for $g = 2$ yr, $r \in (0.77, 0.81)$ yr^{-1} at $\eta = 0.914$ (the uncalibrated Candidate-A baseline; Appendix A) with a delay interval of approximately 2.6–7.8 yr; for $g = 1$ yr, the high-r interval is approximately 1.565–1.585 yr^{-1} with a delay interval of 1.6–3.5 yr. These are finite-grid trajectory summaries, not closed analytical stability regions.

A τ_0 decomposition separates, within the delayed-recruitment operator, cohort-driven from institutional-delay-induced crossings: at $g = 5$ yr and $\eta = 0.914$ the raw r-window [0.0076, 0.3705] narrows to the institutional-only [0.2660, 0.3285], and fish-range crossings from 12 to 8, under τ_0-stable classification by nonlinear ground truth. The separation is within one operator; it does not isolate the cross-operator effect compared in Section 4.1. The $g = 5$ middle band overlaps slow cod-class growth rates but remains confounded with cohort resonance. That operator differs from Section 3.7's four-state loop. Full windows, the twelve-cell decomposition grid, and the middle-band table are tabulated in Supplementary S9.2.

Multiplicative assessment-error experiments preserve the anchovy-class trajectory region through 30% error and produce no noise-induced persistent tail oscillation at annual review in the tested ensemble. This is a robustness summary for the specified multiplicative perturbation only. The archived diagnostics carry observable-specific dominant peaks — approximately 4 yr in anchovy-class biomass with 12 yr in effort, and approximately 8 yr in sprat-class biomass with 60 yr in effort. These are reported only as observable-specific dominant peaks over the analysed windows: components of one stationary periodic orbit cannot have different fundamental periods, and harmonics, subharmonics, modulation, or transient spectral content are unresolved; no decomposition has established whether the baseline term, signal regularisation, or another controller component dominates these responses. The archived amplitudes indicate percent-scale biomass excursions and order-one effort excursions relative to equilibrium. The exact percentages are not used as effect-size estimates, because the archived record does not distinguish peak-to-peak range from half-range and the approach to the large effort response contains a long transient.

3.4 The operator spectra

The logistic hold-map core permits a closed-form expression for the held logistic flow; the linearised review map is then constructed from this flow, the signal dynamics, and the update derivatives. With effort held at E, the logistic flow over one review interval is, for $a(E) = r - qE \neq 0$,

$$N(t_{n+1}^-) = \frac{a(E)\, N(t_n^-)\, e^{a(E)T_r}}{a(E) + \frac{r}{K} N(t_n^-)\big(e^{a(E)T_r} - 1\big)},$$

and $N(t_n^-)\big/\big(1 + \frac{r}{K} N(t_n^-) T_r\big)$ for $a(E) = 0$. In computations, the limit branch is used when $|a(E)| < 10^{-12}$. The signal flow enters through equation (2) with this forcing, and the linearised pre-review monodromy is assembled from the held-flow state transition, the assessment

derivative at $Z(t_{n+1}^{-})$, and the update derivative of equation (4). The exact held-assessment controller — the separating comparator that isolates the Euler step — is the exponential update (the exact update; its review map the exact map) $e_{n+1} = e^{C_E T_r} e_n + \frac{e^{C_E T_r} - 1}{C_E} C_Z z_n$ for $C_E \neq 0$, and $e_{n+1} = e_n + T_r C_Z z_n$ for $C_E = 0$, with $e_n = E_n - E^*$ and $z_n = Z_n - Z^*$ the deviations from the compared fixed point — the exact solution, over one review interval, of the linearised effort law $\dot{e} = C_E e + C_Z z$ with the flowed end-of-period assessment z held, where C_E and C_Z are the coefficients of that linearisation (the partial derivatives of F_B with respect to E and Z at the compared fixed point; unrelated to the removals C of the cod case), and equation (4)'s increment is the forward-Euler discretisation of the same linear object. This comparator is analytical, not a proposed institutional rule. Comparing its monodromy with the forward-Euler map's separates review cadence, zero-order holding, and Euler discretisation. That comparison is reported here: executed and verified on the companion delay study's identical hold map (Abaee, 2026; same effort-law bracket, same softplus signal, same forward-Euler monodromy reproducing the 47.536 yr crossing reported here), the exact-update annual spectral radius is $\rho = 1.00035$ (Euler 1.00055); the 47.536 yr and 79.143 yr crossings are command-step artefacts of the forward-Euler update; the exact map's single unit-circle crossing is ≈ 6.5 yr; and the restabilising direction is preserved.

On the logistic hold-map core, the undelayed equilibrium is already unstable, so annual review is unstable. On the computed crossing set, the sampled equilibrium has a complex unit-circle crossing — the spectral signature of a Neimark–Sacker bifurcation, nonlinear non-degeneracy not verified — at $T_r^{\mathrm{UC}} = 47.536$ yr under the forward-Euler command step and at ≈ 6.5 yr under the exact update, with the same restabilising direction. For an institution implementing the incremental Euler rule these crossings are dynamically real; they are artefacts only relative to the exact update.

The complete crossing record. On the logistic hold-map core the unit-circle crossing record over $[0.2, 200]$ yr is complete (Figure 2). It is computed from the linearised pre-review monodromy on a 200,001-point scan with bisection refinement (deterministic; it reproduces the numbers reported here on the range they cover). The forward-Euler update crosses twice on the extractive channel — a complex pair at $T_r = 47.536$ yr (unstable $\to$ stable) and a real -1 multiplier at 79.143 yr (stable $\to$ unstable), stable band $[47.536, 79.143]$ yr — and once on the protective channel — a real -1 at 2.306 yr (stable $\to$ unstable), stable only on $[0.2, 2.306]$ yr. The exact update crosses once on the extractive channel — a complex pair at 6.501 yr (unstable $\to$ stable) at the baseline parameter vector (Table 3), stable on $[6.501, 200]$ yr — and never on the protective channel, which is stable at every tested interval with maximum spectral radius 0.9967. Both Euler crossings of the extractive channel are therefore command-step artefacts relative to the exact map's single crossing, and the Euler protective instability beyond 2.306 yr is likewise an artefact: the exact protective map is stable throughout.

Spectral margins of the annual-review verdict. The annual-review instability is a near-unit-circle record, and its margins are part of the record. The exact update's annual spectral radius is $\rho = 1.00035$ — a modulus exceeding unity by 3.5×10^{-4} — with the forward-Euler update at 1.00055 and the protective channel at 0.9838 (maximum 0.9967 over the tested range). The multiplier types are reported with the crossings — a complex pair at the exact map's single 6.501 yr crossing (the Neimark–Sacker signature) and at the Euler extractive crossing (47.536 yr), and real -1 multipliers at the Euler crossings (79.143 yr extractive, 2.306 yr protective) — while the dominant multiplier's angle θ at the crossing and the continuous eigenvalue λ to which Section 3.2's transfer relation refers are listed in Table 3 (Appendix A). At a margin of this size the verdict is conditional on the monodromy's numerical construction — the stated precision record is the 200,001-point scan with bisection refinement; the linearisation construction and floating-point detail are archive items — and on the parameter vector listed in Table 3. What the margin does not affect is the ordering across updates and channels (exact 1.00035 below Euler 1.00055; protective 0.9838 below both) or the crossing directions;

the annual-instability verdict is read at that status. Near-unit-circle behaviour of this size is read as high sensitivity of the verdict to the parameter vector, not as a robust claim about annual-review institutions in general.

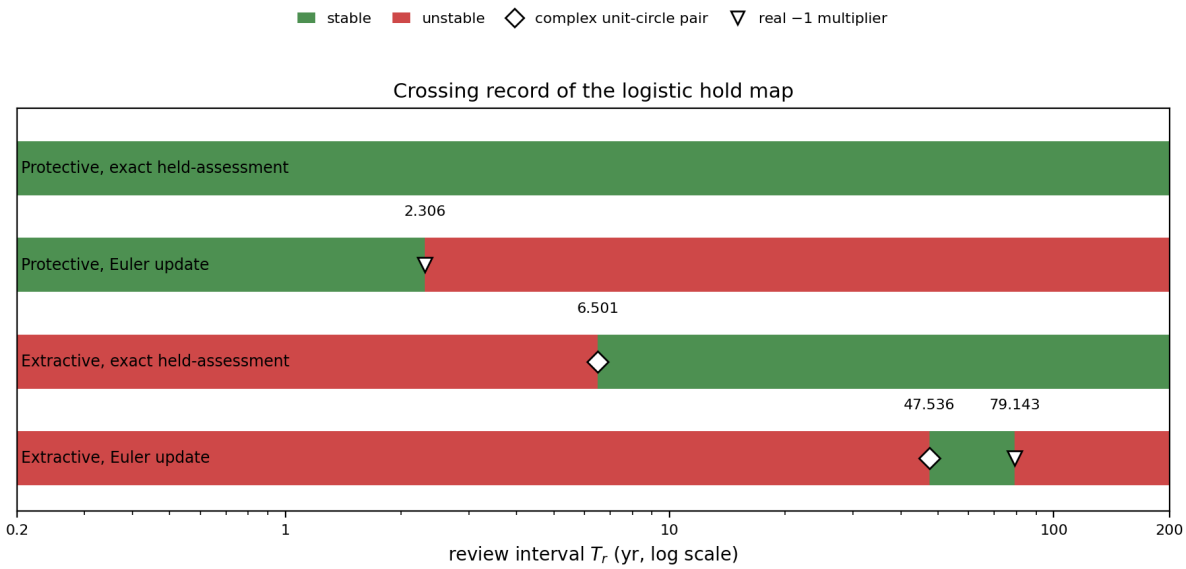


Figure 3: Crossing record of the logistic hold map: stable (green) and unstable (red) review-interval ranges for the four update × channel combinations (forward-Euler and exact updates on the extractive and protective channels), with crossing markers (diamond: complex unit-circle pair; triangle: real −1 multiplier). Annual review is nominally unstable on both extractive updates ($\rho = 1.00035$ exact, 1.00055 Euler; protective 0.9838), conditional on the numerical construction and parameter vector (Spectral margins paragraph); the Euler-reported 47.536 and 79.143 yr crossings are command-step artefacts relative to the exact update's single 6.501 yr crossing, and the protective Euler crossing at 2.306 yr is likewise an artefact — the exact protective map is stable at every tested interval (maximum spectral radius 0.9967).

The one-plant operator contrast. With the crossing record complete, the operator comparison can be made on one plant rather than across plants. The same extractive loop under continuous delay carries the interval-enclosed spectral crossing pair of Part I at 3.666 and 150.358 yr; the sampled operator's exact crossing sits at 6.501 yr. The continuous-delay window is delay-stabilising, not delay-destabilising: the undelayed loop is already unstable, the lower crossing at $\tau_- = 3.666$ yr is stabilising, the upper crossing at $\tau_+ = 150.358$ yr destabilising, and the loop is linearly unstable for $0 < \tau < \tau_-$ (Abaee, 2026). Annual review is nominally unstable under sampling ($\rho = 1.00035$, conditional on the numerical construction and parameter vector as stated above), and the same loop under continuous delay is likewise unstable at $\tau = 1$ yr; at $T_r = 8$ yr both operators are stable (the sampled loop restabilised above its 6.501-yr crossing, the continuous loop inside its stabilised window). The operator contrast on one plant is an edge contrast: the sampled operator's restabilising crossing (6.501 yr) sits above the continuous operator's stabilising entry (3.666 yr), and the sampled loop's stability persists to the 200-yr scan ceiling where the continuous loop has already destabilised at τ_+ — the operator moves both edges of the stability window on the same plant, and the cross-plant observation of Section 2.3 is a same-plant one.

The undelayed limit, reconciled explicitly. Three statements about the undelayed loop must be read together. Section 2.3 states the continuous-delay equation and the logistic hold map to be the same feedback loop under two delay operators; Section 3.2 states that under hyperbolicity the sampled multipliers satisfy $\mu_j(T_r) = 1 + T_r\lambda_j + O(T_r^2)$, so that small-$T_r$ sampled stability follows the sign of $\mathrm{Re}\,\lambda_j$; and this section records that the undelayed equilibrium of the hold-map core is already unstable (annual $\rho = 1.00035$). The continuous-

delay calculation fixes the remaining sign: the undelayed linearisation of the same loop violates the Routh–Hurwitz condition, so $\mathrm{Re}\,\lambda > 0$ — the direction the sampled record suggests through Section 3.2's relation, with the sampled instability persisting down to $T_r = 0.2$ yr. Under that sign the records close at the zero-delay limit: the loop is unstable under both operators at zero delay and at $\tau = 1$ yr, and the stabilising crossing at 3.666 yr is the first crossing (there are no crossings in $(0, \tau_-)$; this even count is consistent with the stability-switch principle because the stability state does not change between zero delay and the first crossing) — no additional crossings are required. On this plant, the two operators differ only in where they place the window's edges, as the one-plant contrast above shows. The eigenvalue λ is listed in Table 3; it is a derived eigenvalue, not a parameter, and its sign is read from the companion's undelayed record, and the operator-scoped records stand as stated — annual instability under sampling, instability under continuous delay at $\tau = 1$ yr, and the same-plant edge contrast — with Section 2.3's same-loop statement and Section 3.2's transfer statement consistent with both records.

The protective controller on the same maps. The protective comparator run (sign-reversed response; linearised gains $C_E = -0.850336$, $C_Z = -1.661702$) on the same plant gives exact-update spectral radius 0.9838 at annual review and stability at every $T_r \in [0.2, 200]$ yr (maximum 0.9967). The Euler-run protective instability is an artefact band, not a property of the protective architecture.

Executed comparator evaluation at model level. The protective and fixed-plan comparators are run through the closed loop on the logistic hold-map core (nonlinear plant, $N_0 = 0.95\,N^*$, 800 reviews, metrics on the last 60%, multiplicative assessment error $\{0, 0.3\}$, seeds fixed; the spectral record remains the authoritative result, and the prospective real-system design of Section 4.5 remains unexecuted). Under 30% assessment error with ten seeds per cell: the extractive Euler core crashes to extinction in all ten seeds at $T_r = 8$, 12, and 20 yr, and at annual review its mean depletion frequency (fraction of post-transient reviews with $N < N^*/2$) is 0.22 with mean minimum stock $0.22\,N^*$; the extractive exact update never crashes, with mean depletion falling from 0.07 ($T_r = 1$) to 0.001 ($T_r = 20$); the protective exact update holds the equilibrium at every interval (depletion 0, mean minimum $0.89\,N^*$, rms stock deviation $\leq 0.001K$); the protective Euler update crashes in all ten seeds at $T_r = 20$ yr and shows depletion 0.9996 at $T_r = 12$ — the artefact operating at the trajectory level; in the deterministic baseline with fixed effort set at the equilibrium value, the fixed plan remains at equilibrium — a baseline rest point, not a robust performance result. The linear trajectory distortion between the Euler and exact updates grows with T_r (scaled-norm RMSD ≈ 2 at $T_r = 0.5$ yr, 10^4–10^5 at $T_r = 5$–40 yr, with the Euler path diverging in its artefact bands): the command step is not a small distortion. The stage-map inputs to this evaluation are the archived, unreproduced records of Section 3.3 — annual stability, the anchovy-class 3–4 yr and sprat-class 6–12 yr windows, and robustness to 30% multiplicative assessment error — read at that section's status, not as results of the reconstructed object.

Two consequences follow. First, in the reported simulations relative effort excursions were larger than relative biomass excursions, and any quota-utilisation reading of the signal would require a quota, realised catch, and compliance model, which are not part of the model core. Second, no qualifying example was found in the case search of Section 2.6 of a real small-pelagic system operating a responsive 3–4 yr review, so the window prediction is untested, not falsified. The corresponding qualitative test for slow-regenerating resources is whether a responsive multi-year plan, adjusted against measured change, produces large-amplitude extraction cycles that a frozen multi-year cap does not (Section 4.5).

The stage-map reconstruction record. The stage-output values carry a stated limitation: the stage map's computational record — solver configuration, initial histories, stage registration — is not attached, so those values are not reproducible numerical propositions (Section 2.2). To supply the stage operator's boundary record, a new stage map was specified and pre-registered (plan dated and fixed 2026-09-01, before any run; parameter values and compar-

ison criteria fixed in the plan), and its complete multiplier and trajectory records are reported here. The archived stage-output records are trajectory-classified only (Section 2.2). The reconstruction is a newly specified object: it is not claimed to be the object that produced the archived windows, and the comparison below is a post-hoc consistency check, not a validation of those windows.

Plant. A two-stage delayed-recruitment map with annual steps, $A_{t+1} = s_A A_t + s_J J_t - qEA_t$, $J_{t+1} = f(A_{t-\tau})$ with Beverton–Holt recruitment $f(A) = \alpha A/(1+\beta A)$, $s_A = s_J = e^{-M}$, and the pair (α, β) fixed by steepness and scale: $\beta = (c-1)/A_0$, $\alpha = c(1-s_A)/s_J$, $c = 4h/(1-h)$, $A_0 = 100$ — a scale convention matching the logistic core's carrying capacity. Surplus production $S(A) = s_J f(A) - (1-s_A)A$ replaces the logistic surplus in the deficit signal of equation (2). Every parameter is a cited literature value or a stated convention; none was chosen with reference to the archived windows. The class parameter sets: anchovy $M = 0.90$ yr^{-1} and recruitment delay $\tau = 1$ yr (Pauly and Tsukayama, 1987); sprat $M = 0.40$ yr^{-1} (mid-range of the Baltic SMS keyrun values, 0.3–0.8 yr^{-1}; ICES, 2020) and $\tau = 2$ yr (age at 50% maturity in the Baltic sprat assessment); cod $M = 0.20$ yr^{-1} (the fixed cod-assessment convention; ICES, 2021) and $\tau = 5$ yr (northern cod age at 50% maturity; DFO, 2011); slow-stock $M = 0.045$ yr^{-1} and $\tau = 25$ yr (orange roughy; Branch, 2001). Steepness $h = 0.75$ throughout — the standard default convention when no stock-specific estimate is available — with a sensitivity layer $h \in \{0.6, 0.9\}$. The controller of Section 2.1 is unchanged: equations (3)–(4) in discrete form with T_r varied over $\{1, \ldots, 50\}$ yr at annual internal steps, softplus sharpness $k = 10$ (the comparator machinery's value), contemporaneous exact assessment, and catchability $q = 0.001$ (the hold-map core's value) with a sensitivity case $q = 0.1$ (the archived stage record states no q; both values are reported for the reconstruction). The derivative construction is central finite differences of the review map at the fixed point (step 10^{-6}, chain-rule cross-check to 10^{-4}), and the scan grid is $T_r \in \{1, \ldots, 50\}$ yr at annual internal steps — finite-grid by construction, as the archived record itself was. The extractive channel is primary; the protective channel is computed with quota-tracking gains of Section 2.1 in the linearised review map, the same convention as the logistic protective record.

Records. At the specified controller value every class is stable at annual review — spectral radii $\rho(1) = 0.895$ (anchovy), 0.923 (sprat), 0.956 (cod), 0.994 (slow-stock) — reproducing the archived stage map's annual stability at the tested response value. The protective channel is stable everywhere on the grid (maximum spectral radius 0.968), mirroring the logistic protective record. The extractive channel's only instability at the specified value is a long-horizon band from $T_r = 34$–35 yr to the grid's end for the three faster classes (maximum spectral radius ≈ 1.98 at $T_r = 50$), with no crossings anywhere for the slow-stock class ($\rho(50) = 0.67$); the band is insensitive to the steepness layer. The trajectory classification (2000 review-steps from the specified initial condition — plant at the unfished equilibrium, memory filled, effort at half the equilibrium — tail of 500, relative tail standard-deviation thresholds 2% and 0.1%) agrees with the spectral record for the three faster classes, which converge at every $T_r \in [1, 20]$. In the multiplier-unstable long-horizon band ($T_r \geq 34$ yr) the faster-class trajectories remain biomass-converged or weak (relative tail standard deviation at most 0.6%) while effort excursions grow from approximately 7–15% at band entry to approximately 190% at $T_r = 50$: the instability manifests in the effort channel, to which the classifier — a biomass-tail threshold — is insensitive. The slow-stock class does not agree with its own multiplier record: it shows a persistent oscillation at $T_r = 10$ yr only (relative tail standard deviation 5.5%; dominant period ≈ 24 yr in both stock and effort; effort excursion 537% against a biomass excursion of 18%), converging at every other grid point, against a multiplier record with no crossings anywhere on the grid. A persistent oscillation at a linearly stable fixed point is bistability, a non-decayed transient, or a classification-threshold artefact; the record does not distinguish among them, and the cell is reported as a disagreement between the reconstruction's two records, not as an agreement, and is not used inferentially. Under 30% multiplicative assessment error the

anchovy and sprat cells remain converged (maximum relative tail standard deviation 0.0009); the cod cells weaken at $T_r = 10$ and 20 yr (converged $\rightarrow$ weak, maximum 0.0025) while its annual-review cell stays converged (0.0004); and the slow-stock class oscillates persistently at the long-horizon cells $T_r = 30$–50 yr (2.1–2.6%, against a stable multiplier record without error — cells read as noise-driven variance rather than oscillation, since the 2% relative tail threshold is not noise-adjusted) while its $T_r = 10$ cell weakens from its error-free oscillation and its $T_r = 20$ cell weakens (0.017). The catchability sensitivity analysis is informative: at $q = 0.1$ every class is unstable at annual review ($\rho(1) \geq 1.29$); the three faster classes destabilise on $[1, 18{-}31]$ yr, restabilise, and re-enter an unstable band at 36–42 yr, while the slow-stock class is unstable across the entire grid ($\rho(50) = 7.8$) — the reconstructed stage loop's stability profile depends strongly on the catchability scale, which is unspecified for this plant.

Comparison with the archived windows (criteria pre-registered; verdicts from the first complete run):

Criterion (archive)	Reconstruction	Verdict
anchovy persistent oscillation at $T_r = 3$–4 yr	converged at both	MISMATCH
anchovy weak response at $T_r = 2$ yr	converged	MISMATCH
anchovy annual convergence	$\rho(1) = 0.895$	MATCH
sprat persistent oscillation at $T_r = 6$–12 yr	converged throughout	MISMATCH
cod convergence for every $T_r \in [1, 20]$	converged throughout	MATCH
slow-stock oscillation below 20–30 yr, convergence at 30–50 yr	oscillation at $T_r = 10$, convergence at 30–50	MATCH

Reading. The reconstruction reproduces the archived record's qualitative features — annual stability on the stage map, the protective channel's stability, cod-class convergence over the 1–20 yr grid, the slow-stock oscillation-then-convergence pattern, and the effort-versus-biomass excursion ordering — but not the anchovy 3–4 yr or the sprat 6–12 yr response regions: on this specified plant family those classes converge at every review interval, and the only instability the reconstructed loop produces is its own long-horizon band, entering at 34–35 yr and extending to the grid's end, which has no counterpart among the archived windows. The comparison, however, is uninformative rather than a non-reproduction. The stage map states no catchability; the table is computed at the imported value $q = 0.001$; and at the sensitivity case $q = 0.1$ every verdict in the table flips — every class unstable at annual review ($\rho(1) \geq 1.29$), the slow-stock class unstable across the entire grid — so a match or a mismatch at either value is a statement about this specified family at an unspecified scale, not an adjudication of the archived values, whose generating computation is not available. The reconstruction therefore supplies the stage operator's boundary record as a new, fully specified object, whose comparison with the archived windows is a consistency check rather than a validation.

Three verification records qualify the crossing record. First, a one-at-a-time sensitivity battery perturbs each baseline parameter (r, K, q, $E_{\max}$, η, δ_0, Z_{ref}, Δ_{ref}, τ_m, δ) by $\pm 5\%$ and $\pm 10\%$, reproducing all eight committed values before extension (Supplementary S11). The exact-map crossing is highly sensitive to r, K, q, $E_{\max}$, η, and Δ_{ref} — several $\pm 10\%$ cells remove the crossing or move it past 13 yr — moderately sensitive to τ_m, affected by δ (removed at -10%), and insensitive to δ_0 and Z_{ref} (crossing moves < 0.07 yr). The protective channel is stable in every battery cell (maximum $\rho = 0.9971$), so the channel contrast survives uniform $\pm 10\%$ perturbation. The spectral-radius curves for all four update–channel combinations are Figure 4.

Second, the equilibrium fishing mortality $F^* = qE^*$ organises the q-sensitivity: q and $E_{\max}$ perturbations act on the crossing almost entirely through F^* ($q \times 1.05$ and $E_{\max} \times 1.05$ give $F^* = 0.00219$ with crossings at 13.797 and 13.790 yr respectively; $\times 1.10$ gives 18.377 and 18.367 yr), and a decade-spanning q grid shows review-instability turning on as F^* rises through the baseline 0.00209, with the equilibrium infeasible ($N^* < 0$) once $F^* > r$ (Supplementary S11). Assessed fishing mortality across the 42-stock cohort (median mean-F 0.37) and a 454-stock broader pool (median 0.21) exceeds the illustrative baseline F^* by two orders of magnitude, sitting beyond the model feasibility boundary; quantitative thresholds are therefore scale-conditional (Supplementary S12).

Third, the 6.501 yr crossing is a verified Neimark–Sacker bifurcation of the nonlinear sampled map: the finite-difference Jacobian matches the monodromy to 10^{-6}, the crossing is transversal ($d|\mu|/dT_r = -0.000683$) and non-resonant through fourth order, and the first Lyapunov coefficient is sign-definite across a sevenfold finite-difference step range ($a(0) \approx -0.0008$, supercritical). Long-horizon iterates rotate at the predicted frequency (rotation number 0.027–0.029 against $\theta_0/2\pi = 0.0279$); the invariant circle itself is not directly exhibited because convergence at $|\rho - 1| \sim 10^{-4}$ is too slow for the tested horizons (Supplementary S11).

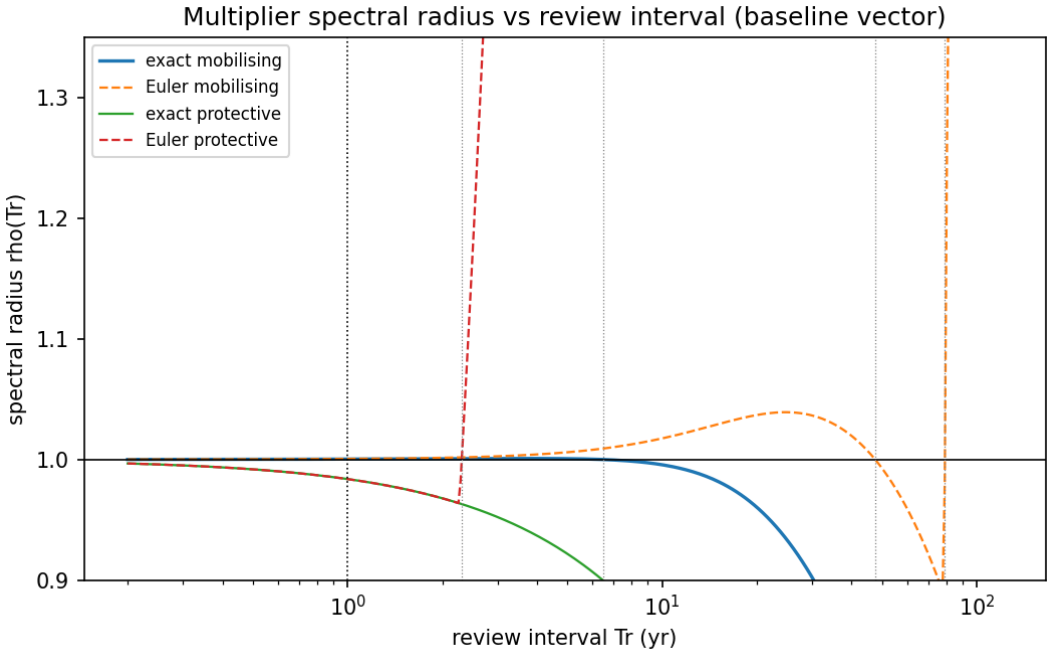


Figure 4: Spectral radius $\rho(T_r)$ against review interval at the baseline parameter vector (Table 3) for the four update–channel combinations. Horizontal line: the unit circle; dotted verticals: the 6.501, 47.536, 79.143, and 2.306 yr crossings and the annual review (1 yr). The exact mobilising curve crosses once; the Euler mobilising curve reports two artefact crossings; the exact protective curve never reaches unity.

3.4.1 Sensitivity of the crossing: why 6.5 is reported as a band

Section 3.4 reports the exact-update mobilising crossing at 6.501 yr. That number is reproducible but it is not identifiable, and reporting it to four significant figures implies a precision the system does not possess. This subsection quantifies the sensitivity and replaces the point with a band.

Why it is ill-conditioned. At the baseline the annual spectral radius is $\rho(1) = 1.00035$ — three and a half parts in ten thousand above unity — and the crossing is where a complex

multiplier pair crosses the unit circle with $\mathrm{d}|\mu|/\mathrm{d}T_r = -6.83 \times 10^{-4}$ per year. Inverting that slope: an error of 10^{-3} in ρ near the crossing displaces the crossing interval by approximately 1.5 yr. Any quantity located by a near-tangential unit-circle crossing of this kind inherits that conditioning.

One-at-a-time sensitivity. Each parameter was perturbed by ±1% and ±2% with all others held at the baseline, and the exact-update mobilising crossing recomputed (Table 10). The crossing moves by more than half its baseline value under a 2% change in K, q or $E_{\max}$.

Table 10: Sensitivity of the exact-update mobilising crossing to one-at-a-time parameter perturbation. Baseline crossing 6.501 yr. The last column is the largest finite crossing shift among the ±2% endpoints, as a percentage of baseline; a non-crossing endpoint is excluded from that percentage. "none" indicates that no unit-circle crossing occurs on $[0.2, 200]$ yr at the tested grid resolution.

	crossing (yr)				worst 2%
parameter	−2%	−1%	+1%	+2%	swing
K (carrying capacity)	none	2.99	8.67	10.37	59.5%
q (catchability)	none	3.44	8.50	10.10	55.4%
$E_{\max}$ (effort cap)	none	3.45	8.50	10.10	55.3%
η (control gain)	5.27	5.92	7.02	7.49	18.9%
r (intrinsic growth)	7.35	6.94	6.04	5.55	14.6%
d_{ref} (reference)	10.78	8.89	2.35	none	65.7%
t_m (memory timescale)	6.60	6.55	6.45	6.39	1.6%
d_0 (baseline mortality)	6.50	6.50	6.51	6.51	0.1%
Z_{ref} (reference)	6.51	6.50	6.50	6.50	0.1%

The verdict itself is fragile, not merely the number. Four of the nine parameters, perturbed by 2% in the directions shown in Table 10, remove the crossing altogether: the loop is stable at *every* tested review interval, so the claim that annual review is unstable and the loop restabilises above a threshold is not merely numerically displaced but qualitatively false. This is the more serious finding, and it concerns the sign of $\rho(1) - 1$, which is 3.5×10^{-4} at baseline.

Joint perturbation. Perturbing the six influential parameters jointly over the corners of a box gives the band:

box	corners with a crossing	crossing range	range as % of baseline
±0.5%	60 of 64	0.87 – 10.67 yr	−87% to +64%
±1%	44 of 64	4.42 – 13.65 yr	−32% to +110%

At ±1%, twenty of sixty-four corners lose the crossing entirely. At ±0.5%, the crossing ranges from under one year to nearly eleven.

What is reported, and what is not. The crossing is therefore reported as *approximately 6.5 yr at the baseline parameter vector, with a plausible range of roughly 1–11 yr under a half-percent joint parameter perturbation, and with the instability verdict itself not robust to a 2% change in four of the parameters.* We do not report a confidence interval: the parameter uncertainties here are not measurements with a stated sampling distribution but conventions, so any interval would be an assumption dressed as an inference. The honest statement is the range swept under stated perturbations, which is what Table 10 and the box above give.

What survives. Three findings of Section 3.4 are unaffected, and they are the ones this paper rests on. (i) The *operator* contrast: the forward-Euler command step reports crossings at 47.536 and 79.143 yr that are artefacts relative to the exact update, and this was verified separately from the crossing location. (ii) The protective channel is stable at every tested interval under the exact update (maximum spectral radius 0.9967), and its smallest stability margin on that grid is approximately 0.0033 ($1 - 0.9967$). At annual review the mobilising instability excess is approximately 0.00035 ($1.00035 - 1$); these two margins refer to different interval sets. (iii) The multiplicity-controlled screen of Section 3 finds no robust institutional cycles, which is a null result and does not depend on the crossing value. The 6.5-year figure is a property of one illustrative parameter vector. It is reported as such, not as a threshold for institutions.

3.5 The selected 42-stock spectral screen

No stock in the screened cohort has a peak in the specified biomass target bands meeting all robustness criteria. Baltic sprat, for example, has a biomass coefficient of variation of 0.40, but its variation is low-frequency and regime-dominated rather than a robust target-band peak. The null carries a three-way restriction: it is not proof of absence, not a comparison of controller signs, and not causal evidence that annual review stabilises anything. On the stage-structured review map (Section 3.3), annual-review stability at every tested response value is consistent with this null; consistency is not a test. The institutional reading is therefore limited: under this screen, target-band cycles are not common in assessed-stock records, and any claim that periodicity itself diagnoses an institutional feedback is unsupported by this cohort. That null verdict is robust: an eleven-variant sensitivity battery — the AR(1) baseline plus ARMA(1,1), trend-stationary-no-detrend, circular block bootstrap at three block lengths, Hodrick–Prescott-cycle and first-difference pretreatments, and piecewise-AR(1) regime surrogates at three break dates, each at the published 200-replicate, seed-7 resolution — returns the BH-adjusted zero count in every variant (smallest nominal p 0.0100, Hodrick–Prescott variant; details and the resolution bound in Supplementary S9.1; the screen's restricted-result status is stated in Section 2.4).

The screen's zero count holds under exploratory alternative bands: a broad 2–10 yr band, management-cadence bands (1–3, 3–8, 8–20 yr), and an 8–80 yr effort-proxy band all return zero Benjamini–Hochberg-significant cells, with the effort-proxy band unresolvable (record span below three upper periods) on all 42 records (Supplementary S11). Figure 5 shows the baseline q-value distribution and per-stock null excess. Exploitation intensity carries no spectral excess once the ENSO confound is removed (Spearman 0.03 with Peru/Chile excluded, against 0.31 before; Supplementary S12). Across a 454-stock broader pool, the archived cohort median buffer-years (1.79) sits at the second percentile of resampled medians, independently re-executed (Supplementary S12).

3.6 Power of the injected-signal screen

On 100–200 yr synthetic records, the sprat-class signal has estimated power 1.0 at noise scale $\sigma = 0.1$ and approximately 0.24–0.58 at $\sigma = 0.3$. The anchovy-class effort signal has power approximately 0.02–0.14 over the tested horizons and noise levels; its longer effort peak and slow amplitude growth can offset its larger relative excursion. These are conditional simulations: no minimum-power guarantee holds per empirical stock, and the 100–200 yr horizons exceed many eligible series. The anchovy-class null is consequently weakly informative under this test, and the sprat-class result is informative only in favourable noise and record-length regimes. The empirical screen is therefore a selected-cohort consistency check and the baseline for a prospective design, not a general test of the extractive mechanism. The sprat-class 1.0 at $\sigma = 0.1$ is trend detection (growth-transient run-up in the 30–120 yr band), not cycle detection, and is fragile at $\sigma = 0.3$.

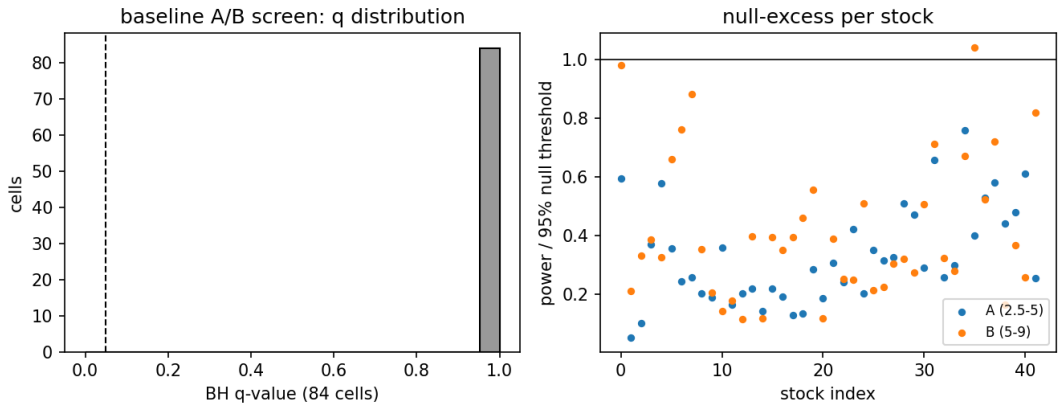


Figure 5: Baseline screen diagnostics. Left: Benjamini–Hochberg q-values over the 84 target-band cells (dashed line: 0.05; no cell significant). Right: per-stock band power relative to the 95% per-stock AR(1) threshold in bands A and B (horizontal line: unity).

The evidentiary separation is complete: the stage-dependent regions, noise experiments, and power estimates do not establish population-wide frequencies or policy effects and cannot be transferred to the extractive hold-map controller or to protective control. The complementary prospective design is specified in Section 4.5 (detail in Supplementary S13.3).

3.7 Structured case search: no qualifying cases

No candidate satisfied all four criteria after primary-source and station-level review. The zero count is an evidential gap, not a disconfirmation: it records that no unconfounded oscillator was found, not that none exists. Bangkok (durably responsive institutional feedback: the groundwater-extraction control regime persisted across multiple review cycles rather than being a one-time cap or ban) and La Mancha Oriental (on the stabilising side before its 2019–2023 extraction relapse) are the closest cases. The unconfounded delay oscillators identified in produced-capital systems (livestock and electricity-capacity cycles) do not contain the autonomously regenerating stock the resource model assumes. The per-system calculations use archived input series and are reported at that status in the Supplementary material (S4); three groups illustrate the screening logic.

Sheridan-6 groundwater (hydrological driver without a feedback rule) and Icelandic cod and haddock under harvest-control rules (rule present, no review-cycle identification) illustrate the confound patterns at case level (Supplementary S4). The four-state prediction is the stage-structured review map's closed loop of four state components (adults A, juveniles J, memory signal Z, held effort E), whose slow-stock class carries the centuries-scale dominant timescales of Section 3.3.

Peruvian anchoveta provides a further discriminator: a robust 3.7 yr catch periodicity with ENSO as candidate driver, evidence of shared periodicity rather than an identified mechanism (Supplementary S4 for the battery, multiplicity family, and confirmatory tests). The subannual review regime lies below the anchovy-class review-interval response region, so the unclassified controller prevents that comparison from testing the mechanism. The 3.7 yr catch periodicity is not a review interval.

Two structural hypotheses may help explain the zero count. The long records needed to test the claim exist primarily for visible crises, and visibility is correlated with a fast non-institutional driver. And on the continuous-delay parameterisation that produces the instability window ($rg \approx 1.5$–1.6) the predicted period is centuries — longer than any institution has held a fixed response rule — while the 3–12 yr windows belong to the stage-structured map. The two period scales are properties of different operators and should not be read as one prediction.

The visible-record pattern — long series repeatedly associated with climate variability, cohort effects, infrastructure changes, or emergency interventions — suggests, without establishing, a selection mechanism in which systems receive intensive monitoring after complex crises; no causal identification of selection bias is claimed.

One documented system sits at the window's edge rather than inside it: the spruce budworm 30–40 yr outbreak cycle is endogenous predator–prey outbreak dynamics rather than institutionally driven (Supplementary S4 for the rate mapping and window comparison). The system corroborates the mechanism class, not the institutional claim.

3.8 The northern cod two-window split

The case's positive content is a descriptive partition, offered as a consistency check of the assessment-window mechanics rather than an empirical rejection of scalar autonomous or single-driver dynamics. **The data split the phenomenon into two events — the crash interpretation is formulation-dependent, and the post-collapse dynamics expose a second identification problem not resolved by the mortality-allocation comparison. The split is the positive content, not a new mechanism.**

The crash window (1991–1995) is documented by the assessment table (DFO CSAS SAR 2016/026, Table A2; Table 2): spawning stock biomass falls from 735 to 10 kt across the five calendar years 1991–1995 while estimated natural mortality reaches 2.2–2.6 yr^{-1}, roughly ten times pre-collapse levels (the pre-collapse reference period, estimator, and uncertainty interval for this ratio are documented with the assessment table; Table 2 begins in 1991). The displayed values are rounded values from the assessment table . The interpretation of the crash is a formulation attribution, not a causal claim. The NCAM M-shift formulation allocates most estimated mortality to natural death, and NCAM's M is an estimated unobserved-death component conditional on model structure — assessment-framework proceedings explicitly caution that unreported fishing deaths may enter de-facto M (Cadigan, 2016). The constrained-M formulation attributes the crash to unreported catch. The constrained-M quantities are unreproduced (Supplementary S13.2 for the quantities and reproduction requirements): they are listed as open problems and stated here as hypotheses, not results.

Table 2. Northern cod spawning stock biomass (SSB) and estimated instantaneous natural mortality (M) from DFO CSAS SAR 2016/026 (Table A2, NCAM M-shift formulation), rounded from the assessment table. The survival column $\exp(-M)$ is a transformation of the reported M — annual survival conditional on the estimated natural-mortality component alone, since fishing mortality and other modelled removals also apply — not an independently observed survival series.

Year	SSB (kt)	M (yr^{-1})	$\exp(-M)$
1991	735	1.002	0.367
1992	382	2.214	0.109
1993	101	2.575	0.076
1994	31	2.331	0.097
1995	10	0.288	0.750
1996	16.05	0.341	0.711
2000	34.42	0.717	0.488
2004	20.07	0.362	0.696
2005	25.18	0.288	0.750
2010	96.91	0.696	0.499
2015	298.65	0.278	0.757

The non-recovery window is the second event. After the moratorium, the series both rises and falls across tens of thousands of tonnes (Table 2; the full annual record with uncertainty

intervals is the assessment table). The phase-line obstruction of Section 2.7 applies as a model-class diagnostic (not an empirical rejection of scalar autonomous models for northern cod): an error-free trajectory with repeated direction reversals cannot arise from a fixed scalar autonomous model with constant removals. The assessment record does not satisfy the diagnostic's two conditions — the SSB series is an assessment estimate rather than an exact trajectory, and removals were not fixed through the window — so the proposition is a statement about model adequacy for this class, not a direct empirical rejection. Two scope statements govern the result. The reliable contradiction is repeated direction reversal: convergence toward K can be arbitrarily slow near degeneracy, so failure to reach an unspecified biomass by an unspecified deadline is not a contradiction. And the obstruction must be separated from rejection under measurement error, process noise, age structure, migration, time-varying mortality, and state-space observation models — the incompatibility is a property of the exact trajectory class, nothing broader.

The subsequent record does not close the second window. Rose (2026), comparing two surplus-production reconstructions spanning 1983–2023 (Rose and Walters, 2019; DFO, 2024), documents that surplus production and stock growth stalled after 2015, with some years negative, and that the stall-point biomass remains controversial. Structural-equation analyses in the same review attribute the production deficit to capelin limitation and harp seal predation rather than fishing alone. For the present analysis the key point is the persistence of the non-recovery: a decade after the rebound reported by Rose and Rowe (2015), the post-2015 production stall is inconsistent with the simplest fixed-parameter surplus-production interpretation considered here, which is precisely what the split's second window — a second identification problem not resolved by the mortality-allocation comparison — states. Read through Lemma 2.2(ii), the seal-predation attribution carries a directional consequence: to the extent predation acts as additional mortality on the stock, the effective threshold facing recovery lies above $\mathfrak{s}$, so predation raises the threshold for re-establishment rather than merely slowing convergence. No predation-mortality rate is estimated here; the lemma is used for the direction of the effect, not its magnitude.

The ecosystem context (Supplementary S5) is background only: the harp seal increase and capelin decline in the mass-balance record (Tam and Bundy, 2019) are descriptive inputs, not causal tests.

Figure 6 plots the Table-2 SSB and mortality series against the two windows.

Table 5 collects the Section 3 results with their evidential status; Supplementary S1 is the complete inventory.

Table 5. Results at a glance: each Section 3 result with its evidential status.

Forward invariance (Section 3.1)	theorem
Rapid-review consistency (Section 3.2)	conditional theorem (finite horizons)
Response regions (Section 3.3)	archived records (provisional)
Crossing record; battery, F-grid, NS records (Section 3.4)	re-execution-verified computation; nominal-tier records (S11)
Screen null; alternative bands; F-excess; ADH (Section 3.5)	re-execution-verified zero count; nominal-tier records (S9.1, S11, S12)
Power analysis (Section 3.6)	independently re-executed computation
Case search (Section 3.7)	executed search (evidential gap, not disconfirmation)
Cod two-window split (Section 3.8)	descriptive partition; hypotheses, not results

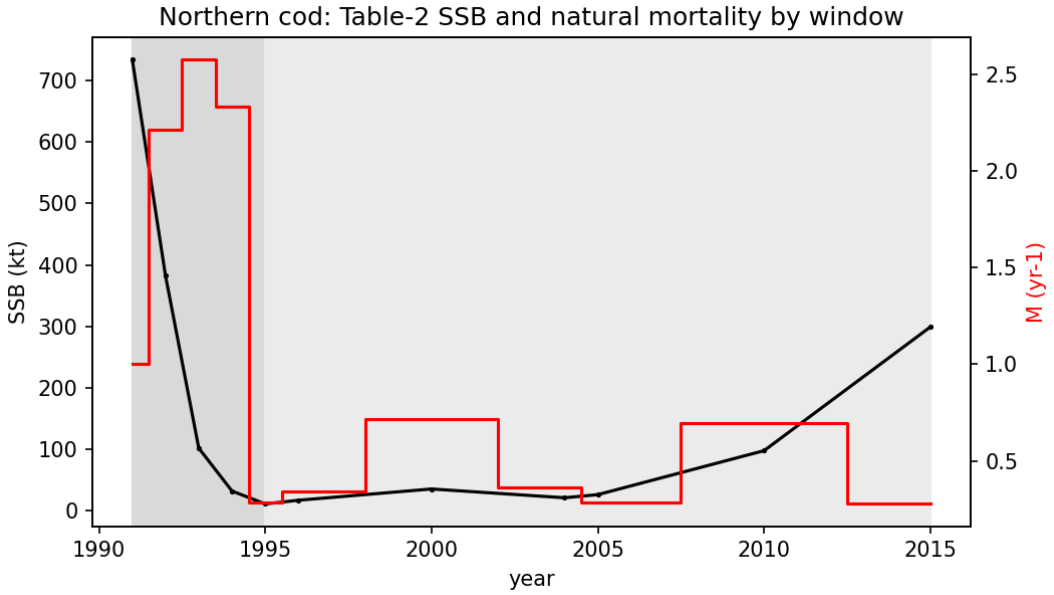


Figure 6: Northern cod Table-2 series by window. Black: spawning stock biomass; red steps: instantaneous natural mortality; shaded: the crash (1991–1995) and non-recovery (1995–2015) windows.

4 Discussion

4.1 Architecture substitution: what changes when the operator changes

The operator-spectra results of Section 3.4 are the paper's core methodological finding. Stability windows are operator-specific: they do not transfer across review-map operators. The same feedback loop — surplus production, an institutional deficit signal, an effort law — carries an archived, unreproduced instability record near 3–4 yr of review under the stage-structured map, convergence over a 1–20 yr grid under that map's cod-class parameterisation, and instability at annual review with a complex crossing near 6.5 yr under the exact logistic hold map (full records, the Euler command-step artefact, and the q-sensitivity caveat in Section 3.4 and its Reading; the plant–operator confound is stated alongside and the operator effect is not claimed isolated).

The complete crossing record sharpens the operator finding: on one plant the exact and Euler updates disagree on the number and location of crossings (Section 3.4). The review cadence is a local spectral design parameter, and the command step is a second, independent one. None of these statements can be obtained from the continuous-delay equation, and none transfers to it. Any empirical or theoretical claim about institutional feedback must therefore state its operator: a stability window located under continuous delay is not a prediction about an institution that reviews periodically, and an annual-review result does not generalise to multi-year plans. The governance architecture itself is a testable component of the hypothesis, not a re-parameterisation detail.

4.2 What the null does and does not establish

The 42-stock screen returns a spectral null, and its value lies in what it refutes: the claim, implicit in treating periodicity as evidence, that robust target-band cycles are common in assessed-stock records. It does not establish the contrary. With anchovy-class power between 0.02 and 0.14 across tested noise and record-length regimes, the screen cannot adjudicate the extractive mechanism for that class; the sprat-class result is informative only in favourable regimes; and no screen of retrospective series can compare controller signs, because the sign is not in the

data. This is the same null discipline that the skeptical re-analysis of the Russell Cycle applied to climate-forced cycles (McManus, Licandro, and Coombs, 2016), transferred to governance-feedback claims: a diagnostic is not a causal claim, and no accumulation of diagnostics converts to causal evidence.

4.3 The cod case at its exact status

The cod case contributes a descriptive partition, not a mechanism: crash-window mortality is formulation-dependent (Section 3.8), and the post-collapse record is not resolved by either mortality-allocation formulation, with the post-2015 production stall persisting under independent reconstructions (Rose, 2026). The obstruction mathematics applies within the scope stated in Sections 3.8 and 4.7(vii). What the case supplies is a falsification benchmark: a well-documented collapse-and-stall record against which prospective designs can be powered and scored.

4.4 A falsification standard for institutional-feedback claims

The model family gains empirical content by specifying the outcomes that count against its mechanism and parameterisation. Five outcomes:

1. If independently measured implementation and deployment lags do not covary with the instability bands predicted under fitted resource parameters, the quantitative delay claim is weakened.
2. If an observed oscillation remains after climate, cohort, and regime forcing are modelled, but its phase relation between decline signal, effort, and biomass contradicts the controller's causal ordering, the mechanism is rejected for that case.
3. If sampled-data analysis removes a continuous-delay band under realistic review rules, the continuous model cannot be used for that institution.
4. If a protective controller dominates the extractive controller across uncertainty without inducing the claimed volatility, no anti-regulation conclusion may be drawn.
5. If realistic record lengths have low power for the predicted signal, field spectral nulls cannot adjudicate the mechanism; prospective or experimental evidence is required. The Icelandic cod case illustrates the bind: its post-1995 variability (coefficient of variation 0.37–0.39; no dominant spectral peak, Supplementary S11) overlaps the continuous-delay band (9.9–20.3 yr) in timescale yet carries no institutional signature, because a competing environmental driver is empirically supported for the same outcome (criterion (iii); Supplementary S4).

4.5 Prospective designs

The retrospective evidence does not identify the institutional mechanism; the constructive content of this paper is the programme that could. Retrospective evidence can still discipline the mechanism where a prespecified contrast exists: the T_r-ranked cross-sectional test across the 42-stock cohort is executed in Supplementary S10 and returns a controlled null — no window-approach gradient, with $T_r \approx 1$ dormancy holding at 0/42 and 1/42 flags in the two bands. Five designs are specified as preregistration targets; none has been executed, and each is intended for preregistration before outcome inspection — no registration identifier or archived protocol exists yet, and none is claimed.

Governance-event panels. For each resource–jurisdiction unit, a source-linked event record dates each stage from raw observation to ecological response, with date uncertainty kept distinct rather than collapsed to one lag (full specification in Supplementary S13.3). The cod case supplies two dated decision events — the moratorium announcement of 2 July 1992 and the reopening of 26 June 2024 with an 18 kt total allowable catch — with two negative constraints:

no governance lead is inferred from annual biomass data, and a fast response does not establish adequacy.

Quasi-experimental timing. Event-study, difference-in-differences, synthetic-control, interrupted-time-series, or state-space intervention designs on schedule variation plausibly independent of the outcome innovation, with preregistered estimands (full specification in Supplementary S13.3). Two coding rules are binding: a change in implementation lag is not coded as a change in T_r, and a nominal schedule change is not a treatment unless it changes a responsive decision opportunity.

Out-of-sample mechanism comparison and the displacement discipline. Held-out comparison across environmental-forcing, cohort-resonance, institutional-delay (specified controller sign), combined, and null time-series models, with frozen splits and declared outcomes (full specification in Supplementary S13.3). The displacement rule: a delay explanation is weakened when it cannot reproduce the pre-registered phase ordering, and displaced when a competitor predicts the held-out observations better.

Controlled and randomized human-in-the-loop experiments. Randomised review cadence and feedback timing or sign in a common simulated renewable-resource environment, with the gain–phase signature as the diagnostic target (full specification in Supplementary S13.3). The randomisation described is a laboratory or advisory-interface design, not a field intervention on a live stock; mechanism-class precedent exists (Costantino, Cushing, Dennis, and Desharnais, 1995; Moxnes, 1998) but is not validation of the institutional equations.

Closed-loop management strategy evaluation. Policy comparison in a closed loop keeping six uncertainty classes distinct (the management-procedure tradition: Punt and Donovan, 2007), crossing review timing, delays, controller sign, observation error, parameter draws, operating-model class, and process-noise regime (full specification in Supplementary S13.3). At minimum, responsive extractive, responsive protective, and fixed-plan controllers are compared; a conclusion that compares only two review intervals inside the extractive class cannot be generalised to governance as a whole. Three wider empirical hypotheses of the programme's general theory — observation aggregation, governance phase ordering, and substitution certificate — are adopted here only as motivation, and otherwise explicitly deferred: governance phase ordering supplies the gain–phase diagnostic target the designs above inherit, while testing any of the three requires prospective, adequately powered designs this paper specifies but does not execute.

4.6 Distributive constraints where reproducible

The social side of the cod case is presented at measurement level (Appendix B): the constituency-definition mismatch table (licence holders against census-subdivision residents; fishing income against all-resident sector income; the unspecified floor), the Statistics Canada community-level series (tables 38-10-0167-01 and 38-10-0168-01), the measured-floor construction (I_k, c_k) with its componentwise non-decline rule, and the admitted world-hook instruments are given in Appendix B, with the full instrument detail and the unreproduced pipeline register in the Supplementary material (S6). What the case does not measure is the point: no licence-holder panel exists, the community series is not one, and no measured floor is operational — the distributive constraints remain stated rather than resolved (Section 4.7 (viii)).

4.7 Limitations

(i) Diagnostics are not causal claims: the spectral null, the response regions, the power values, the case calculations, and the cod split carry their stated types and no more.

(ii) The stage-structured response regions are exploratory finite-grid, trajectory-classified records pending complete stage registration and multiplier analysis; the logistic hold-map

multiplier record is complete (Section 3.4), and the stage operator's multiplier and trajectory record is supplied by the reconstruction of Section 3.4 — a newly specified object whose comparison with the archived windows is a consistency check, not a validation. No Poincaré-map multiplier classification is claimed for the archived stage-output values, and they are not reproducible numerical propositions until the original computational record is attached. The effort scale is uncalibrated: results are reported at imported catchability values, and reparameterising the controller in fishing mortality ($F = qE$) is the stated next step for the stage operator.

(iii) The two review-map operators are distinct: statements computed on the hold map, on the stage-structured map, and on the continuous-delay equation do not transfer to one another.

(iv) The screen is a selected-cohort consistency check whose power is high only in favourable noise and record-length regimes; the null is not proof of absence and adjudicates nothing about controller sign; the AR(1) null may understate low-frequency power. The eleven-variant battery of Supplementary S9.1 holds the zero count under ARMA(1,1), trend-stationary, block-bootstrap, detrending, and regime-shift nulls; the remaining masked-peak risk runs via overestimated null power (unmodelled observation-error or retrospective-bias variance raising the threshold), not the underestimation above.

(v) The structured case search does not independently disconfirm the mechanism: finding no eligible cases under these criteria is not evidence against it.

(vi) The cod case establishes a descriptive partition, and no mechanism is identified.

(vii) The obstruction mathematics concerns exact trajectories of the fixed-parameter, fixed-removals autonomous class and is not a rejection under measurement error, process noise, age structure, migration, time-varying mortality, or state-space observation models.

(viii) The social object is not operationalised: the human series is not assembled, the normative floors are unoperationalised, and documenting the gap is the finding.

(ix) The prospective designs are preregistration targets, not executed studies, and they do not retroactively change the status of the retrospective results.

4.8 Design principles for the governance clock

The results support four design principles for the decision clock (Box 2).

Box 2. Design principles for the governance clock.

Principle	Content
Treat timing as a control variable	The review interval is optimised alongside harvest limits in management strategy evaluation; annual review is not assumed stabilising.
Match the rule to the clock	Protective rules tolerate institutional rigidity; extractive rules require frequent review to contain destabilisation.
Decouple observation from decision	High-frequency monitoring does not require high-frequency revision; multi-year plans are stable under protective rules.
Design for structural breaks	Scheduled review is paired with asynchronous emergency triggers (circuit breakers) that bypass the review clock during regime shifts.

Implications for management design. Eight implications follow for review-interval choice.

(1) The review interval is a control variable: stability is computed per interval, not per loop (Section 3.4).

(2) Annual review is not automatically stabilising: the baseline annual verdict is unstable with a near-unit-circle margin (Section 3.4).

(3) Protective rules are more robust across review intervals than extractive rules: the exact protective map is stable at every tested interval (Section 3.4).
(4) Fixed and responsive plans are not equivalent instruments: a held multi-year command and an annually revised rule are different operators (Sections 2.6 and 4.4).
(5) Emergency triggers are modelled separately from scheduled reviews: structural breaks between reviews are invisible to the fixed clock (Sections 3.8 and 4.5).
(6) Management strategy evaluation treats the review interval as an explicit factor alongside the control rule (Section 4.5).
(7) Spectral periodicity is not evidence of institutional cycles: the screen's null carries no controller-sign information (Sections 3.5–3.7).
(8) Observation, assessment, decision, deployment, and response lags are not collapsed into one governance lag (Section 2.1).

Beyond fisheries, the framework is reusable as a modelling discipline: it answers how periodic governance should be represented in ecological models, how the stages of the advice chain should be kept distinct, and how modellers can test whether a stability claim survives a change of decision operator.

The logistic core is illustrative; the structural results are the operator-dependence of the stability verdict, the finite-horizon scope of the rapid-review limit, and the non-equivalence of one-step and exact updates as a caution within the studied class.

Sample-and-hold architectures beyond fisheries face the same clock-mismatch failure mode: five-yearly climate pledges held between global stocktakes (UNFCCC, 2015) align a political review clock with ecological and environmental clocks only by design, not by default.

5 Conclusion

Periodic review is the operating architecture of fisheries governance, and it is not well represented by either of its standard formalisations. The sample-and-hold model specified here makes the governance clock an explicit object: seven time objects, a projected explicit controller, and a spectrum computed per operator rather than per loop. Its empirical layer contains one case whose positive content is a descriptive split — the northern cod two-window partition — and a falsification discipline for the rest: a selected-cohort screen with no target-band discoveries and stated power, a zero-count case search with its structural hypotheses, and a falsification standard with five outcomes that count against the mechanism. The mechanism is falsifiable, and the five prospective designs specify how it would be falsified, comparatively supported, or left unresolved. Until those designs are executed, the empirical layer stands as stated here: a well-posed mechanism, an exploratory computational record, limited screen power, a case search that does not disconfirm, and one descriptive split. The decision clock — the timing and form of policy revision — is therefore a design variable with stability consequences.

Appendix A. Registration and reproducibility record (consolidated)

This appendix collects the registration and reproducibility requirements stated in Sections 2.2, 2.4, 2.5, 3.3, and 3.4. The convention: an analysis is pre-registered when its plan is dated and fixed before any run, as the stage-map reconstruction of Section 3.4 is; computational artifacts (solver configuration, seeds, identifiers, calibration records) are archived with the materials. The consolidated requirements:

- The solver configuration and initial histories are registration requirements; until that computational record is complete, the stage-output values carry provisional status (Sections 2.2 and 3.3).
- The RAM stock identifiers and the eligibility table are archived with the computational materials (Section 2.4).

- The full null-calibration record — AR(1) coefficient estimation, detrending inside each null replicate, missing-data treatment, and the number of Monte Carlo replicates — is archived with the computational materials (Section 2.4).
- The simulation code and seeds are archived with the computational materials (Section 2.5).

Two further consolidated records. First, the logistic-core parameter vector (r, K, $E_{\max}$, δ_0, Z_{ref}, Δ_{ref}, τ_m; $\delta = \log 2/10$ is stated in Section 2.1) and the linearised fixed point (N^*, E^*, Z^*) with its interiority to the nonsmooth regions of Φ_k and $\Pi_{[0,E_{\max}]}$ are printed in Table 3, with the code constants block pinned per row; the monodromy's numerical construction and the continuous eigenvalue λ and crossing angle θ of Section 3.4's records remain part of the computational record. Second, the Data availability statement records the deposit status of the same requirements (including the reconstruction's dated plan, campaign code, and output tables); the prospective designs of Section 4.5 remain preregistration targets with no registration identifier or archived protocol. The full discharge register is Supplementary S8.

Table 3. Parameter values for the logistic hold-map core and the stage-structured reconstruction. No value here is newly computed except the linearised-gains, crossing-angle, eigenvalue, and solver rows, which are computed in Supplementary S11; the baseline-core vector is printed here, with the code constants block pinned per row, and values not stated remain in the computational record (this appendix). Time is in years; biomass, effort, and signal scales are bare model units (K, $E_{\max}$, Z_{ref}, and Δ_{ref} set the scales).

Object	Parameter	Value as stated	Stated at
Logistic hold-map core	catchability q	0.001	Section 3.4 (stage Plant paragraph, as the hold-map core's value)
Logistic hold-map core	softplus sharpness k	10	Section 3.4 (Plant paragraph)
Logistic hold-map core	effort-response coefficient η	0.914	This table; constants block `droop_test.py` L49–58 @ 24c980cd; inherited illustrative baseline (Candidate-A value from the companion delay study, whose institutional coefficients are explicitly uncalibrated: Abaee, 2026); the delay-Hopf window opens between $\eta = 0.5$ and 0.7 and the $r = 0.02$ inclusion threshold sits at $\eta^* \approx 0.85$, so 0.914 is neither threshold (basis runs deposited; Supplementary S9.2)
Logistic hold-map core	growth rate r (yr^{-1})	0.02 (validation argument)	This table; constants block `droop_test.py` L49–58 @ 24c980cd
Logistic hold-map core	carrying capacity K	100	This table; constants block `droop_test.py` L49–58 @ 24c980cd

Object	Parameter	Value as stated	Stated at
Logistic hold-map core	effort ceiling $E_{\max}$	30	This table; constants block `droop_test.py` L49–58 @ 24c980cd
Logistic hold-map core	effort-law gain δ_0	0.01	This table; constants block `droop_test.py` L49–58 @ 24c980cd
Logistic hold-map core	reference deficit Z_{ref}	1	This table; constants block `droop_test.py` L49–58 @ 24c980cd
Logistic hold-map core	reference scale Δ_{ref}	1	This table; constants block `droop_test.py` L49–58 @ 24c980cd
Logistic hold-map core	memory timescale τ_m (yr)	5	This table; constants block `droop_test.py` L49–58 @ 24c980cd
Logistic hold-map core	memory shift δ	$\log 2/10 \approx 0.069$ (equilibrium memory level)	Section 2.1
Logistic hold-map core	linearised fixed point (N^*, E^*, Z^*); interiority	$(89.55188, \delta, 2.08962)$; $E^* \in (0, E_{\max})$, $\Phi_k(0) = \delta$, $sp'_k(0) = 1/2$	This table; equilibrium routine `base_equilibrium`, `droop_test.py` L86–93 @ 24c980cd; closed form in Supplementary S2.3
Logistic hold-map core	crossing record	47.536, 79.143, 2.306, 6.501 yr; $\rho = 1.00035$, 1.00055, 0.9838, 0.9967	Section 3.4
Logistic hold-map core	scan construction	200,001-point scan with bisection over $[0.2, 200]$ yr	Section 3.4
Logistic hold-map core	linearised mobilising gains (C_E, C_Z)	$(-0.059518, +1.785019)$	Supplementary S11 (sensitivity battery)
Logistic hold-map core	crossing angle θ_0 (exact map, 6.501 yr)	0.175494 rad	Supplementary S11
Logistic hold-map core	continuous eigenvalues λ (undelayed loop)	-0.278147, $+0.000360 \pm 0.027683i$	Supplementary S11
Logistic hold-map core	derivative construction	closed-form monodromy (no finite differences)	Section 3.4
Logistic hold-map core	solver; precision; grid check	matrix exponential (`expm`); IEEE-754 double; 2k/20k/200k scans identical; 50-digit certificate (S12)	Supplementary S11
Stage reconstruction	class natural mortality M and recruitment delay τ	anchovy (0.90, 1), sprat (0.40, 2), cod (0.20, 5), slow-stock (0.045, 25) yr	Section 3.4 (Plant paragraph, with sources)
Stage reconstruction	steepness h	0.75; sensitivity layer $\{0.6, 0.9\}$	Section 3.4
Stage reconstruction	scale and survival	$A_0 = 100$; $s_A = s_J = e^{-M}$; $\beta = (c-1)/A_0$, $\alpha = c(1-s_A)/s_J$, $c = 4h/(1-h)$	Section 3.4
Stage reconstruction	catchability q	0.001; sensitivity 0.1	Section 3.4
Stage reconstruction	softplus sharpness k	10	Section 3.4

Object	Parameter	Value as stated	Stated at
Stage reconstruction	derivative construction	central finite differences, step 10^{-6}, chain-rule cross-check to 10^{-4}	Section 3.4
Stage reconstruction	scan grid	$T_r \in \{1, \ldots, 50\}$ yr at annual internal steps	Section 3.4
Stage reconstruction	trajectory classification	2000 review-steps, tail 500, relative tail standard-deviation thresholds 2% and 0.1%; initial condition at the unfished plant equilibrium, memory filled, effort at half the equilibrium	Section 3.4

Table 4. Notation: every symbol with its definition site. No row introduces a symbol; rows without a site would be gaps, and all rows resolve.

Symbol	Meaning	Defined at
N	resource stock	Section 2.1
Z	institutional deficit signal	Section 2.1
E	extraction effort	Section 2.1
$\mathcal{P}_{T_r}$	review Poincaré map (sampled flow at review instants)	Section 2.1
$D\mathcal{P}_{T_r}(X^*)$	monodromy matrix (Jacobian at the fixed point)	Section 2.1
Π	projection onto $[0, E_{\max}]$	Section 2.1
Φ, Φ_k	signal map; softplus realisation (shift δ, sharpness k)	Section 2.1
δ; δ_0	equilibrium memory level $\log 2/10$; effort-law gain (unrelated)	Section 2.1
F_B; F_B^{prot}	extractive effort law; protective comparator	Section 2.1
$\mathcal{A}_n$; $\mathcal{O}$	assessment operator; observation operator	Section 2.1
$S(\cdot)$; S	surplus production (control sections); spawning stock biomass (cod sections, Table 2)	Section 2.1; Sections 2.7, 3.8
g	maturation delay (delayed-recruitment parameterisation)	Section 3.3
C_E, C_Z	linearised effort-law coefficients (per channel; protective values in Section 2.1)	Section 3.4 (Abaee, 2026, equation (3))
A, J, Z, E	four-state loop: adults, juveniles, memory signal, held effort	Section 3.7
$\mathfrak{s}$, K, $C(t)$, M, M_x, F	cod case: unstable threshold, unexploited capacity, removals, natural mortality, extra mortality, fishing mortality	Sections 2.7, 3.8; Table 2

Appendix B. Distributive constraints where reproducible

Where the social side of the cod case can be presented, it is presented at measurement level. The worst-off relevant population is a modelling choice — a named constituency — and the candidate constituency for 2J3KL is the registered inshore harvesters and licence holders. The measurement record against that candidate is the following mismatch table, each row stated rather than resolved:

Object	Candidate definition	Available measure	Mismatch	Required data
Population	registered licence holders	census-subdivision residents	not the same population	licence microdata
Income	fishing income	all-resident sector income (incl. aquaculture, processing)	includes non-fishing income	administrative income
Floor	specified cutoff (I_k, c_k)	none operational	not specified	specified instrument and cutoff

The available community-level series (Statistics Canada tables 38-10-0167-01 and 38-10-0168-01; mean income from fishing falling from 32.2% to 25.6% across 43 fishing-dependent census subdivisions, 2016–2021) is all-resident employment income including aquaculture and processing. The constituency must therefore either be redefined to the census-subdivision populations, or licence-holder microdata or administrative data must be obtained — the community table is not a licence-holder panel. A measured floor is a pair (I_k, c_k) — an instrument and a cutoff — a measurement object distinct from the norm that motivates it, and the componentwise rule is that the arrangement fails as soon as any measured margin $m_k(G,t) = I_k(G,t) - c_k$ is negative; the conjunction admits no compensatory master scalar. Non-decline is a normative rule, not automatically an empirically justified floor: baseline, cohort, inflation and purchasing-power treatment, uncertainty, attrition, acceptable variation, authority, and structural diversification all must be specified before a floor measures anything. The instruments admitted as world-hooks (the global Multidimensional Poverty Index of Alkire and Foster, 2011; the World Bank Poverty and Inequality Platform) carry release and vintage locks, and what they cannot capture — who pays for persistence, who may use remaining variety, whose knowledge counts, future persons — is stated as a boundary and left open rather than filled with a scalar score. The full instrument detail and the unreproduced pipeline register are in the Supplementary material (S6).

Cross-regime conclusion

The two parts address a shared governance question — how the timing and form of institutional response affect deployed effort and local stability — but they formalize distinct operators, not one interchangeable latency parameter. Part I studies a continuous response delay τ; Part II studies a periodically updated sample-and-hold map with review interval T_r. Read together, the results support a comparison of operators, while delimiting what cannot be transferred between them.

Two latency operators, two scoped comparisons. At the declared baseline parameterization of the continuous-feedback system, stability depends on where the response lag enters and on the feedback rule: intermediate lag stabilises the mobilising equilibrium, while the protective law satisfies a delay-independent local stability condition. In the sampled-review system, the review interval affects the monodromy spectrum; at the stated baseline the exact held-measurement map and one-step Euler command map have different crossing records. The response lag τ and review interval T_r are not interchangeable parameters, and a numerical threshold in either model cannot be carried to the other.

Representation and uncertainty. The demonstrated representation effect is the operator-specific local spectral contrast, especially the artefact crossings reported by Euler but not by the exact sampled map. This result follows from the characteristic-equation and monodromy analyses, independently of the unresolved continuous-delay periodic-orbit folds. The approximately 6.5-yr exact-map crossing is reproducible across the tested discretisation conventions

(6.50–6.73 yr) but moves from 0.87 to 10.67 yr under a half-percent joint parameter perturbation and disappears in twenty of sixty-four one-percent corners. Scheme agreement therefore does not imply parameter robustness.

Interpretive limits. The protective law remains locally stable at the stated continuous-feedback parameterization and throughout the sampled intervals tested; neither observation proves a universal ranking of institutional rules. The numerical five-regime topology is conditional on the specified core, tested histories and finite collocation records; a validated fold theorem for the continuous delay equation remains open. The empirical stock screen's null and the absence of an unconfounded oscillator in the case search show that periodicity by itself does not identify institutional feedback.

Design implication. A stability claim for a periodically reviewed resource should state its feedback rule, review operator, parameter range and disturbance assumptions. Stability boundaries computed at a declared baseline should be checked on the operator the institution actually implements and stress-tested against parameter uncertainty before they are used as design thresholds.

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Supplementary material

The cross-candidate interval-enclosure table is in the main text. Supplementary S1 gives the gated Candidate A derivation, its verified root intervals in $x = \omega^2$, and the branch-safe phase evaluations; the delay supplement also gives the collocation formulation details (mesh $m = 64/96/128$ and the small-branch computation), the fold-certificate gap documentation (the open continuum routes — the free-τ enclosure, the transversality and curvature conditions, and the continuous-delay lift — component by component; S3 and S12 distinguish finite-collocation enclosures from the open continuous-delay lift), the variant registry (the response-sign hypotheses and the registered variants with their obligations), the S7 evidence-tier synopsis, the delayed-recruitment registration records (S9), and the expanded proofs of Corol-

lary 5.1 and Proposition 6.1 (S10, which labels these two objects Corollary 3 and Proposition 5; the label mapping is given in S12) are provided in the unchanged v49 companion file `paper08_governance_delay_v49_supplementary_delay.md` (S1–S12), together with the MPF material of S11 (the η_{crit} sweep and pair-birth structure, the slow-fast intermittency diagnostics, and the sigmoid-gated effort screen record).

Supplementary material for the sampled-governance channel is deposited with this article in the unchanged v49 companion file `paper08_governance_delay_v49_supplementary_governance.md`: the statement inventory with the status of every main-text statement (S1), the remaining proofs and identification results, including the dimensionless identifiability chart (S2), the epistemic-layer results (S3), the case-screening records at full detail (S4), the ecosystem context and toy tests (S5), the distributive-layer detail (S6), the empirical hypotheses with specified tests (S7), the reproducibility register (S8), and the screen-sensitivity battery and stage-scan decomposition records (S9), and the T_r-ranked test record (S10). The register itself accompanies the article; the computational artifacts it inventories (stage registration, solver configuration, query log, nonlinear exact-update comparison) are registration requirements tracked in the register. The stage operator's boundary record is supplied by the pre-registered reconstruction of Section 3.4 — a newly specified object with archived plan, code, and outputs — while the archived stage-output values keep their provisional status until the original computational record is attached.

Declarations

Data availability

All computations are deterministic and re-executable from the registered model registry and runners; the interval pipeline reproduces the displayed enclosures exactly with outward rounding. The five-regime continuation records of Section 8.2 (branch, Floquet, basin, solver, and environment archives; 148 branch records and 87 basin runs) are deposited at `research_program/validated_computations/p4_five_regime_campaign/` with their pre-registered plan and execution report; the rebuilt fold-certification stage sits beside the A025 pipeline at `research_program/validated_computations/a025_fold/`; the second-fold records of Section 8.2 (200 branch records, 45 basin runs, the interval-Krawczyk certificate and its preserved frozen-box failure record) are deposited at `research_program/validated_computations/a025_second_fold/` with their pre-registered plan and execution report. The parameter tables, the certified enclosures, and the numerical protocols are available from the authors and are deposited with the supplementary material. The recovered stage-analysis machinery of Section 9 is archived verbatim with the deposited material and re-executable; its recovery audit and the 21-gate registration campaign that certifies the Section 9 records (gate log and fine-map table, Supplementary S9) are deposited with the supplementary material.

The stock and assessment series analysed in the spectral screen are drawn from the RAM Legacy Stock Assessment Database v4.66 (public release). The northern cod assessment values are published in DFO CSAS SAR 2016/026. The computational record of the response-region analyses (solver configuration, initial histories, stage registration), the nonlinear exact-update comparison of Section 3.4 are registration requirements; the case-screening table and query log are deposited with the article; the corresponding stage-output values carry provisional status until those artifacts are attached. The analysis datasets (RAM stock identifiers and eligibility table; processed spectral series; case-screening table and query log; T_r-ranked test table; ICES cod and haddock series; η-basis window runs) and the analysis code (spectral-screen routines; power-simulation code and seeds; sensitivity-battery code and logs; case-screen, T_r-ranked test, Icelandic-cod audit, and η-basis scripts with their logs) are deposited with the article. The stage-

map reconstruction of Section 3.4 is fully specified and will be deposited with the article: the dated preregistration plan (parameter table, comparison criteria), the campaign code, and the five output tables (equilibria, multiplier record, trajectory classification, robustness experiment, steepness sensitivity). Community-level social values are drawn from Statistics Canada tables 38-10-0167-01 and 38-10-0168-01 with the population-mismatch caveat stated in the text.

Declaration of competing interest

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Author contributions

A.A. conceptualized the entire work, wrote, reviewed and edited the manuscript.

AI declaration

GLM (Z.ai), Qwen (Alibaba Cloud) and DeepSeek AI assisted with drafting and iterative review. The author reviewed and edited the manuscript and takes responsibility for its content.